

Exhibit 103

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

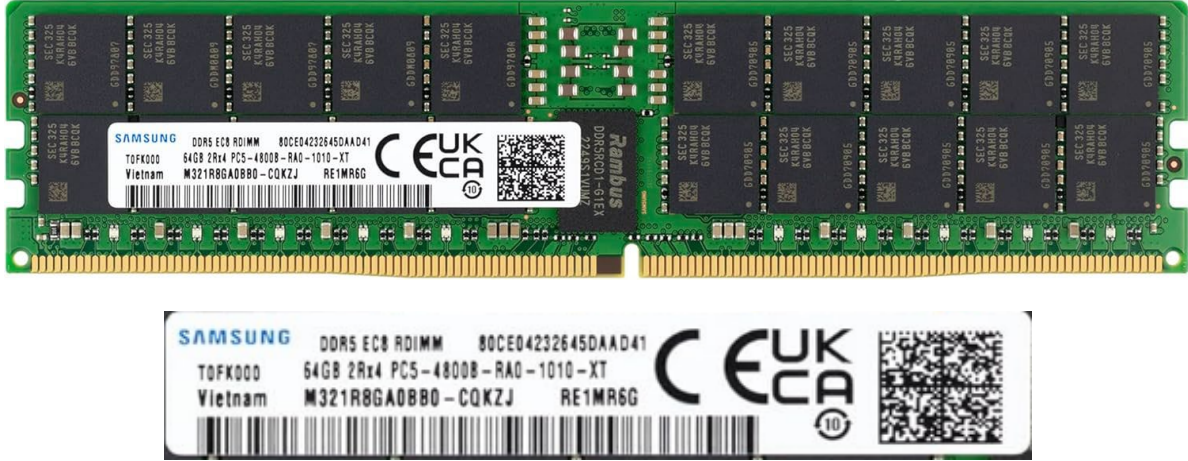
EXHIBIT 103

Domestic Industry Claim Chart for Claims of U.S. Patent No. 12,675,407 (the “’407 Patent”) Based on Samsung Practicing ’407 Products

As shown in the chart below, the Samsung Practicing ’407 Products, which include Samsung which include Samsung DDR5 RDIMMS and MRDIMMS, practice at least claim 1 of the ’407 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Samsung Practicing ’407 Products do not practice the ’407 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Samsung Practicing ’407 Products, such differences are insubstantial and the Samsung Practicing ’407 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Samsung Practicing ’407 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Samsung Practicing ’407 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
[1.pre] A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising:	To the extent the preamble is limiting, each of the Samsung Practicing '407 Products,¹ such as Samsung's imported Registered DIMMs ("RDIMMs") includes a memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines. Examples of Samsung DIMMs are shown below. The Samsung Practicing '407 Products each include a "memory module:"  Ex. CC-116 (Samsung 4800MHz PC5-38400): Samsung 64GB DDR5 4800MHz PC5-38400 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0BB0-CQK.

¹ Analysis for certain limitations below uses RDIMM format as an example, but based on currently available information MRDIMMs would operate in materially the same way in all relevant parts. See JEDEC Press Release (planned features for MRDIMMs include "Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption").

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	A photograph of a Samsung 64GB DDR5 5600MHz PC5-44800 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0PB0-CWM. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number W00J120, and a barcode. The label also includes the text "DDR5 EC8 RDIMM 80CE01234005418888" and "64GB 2Rx4 PC5-5600B-RA0-1010-XT". Ex. CC-117 (Samsung 5600MHz PC5-44800): Samsung 64GB DDR5 5600MHz PC5-44800 ECC RDIMM 2Rx4 (EC8 10x4) Dual Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R8GA0PB0-CWM. A photograph of a Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number K08ND10, and a barcode. The label also includes the text "DDR5 EC8 RDIMM 80CE03251213251911" and "64GB 2Rx4 PC5-4800B-RA0-1010-XT". Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).

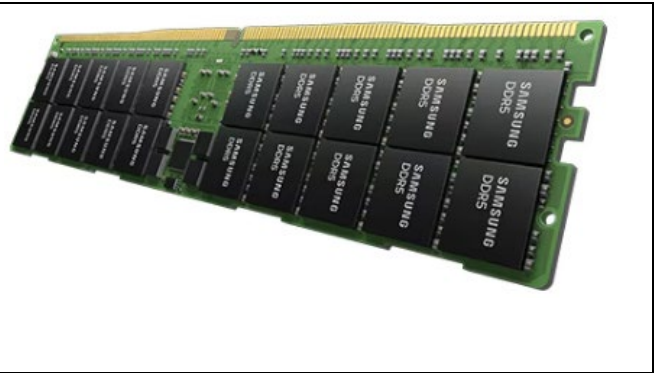
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number W07FD10, and a barcode. The label also includes the text "DDR5 ECR RDIMM", "80CE01251501799891", "Korea", and "M321R8GA0PB2-CCPWF". The module is populated with 16 black memory chips, each labeled "SEC513 K8A0H8 6VP8CCP". Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP). A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module, similar to the one above. The label includes the text "DDR5 ECR RDIMM", "80CE01251501799891", "Korea", and "M321R8GA0PB2-CCPWF". The module is populated with 16 black memory chips, each labeled "SEC513 K8A0H8 6VP8CCP".

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The top photograph shows a Samsung MRDIMM module from a top-down perspective, highlighting its green printed circuit board (PCB) and the gold-plated pins along the bottom edge. The module is populated with numerous dark blue memory chips. The bottom photograph is a close-up of the white label affixed to the module. The label contains the following information: 'SAMSUNG' in blue, 'DDR5 E8 MRDIMM', the part number 'M327R8GA0EB0-CLVXB', and 'MB1M11G'. It also includes a barcode, a QR code, and the text 'W068040', '64GB 2Rx4 PC5-80/56B-MA0-1211-XT', 'Korea', and the 'EUK' logo. An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. <i>See</i> Exs. CC-111, CC-112, CC-113. “DIMM” in RDIMM and MRDIMM stands for dual inline memory module. For example, Samsung’s web pages for its RDIMM, LRDIMM, and UDIMM products are preceded by an overview page labeled “memory modules for extensive use” that explains “Samsung’s memory modules are designed for a wide range of applications to deliver the best performance with low power requirements.” Ex. CC-95 (Samsung DRAM Modules). <i>See also</i> Ex. CC-114 (“Samsung has developed a Multi-Ranked Buffered Dual In-Line Memory Module (MCRDIMM) This module”); Ex. CC-94 (“Module Handling Guide” covering DDR5 UDIMMs, MRDIMMs, RDIMMs, and SO-DIMMs). The memory module of the Samsung Practicing ’407 Products are further operable in a computer system having a memory controller and a system bus. For example:

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Claim 1	Evidence of Use
	Samsung's web site advertises various dual in-line memory modules "for a wide range of applications." Ex. CC-95 (Samsung DRAM Modules). For instance, Samsung advertises registered DIMMs ("RDIMMs"), including ones with ECC capability, as suitable for server applications. <i>Id.</i> RDIMM Registered DIMM Include a register for enhancing the clock, command and control signals Error correction is available by adding 8-bit parity signals Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration Application : Server  Additionally, the Samsung Practicing '407 Products are JEDEC compliant, further demonstrating they are memory modules operable in computer systems having memory controllers and system busses. <i>See, e.g.:</i> Samsung first entered the DDR5 PMIC market in 2017. After making this move, the company released its first PMIC for DDR5 DRAM modules in 2021. Previously, the mainboard supplied power to the module through the PMIC. With the transition to DDR5, however, JEDEC has mandated that the PMIC must be embedded internally within the module. As a leader in the field, Samsung manufactures products that abide by JEDEC standards and consistently aims for the best efficiency in the industry, as well as robust design. This approach is not limited to PMICs, however. The company is also expanding the lineup of products necessary for DDR5 modules, making the goal of providing customers with a total solution a top priority. Ex. CC-104 (Samsung Optimized DDR5 DIMM Solutions) at 1.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

1.0 DDR5 REGISTERED DIMM ORDERING INFORMATION

[Table 1] Ordering Information

Part Number ¹⁾	Density	Organization	Component Composition ²⁾	Number of Rank	Height
M321R2GA3BB0-CQK##	16GB	2Gx80	2Gx8(K4RAH086VB-BCQK) * 10	1	31.25mm±0.15mm
M321R4GA3BB0-CQK##	32GB	4Gx80	2Gx8(K4RAH086VB-BCQK) * 20	2	31.25mm±0.15mm
M321R4GA0BB0-CQK##	32GB	4Gx80	4Gx4(K4RAH046VB-BCQK) * 20	1	31.25mm±0.15mm
M321R8GA0BB0-CQK##	64GB	8Gx80	4Gx4(K4RAH046VB-BCQK) * 40	2	31.25mm±0.15mm

NOTE :

1) "##" : Bucket field for PMIC/RCD information.

2) "QK" : 4400Mbps 36-36-36, 4800Mbps 40-39-39.

2.0 KEY FEATURES

[Table 2] Key Parameters for 4800Mbps Speed Bin

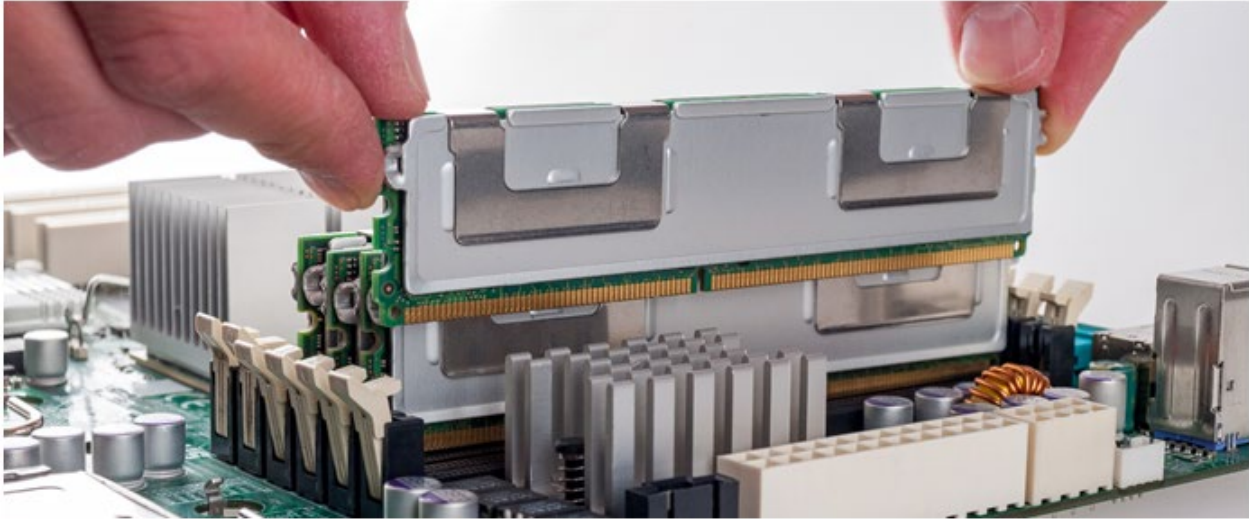
Speed	DDR5-4400	DDR5-4800	Unit
	36-36-36	40-39-39	
tCK(min)	0.454	0.416	ns
CAS Latency	36	40	nCK
tRCD(min)	16.000	16.000	ns
tRP(min)	16.000	16.000	ns
tRAS(min)	32.000	32.000	ns
tRC(min)	48.000	48.000	ns

• JEDEC standard compliant

- VDD = VDDQ = 1.1V (1.067V(-3%) ~ 1.166V(+6%)).
- VPP = 1.8V (1.746 (-3%) ~ 1.908 (+6%)).
- 2200MHz f_{CK} for 4400Mb/sec/pin, 2400MHz f_{CK} for 4800Mb/sec/pin.
- 32 Banks (8 Bank Groups).
- Programmable CAS Latency (posted CAS): 22,26,28,30,32,36,40,42.
- Programmable CAS Write Latency (CWL) = RL-2.
- 16-bit prefetch.
- Burst Length: 16 by default. 8 with tCCD=8, which does not allow gapless READ or WRITE, where BC8 and BL32 refer to CA5 BL*=L.
- Bi-directional Differential Data-Strobe.
- Internal ZQ calibration via Multi-Purpose Command (MPC) - ZQcal start and ZQcal Latch.
- On Die Termination (ODT) via Mode Register setting : RTT_PARK, RTT_WR, RTT_NOM_WR, RTT_NOM_RD.
- Average Refresh period 3.9us at lower than T_{CASE} 85°C, 1.95us at 85°C < T_{CASE} ≤ 95°C.
- Connectivity Test Mode (TEN) is supported.
- Asynchronous Reset.

Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 5.

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Claim 1	Evidence of Use
	This standard defines the electrical and mechanical requirements for 288-position, 1.1 Volt (VDD and VDDQ), DDR5 Registered (RDIMM), Double Data Rate (DDR), Synchronous DRAM Dual In-Line Memory Modules (DIMM). These Registered DDR5 SDRAM DIMMs are intended for use in server, workstation, and database environments. Ex. CC-88 DDR5 Registered Dual Inline Memory Module (RDIMM) Common Standard, JESD305B.01 v. 3.00, JEDEC, (Feb. 2026) (“JEDEC 305B”) at 1. As seen above, Samsung’s RDIMMs each include edge connections with gold fingers which are configured to fit into a memory slot connector of a server computer’s motherboard (system board) for communication with a memory controller via a system bus. <i>See also, e.g.,</i>  Ex. CC-120 (Samsung, “Memory VR changes in the system by DDR5 adoption”) (showing a DIMM being inserted into a memory slot having a system bus on a motherboard of a computer system).

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Claim 1	Evidence of Use
	Ex. CC-70 (DDR5 Memory Channel Scaling Performance). System busses with which the Samsung Practicing '407 Products are operable each include one or more clock signal lines, control/address (C/A) signal lines and data signal lines. For example, Samsung's web site described its RDIMMs as each "[i]nclud[ing] a register for enhancing the clock, command and control signals." <i>See</i> Ex. CC-95 (Samsung DRAM Modules): RDIMM Registered DIMM Include a register for enhancing the clock, command and control signals Error correction is available by adding 8-bit parity signals Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration Application : Server

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Claim 1	Evidence of Use
	This indicates that the system bus includes clock, command and control signals. See also, e.g., Ex. CC-95 (Samsung DRAM Modules): DDR5 Module RDIMM MRDIMM UDIMM ECC UDIMM SODIMM ECC SODIMM  RDIMM Registered DIMM Samsung RDIMM uses a register to buffer command and control signals, improving signal integrity and system stability for server platforms. With ECC support and x4/x8 organizations across single- and multi-rank options, RDIMM provides dependable performance for enterprise workloads. Capacity Up to 256 GB Speed Up to 8,000 Mbps Voltage 1.1 V
	The system bus also includes data signal lines, as illustrated by an exemplary Samsung’s RDIMM datasheet, showing interconnects between a host (memory system’s memory controller) and a DRAM device (e.g., one on an RDIMM) for DQ (data) signals, as well as clock and data strobe (DQS) signals:

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Claim 1	Evidence of Use
	Figure 10. Example of DDR5 Memory Interconnect (Ex. CC-89 at 42, Fig. 10). As another example, a representative Samsung RDIMM datasheet show that the RDIMM includes CA and CS control/address signal pins, CK_t/c clock signal pins, and DQ and CB data signal pins:

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																		
	[Table 5] PIN Description <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>CA[6:0]_A CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </table> <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>SDRAM data strobes (positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>SDRAM data strobes (negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </table> Ex. CC-89 at 9.	Pin Name	Description	CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus	CS[1:0]_A CS[1:0]_B	SDRAM Chip Select	PAR_A PAR_B	SDRAM Parity input	CK_t	SDRAM Clocks (true/positive)	CK_c	SDRAM Clocks (complement/negative)	ALERT_n	SDRAM alert for CRC error	RESET_n	Set DRAM to known state	PCAMP	Control and Monitor Port	HSCL	I2C/I3C-Basic Host Sideband Bus Clock	HSDA	I2C/I3C-Basic Host Sideband Bus Data	HSA	I2C/I3C-Basic Host Sideband Bus Address	LBDQ	Loopback data output:	Pin Name	Description	DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B	CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B	DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)	DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.	VIN_BULK	DIMM Power Supply from system to PMIC	VIN_MGMT	DIMM Power Supply from system to PMIC	VSS	Power supply return (ground)	RFU	Reserved for future use	LBDQS	Loopback data strobe output
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

[Table 4] PIN Configuration

Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side
1	VIN_BULK	145	VIN_BULK	50	VSS	194	DQ31_A	96	CB0_B	240	VSS
2	RFU	146	VIN_BULK	51	CB0_A	195	VSS	97	VSS	241	CB2_B
3	VIN_MGMT	147	PWR_GOOD(FAIL_n)	52	VSS	196	CB2_A	98	CB1_B	242	VSS
4	SCL	148	SAA	53	CB1_A	197	VSS	99	VSS	243	CB3_B
5	SDA	149	RFU	54	VSS	198	CB3_A	100	DQ0_B	244	VSS
6	VSS	150	RFU	55	DQ64_A_t	199	VSS	101	VSS	245	DQ2_B
7	DQ9_A	151	VSS	56	DQ64_A_c	200	DQ69_A_cTDQ69_A_c	102	DQ1_B	246	VSS
8	VSS	152	DQ2_A	57	VSS	201	DQ69_A_tTDQ69_A_t	103	VSS	247	DQ3_B
9	DQ1_A	153	VSS	58	CB4_A	202	VSS	104	DQ60_B_t	248	VSS
10	VSS	154	DQ3_A	59	VSS	203	CB6_A	105	DQ60_B_c	249	DQ65_B_cTDQ65_B_c
11	DQ60_A_t	155	VSS	60	CB5_A	204	VSS	106	VSS	250	DQ65_B_tTDQ65_B_t
12	DQ60_A_c	156	DQ65_A_cTDQ65_A_c	61	VSS	205	CB7_A	107	DQ4_B	251	VSS
13	VSS	157	DQ65_A_tTDQ65_A_t	62	ALERT_n	206	VSS	108	VSS	252	DQ6_B
14	DQ4_A	158	VSS	63	VSS	207	RESET_n	109	DQ6_B	253	VSS
15	VSS	159	DQ6_A	64	CS0_A_n	208	VSS	110	VSS	254	DQ7_B
16	DQ5_A	160	VSS	65	VSS	209	CS1_A_n	111	DQ8_B	255	VSS
17	VSS	161	DQ7_A	66	CA0_A	210	VSS	112	VSS	256	DQ10_B
18	DQ6_A	162	VSS	67	VSS	211	CA1_A	113	DQ9_B	257	VSS
19	VSS	163	DQ10_A	68	CA2_A	212	VSS	114	VSS	258	DQ11_B
20	DQ9_A	164	VSS	69	VSS	213	CA3_A	115	DQ61_B_t	259	VSS
21	VSS	165	DQ11_A	70	CA4_A	214	VSS	116	DQ61_B_c	260	DQ66_B_cTDQ66_B_c
22	DQ61_A_t	166	VSS	71	VSS	215	CA5_A	117	VSS	261	DQ66_B_tTDQ66_B_t
23	DQ61_A_c	167	DQ66_A_cTDQ66_A_c	72	CA6_A	216	VSS	118	DQ12_B	262	VSS
24	VSS	168	DQ66_A_tTDQ66_A_t	73	VSS	217	CK_t	119	VSS	263	DQ14_B
25	DQ12_A	169	VSS	74	PAR_A	218	CK_c	120	DQ13_B	264	VSS
26	VSS	170	DQ14_A	75	VSS	219	VSS	121	VSS	265	DQ15_B
27	DQ13_A	171	VSS	KEY				122	DQ16_B	266	VSS
28	VSS	172	DQ15_A					123	VSS	267	DQ18_B
29	DQ16_A	173	VSS					124	DQ17_B	268	VSS
30	VSS	174	DQ18_A					125	VSS	269	DQ19_B
31	DQ17_A	175	VSS	76	CA0_B	220	RFU	126	VSS	270	VSS
32	VSS	176	DQ19_A	77	VSS	221	CA1_B	126	DQ62_B_t	270	VSS
33	DQ62_A_t	177	VSS	78	CA2_B	222	VSS	127	DQ62_B_c	271	DQ67_B_cTDQ67_B_c
34	DQ62_A_c	178	DQ67_A_cTDQ67_A_c	79	VSS	223	CA3_B	128	VSS	272	DQ67_B_tTDQ67_B_t
35	VSS	179	DQ67_A_tTDQ67_A_t	80	CA4_B	224	VSS	129	DQ20_B	273	VSS
36	DQ20_A	180	VSS	81	VSS	225	CA5_B	130	VSS	274	DQ22_B
37	VSS	181	DQ22_A	82	CA6_B	226	VSS	131	DQ21_B	275	VSS
38	DQ21_A	182	VSS	83	VSS	227	PAR_B	132	VSS	276	DQ23_B
39	VSS	183	DQ23_A	84	CS0_B_n	228	VSS	133	DQ24_B	277	VSS
40	DQ24_A	184	VSS	85	VSS	229	CS1_B_n	134	VSS	278	DQ26_B
41	VSS	185	DQ26_A	86	LBD_RSP_A_n	230	VSS	135	DQ25_B	279	VSS
42	DQ25_A	186	VSS	87	LBS_RSP_B_n	231	RFU	136	VSS	280	DQ27_B
43	VSS	187	DQ27_A	88	VSS	232	RFU	137	DQ63_B_t	281	VSS
44	DQ63_A_t	188	VSS	89	CB4_B	233	VSS	138	DQ63_B_c	282	DQ68_B_cTDQ68_B_c
45	DQ63_A_c	189	DQ68_A_cTDQ68_A_c	90	VSS	234	CB6_B	139	VSS	283	DQ68_B_tTDQ68_B_t
46	VSS	190	DQ68_A_tTDQ68_A_t	91	CB5_B	235	VSS	140	DQ28_B	284	VSS
47	DQ28_A	191	VSS	92	VSS	236	CB7_B	141	VSS	285	DQ30_B
48	VSS	192	DQ30_A	93	DQ69_B_tTDQ69_B_t	237	VSS	142	DQ29_B	286	VSS
49	DQ29_A	193	VSS	94	DQ69_B_cTDQ69_B_c	238	DQ64_B_c	143	VSS	287	DQ31_B
				95	VSS	239	DQ64_B_t	144	RFU	288	VSS

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 6 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	V33	75	219	V33
VIN_MGMT	3	147	PWR_GOOD/FAIL_n	Key			
HSCL	4	148	HSA	CA0_B	76	220	RFU
HSDA	5	149	RFU	VSS	77	221	CA1_B
VSS	6	150	RFU	CA2_B	78	222	VSS
DQ0_A	7	151	VSS	VSS	79	223	CA3_B
VSS	8	152	DQ2_A	CA4_B	80	224	VSS
DQ1_A	9	153	VSS	VSS	81	225	CA5_B
VSS	10	154	DQ3_A	CA6_B	82	226	VSS
DQS0_A_t	11	155	VSS	VSS	83	227	PAR_B
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	CS0_B_n	84	228	VSS
VSS	13	157	DU55_A_t, DU55_A_t DBI0_A_n	VSS	85	229	CS1_B_n
DQ4_A	14	158	VSS	LBD, RSP_A_n	86	230	VSS
VSS	15	159	DQ6_A	LBS, RSP_B_n	87	231	RFU
DQ5_A	16	160	VSS	VSS	88	232	RFU
VSS	17	161	DQ7_A	CB4_B	89	233	VSS
DQ8_A	18	162	VSS	VSS	90	234	CB6_B
VSS	19	163	DQ10_A	CB5_B	91	235	VSS
DQ9_A	20	164	VSS	VSS	92	236	CB7_B
VSS	21	165	DQ11_A	DQS9_B_t, TDQS9_B_t DBI4_B_n	93	237	VSS
DQS1_A_t	22	166	VSS	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	VSS	95	239	DQS4_B_t
VSS	24	168	DQS6_A_t, TDQS6_A_t DBI1_A_n	CB0_B	96	240	VSS
DQ12_A	25	169	VSS	V33	97	241	CD2_D
VSS	26	170	DQ14_A	CB1_B	98	242	VSS
DQ13_A	27	171	VSS	VSS	99	243	CB3_B
VSS	28	172	DQ15_A	DQ0_B	100	244	VSS
DQ16_A	29	173	VSS	VSS	101	245	DQ2_B
VSS	30	174	DQ18_A	DQ1_B	102	246	VSS
DQ17_A	31	175	VSS	VSS	103	247	DQ3_BDQ3_B
VSS	32	176	DQ19_A	DQS0_B_t	104	248	VSS
DQS2_A_t	33	177	VSS	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	VSS	106	250	DQS5_B_t, TDQS5_B_t DBI0_B_n
VSS	35	179	DQS7_A_t, TDQS7_A_t DBI2_n	DQ4_B	107	251	VSS
DQ20_A	36	180	VSS	VSS	108	252	DQ6_B
VSS	37	181	DQ22_A	DQ5_B	109	253	VSS
DQ21_A	38	182	VSS				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

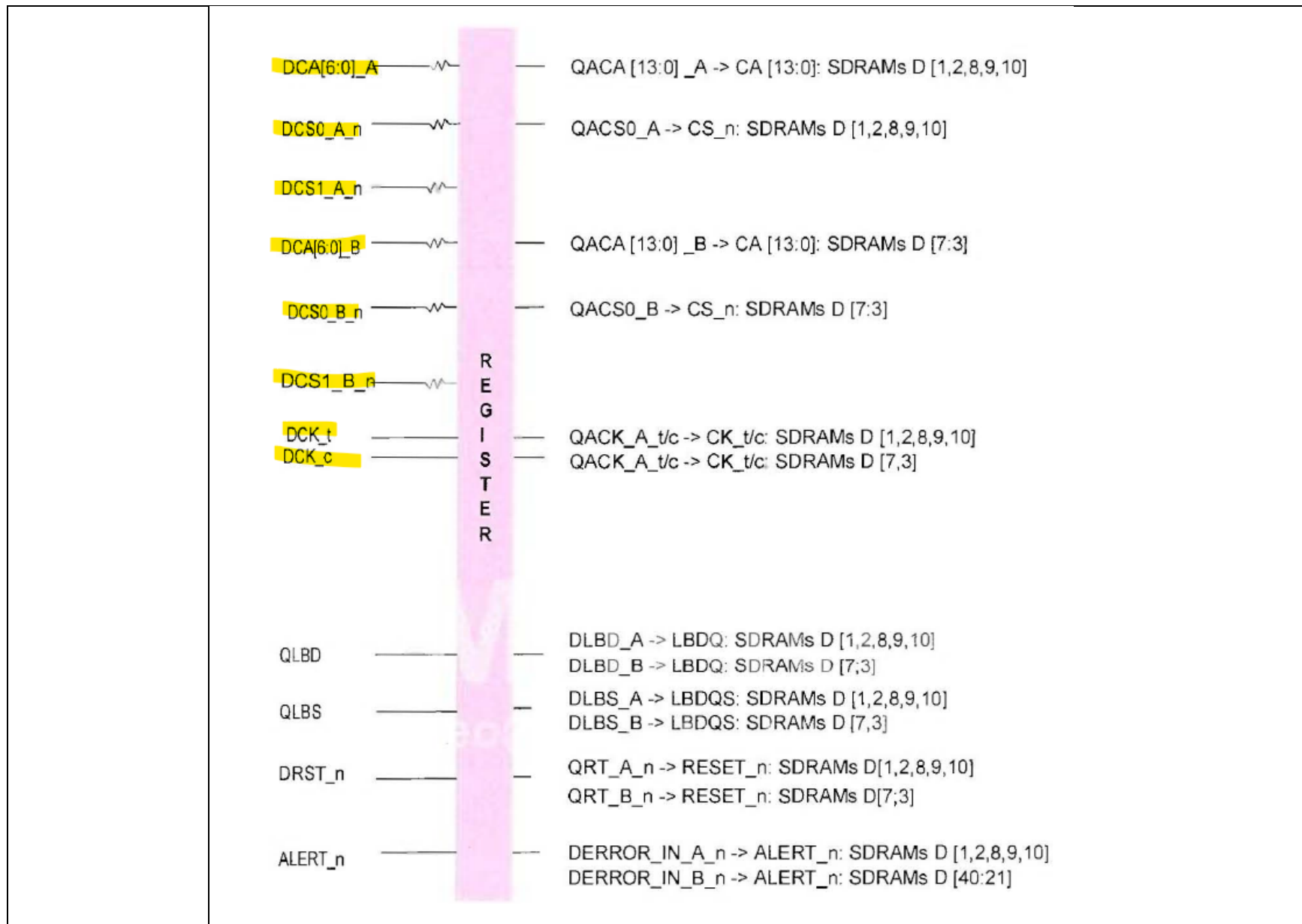
Claim 1	Evidence of Use											
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B				
	DQ24_A	40	184	V33	DQ8_B	111	255	VSS				
	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B				
	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS				
	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B				
	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS				
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c				
	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t				
				DBI3_A_n				DBI1_B_n				
	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS				
	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B				
	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS				
	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B				
	CB0_A	51	195	VSS	DQ16_B	122	266	VSS				
	VSS	52	196	CB2_A	VSS	123	267	DQ18_B				
	CB1_A	53	197	VSS	DQ17_B	124	268	VSS				
	VSS	54	198	CB3_A	VSS	125	269	DQ19_B				
	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	V33				
	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c				
	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t				
				DBI4_A_n				DBI2_B_n				
	CB4_A	58	202	VSS	DQ20_B	129	273	VSS				
	VSS	59	203	CB6_A	VSS	130	274	DQ22_B				
	CB5_A	60	204	VSS	DQ21_B	131	275	VSS				
	VSS	61	205	CB7_A	VSS	132	276	DQ23_B				
	AI FFT_n	62	206	VSS	DQ24_B	133	277	VSS				
	VSS	63	207	RESET_n	VSS	134	278	DQ26_B				
	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS				
	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B				
	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS				
	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c				
	CA2	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t				
	VSS	69	213	CA3_A				DBI3_B_n				
	CA4_A	70	214	VSS	DQ28_B	140	284	VSS				
	VSS	71	215	CA5_A	VSS	141	285	DQ30_B				
	CA6_A	72	216	VSS	DQ29_B	142	286	VSS				
	VSS	73	217	CK_t	VSS	143	287	DQ31_B				
					RFU	144	288	VSS				

Figure 1. Wiring Assignments

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7-8 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)



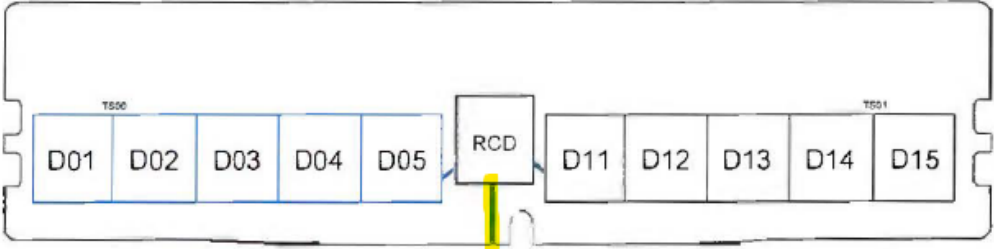
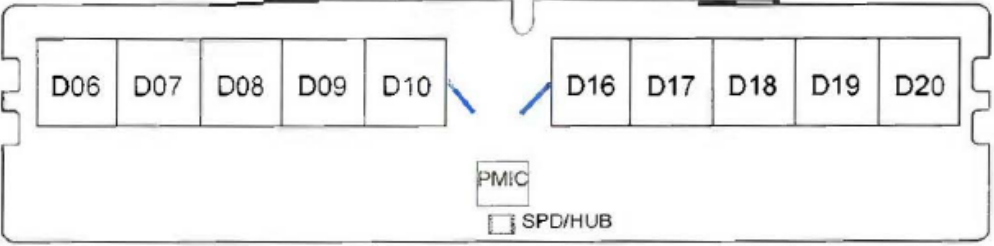
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates the internal architecture of a memory module, showing both the front and back views. The front view features a sequence of components: D1, D2, RCD (Refresh Control Driver), D3, and D4, connected by blue lines representing data paths. The back view shows a sequence of components: D10, D9, D8, D7, D6, and D5, also connected by blue lines. Below the back view, there are additional components: TS00, TS01, PMIC (Power Management IC), and SPD/HUB (Serial Presence Detect/Hub). A yellow line, labeled "Address, Control and Clock lines", runs horizontally across the bottom of the module, connecting to the RCD and the SPD/HUB. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21.

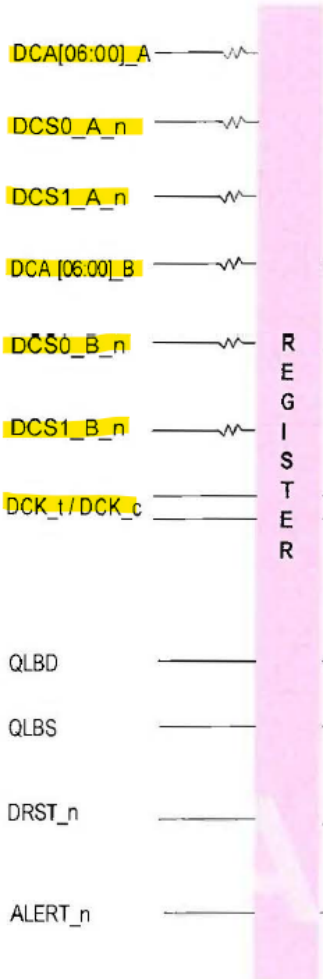
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
DCA[06:0]_A	QACA[13:00]_A -> CA [13:00] : SDRAMs D [10:01]
DCS0_A_n	QACS0_A -> CS_n : SDRAMs D [05:01]
DCS1_A_n	QACS1_A -> CS_n : SDRAMs D [10:06]
DCA[06:00]_B	QACA [13:00]_B -> CA [13:00]: SDRAMs D [20:11]
DCS0_B_n	QACS0_B -> CS_n : SDRAMs D [15:11]
DCS1_B_n	QACS1_B -> CS_n : SDRAMs D [20:16]
DCK_t/DCK_c	QACK_A_t -> CK_t: SDRAMs D [05:01] / QACK_A_c -> CK_c: SDRAMs D [05:01] QCK_A_t -> CK_t: SDRAMs D [10:06] / QCK_A_c -> CK_c: SDRAMs D [10:06] QACK_B_t -> CK_t: SDRAMs D [15:11] / QACK_B_c -> CK_c: SDRAMs D [15:11] QCK_B_t -> CK_t: SDRAMs D [20:16] / QCK_B_c -> CK_c: SDRAMs D [20:16]
QLBD	DLBD_A -> LBDQ: SDRAMs D [10:01] DLBD_B -> LBDQ: SDRAMs D [20:11]
QLBS	DLBS_A -> LBDQS: SDRAMs D [10:01] DLBS_B -> LBDQS: SDRAMs D [20:11]
DRST_n	QRT_A_n -> RESET_n: SDRAMs D [10:01] QRT_B_n -> RESET_n: SDRAMs D [20:11]
ALERT_n	DERROR_IN_A_n -> ALERT_n: SDRAMs D [10:01] DERROR_IN_B_n -> ALERT_n: SDRAMs D [20:11]

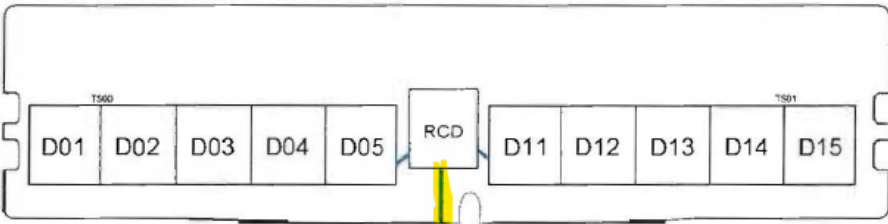
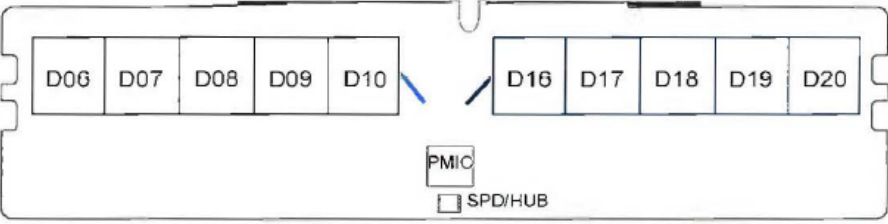
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use	
		QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] QACS0_B -> CS_n: SDRAMsD[15:11] QACS0_B -> CS_n: SDRAMsD[20:16] QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMsD[05:01] QCK_A_t->CK_t: SDRAMsD[10:06] / QCK_A_c->CK_c: SDRAMsD[10:06] QACK_B_t->CK_t: SDRAMsD[15:11] / QACK_B_c->CK_c: SDRAMsD[15:11] QCK_B_t->CK_t: SDRAMsD[20:16] / QCK_B_c->CK_c: SDRAMsD[20:16] DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n->RESET_n: SDRAMsD[20:11] DERROR_IN_A_n->ALERT_n: SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n: SDRAMsD[20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use	
	DCA[6:0]_A	QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16]
	DCS0_A_n	QACS0_A -> CS_n : SDRAMs D [5:1] QBCS0_A -> CS_n : SDRAMs D [10:6]
	DCS1_A_n	QACS1_A -> CS_n : SDRAMs D [15:11] QBCS1_A -> CS_n : SDRAMs D [20:16]
	DCA[6:0]_B	QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36]
	DCS0_B_n	QACS0_B -> CS_n : SDRAMs D [25:21] QBCS0_B -> CS_n : SDRAMs D [30:26]
	DCS1_B_n	QACS1_B -> CS_n : SDRAMs D [35:31] QBCS1_B -> CS_n : SDRAMs D [40:36]
	DCK_t/DCK_c	QACK_A_t -> CK_t : SDRAMs D [5:1] / QACK_A_c -> CK_c : SDRAMs D [5:1] QBCK_A_t -> CK_t : SDRAMs D [10:6] / QBCK_A_c -> CK_c : SDRAMs D [10:6] QCCK_A_t -> CK_t : SDRAMs D [15:11] / QCCK_A_c -> CK_c : SDRAMs D [15:11] QDCK_A_t -> CK_t : SDRAMs D [20:16] / QDCK_A_c -> CK_c : SDRAMs D [20:16] QACK_B_t -> CK_t : SDRAMs D [25:21] / QACK_B_c -> CK_c : SDRAMs D [25:21] QBCK_B_t -> CK_t : SDRAMs D [30:26] / QBCK_B_c -> CK_c : SDRAMs D [30:26] QCCK_B_t -> CK_t : SDRAMs D [35:31] / QCCK_B_c -> CK_c : SDRAMs D [35:31] QDCK_B_t -> CK_t : SDRAMs D [40:36] / QDCK_B_c -> CK_c : SDRAMs D [40:36]
	QLBD	DLBD_A -> LBDQ : SDRAMs D [20:1] DLBD_B -> LBDQ : SDRAMs D [40:21]
	QLBS	DLBS_A -> LBDQS : SDRAMs D [20:1] DLBS_B -> LBDQS : SDRAMs D [40:21]
	DRST_n	QRT_A_n -> RESET_n : SDRAMs D [20:1] QRT_B_n -> RESET_n : SDRAMs D [40:21]
	ALERT_n	DERROR_IN_A_n -> ALERT_n : SDRAMs D [20:1] DERROR_IN_B_n -> ALERT_n : SDRAMs D [40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33. As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant, including the use of a JEDEC compliant 288-pin edge connector used to connect to the system bus. The JEDEC DDR5 RDIMM standard connector pinout and signal description further indicates the system bus of the computer system includes one or more clock signal lines, control/address (C/A) signal lines, and data signal lines. <i>See, e.g.,</i>

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																				
	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)). Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c,</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”) and control/address (C/A) signal lines (e.g., “CA[]_A” and “CA[]_B”)).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	Symbol	Type	I/O Levels	Function	CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.
Pin Name	Description	Pin Name	Description																																		
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B																																		
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B																																		
PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)																																		
CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)																																		
CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.																																		
Symbol	Type	I/O Levels	Function																																		
CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.																																		
CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.																																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																									
	<table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating data signal lines (e.g., “DQ[]_A” and “DQ[]_B”). <table><caption>Table 4 — Input/Output Functional Description (cont'd)</caption><thead><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr></thead><tbody><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></tbody></table> Ex. CC-88 (JEDEC 305B) at 5.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.																							
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use									
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label		
	VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c		
	RFU	2	146	VIN_BULK	VSS	75	219	VSS		
	VIN_MGMT	3	147	PCAMP				Key		
	HSCL	4	148	HSA						
	HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present		
	VSS	6	150	RFU	VSS	77	221	CA1_B		
	DQ0_A	7	151	VSS	CA2_B	78	222	VSS		
	VSS	8	152	DQ2_A	VSS	79	223	CA3_B		
	DQ1_A	9	153	VSS	CA4_B	80	224	VSS		
	VSS	10	154	DQ3_A	VSS	81	225	CA5_B		
	DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS		
	DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B		
	VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS		
	DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n		
	VSS	15	159	DQ6_A	DLBDQ	86	230	VSS		
	DQ5_A	16	160	VSS	DLBDQS	87	231	RFU		
	VSS	17	161	DQ7_A	VSS	88	232	RFU		
	DQ8_A	18	162	VSS	CB4_B	89	233	VSS		
	VSS	19	163	DQ10_A	VSS	90	234	CB6_B		
	DQ9_A	20	164	VSS	CB5_B	91	235	VSS		
	VSS	21	165	DQ11_A	VSS	92	236	CB7_B		
	DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS		
	DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c		
	VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t		
	DQ12_A	25	169	VSS	CB0_B	96	240	VSS		
	VSS	26	170	DQ14_A	VSS	97	241	CB2_B		
	DQ13_A	27	171	VSS	CB1_B	98	242	VSS		
	VSS	28	172	DQ15_A	VSS	99	243	CB3_B		
	DQ16_A	29	173	VSS	DQ0_B	100	244	VSS		
	VSS	30	174	DQ18_A	VSS	101	245	DQ2_B		
	DQ17_A	31	175	VSS	DQ1_B	102	246	VSS		
	VSS	32	176	DQ19_A	VSS	103	247	DQ3_B		
	DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS		
	DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c		
	VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t		
	DQ20_A	36	180	VSS	DQ4_B	107	251	VSS		
	VSS	37	181	DQ22_A	VSS	108	252	DQ6_B		
	DQ21_A	38	182	VSS	DQ5_B	109	253	VSS		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-88 (JEDEC 305B) at 6 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”)).

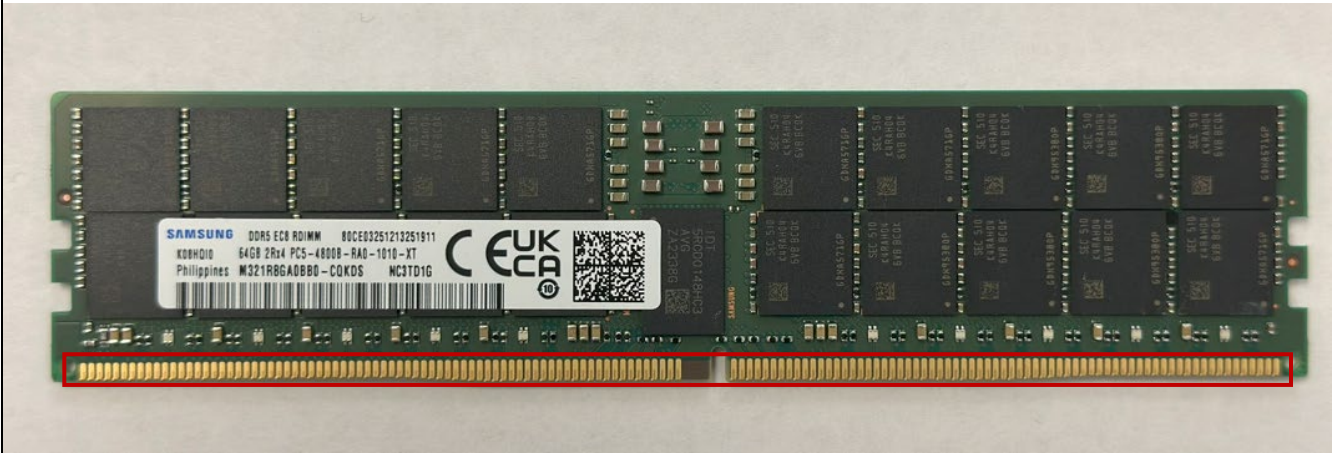
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use									
	Front Side			Back side			Front Side			Back side
	Pin Label	Pin	Pin	Pin Label	Pin Label	Pin	Pin	Pin	Pin Label	Pin Label
	VSS	39	183	DQ23_A		VSS	110	254	DQ7_B	
	DQ24_A	40	184	VSS		DQ8_B	111	255	VSS	
	VSS	41	185	DQ26_A		VSS	112	256	DQ10_B	
	DQ25_A	42	186	VSS		DQ9_B	113	257	VSS	
	VSS	43	187	DQ27_A		VSS	114	258	DQ11_B	
	DQS3_A_t	44	188	VSS		DQS1_B_t	115	259	VSS	
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c		DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c	
	VSS	46	190	DQS8_A_t, TDQS8_A_t		VSS	117	261	DQS6_B_t, TDQS6_B_t	
	DQ28_A	47	191	VSS		DQ12_B	118	262	VSS	
	VSS	48	192	DQ30_A		VSS	119	263	DQ14_B	
	DQ29_A	49	193	VSS		DQ13_B	120	264	VSS	
	VSS	50	194	DQ31_A		VSS	121	265	DQ15_B	
	CB0_A	51	195	VSS		DQ16_B	122	266	VSS	
	VSS	52	196	CB2_A		VSS	123	267	DQ18_B	
	CB1_A	53	197	VSS		DQ17_B	124	268	VSS	
	VSS	54	198	CB3_A		VSS	125	269	DQ19_B	
	DQS4_A_t	55	199	VSS		DQS2_B_t	126	270	VSS	
	DQS4 A c	56	200	DQS9 A c, TDQS9 A c		DQS2 B c	127	271	DQS7 B c, TDQS7 B c	
	VSS	57	201	DQS9_A_t, TDQS9_A_t		VSS	128	272	DQS7_B_t, TDQS7_B_t	
	CB4_A	58	202	VSS		DQ20_B	129	273	VSS	
	VSS	59	203	CB6_A		VSS	130	274	DQ22_B	
	CB5_A	60	204	VSS		DQ21_B	131	275	VSS	
	VSS	61	205	CB7_A		VSS	132	276	DQ23_B	
	ALERT_n	62	206	VSS		DQ24_B	133	277	VSS	
	VSS	63	207	RESET_n		VSS	134	278	DQ26_B	
	CS0_A_n	64	208	VSS		DQ25_B	135	279	VSS	
	VSS	65	209	CS1_A_n		VSS	136	280	DQ27_B	
	CA0_A	66	210	VSS		DQS3_B_t	137	281	VSS	
	VSS	67	211	CA1_A		DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c	
	CA2_A	68	212	VSS		VSS	139	283	DQS8_B_t, TDQS8_B_t	
	VSS	69	213	CA3 A		DQ28 B	140	284	VSS	
	CA4_A	70	214	VSS		VSS	141	285	DQ30_B	
	VSS	71	215	CA5_A		DQ29_B	142	286	VSS	
	CA6_A	72	216	VSS		VSS	143	287	DQ31_B	
	VSS	73	217	CK_t		RFU	144	288	VSS	
NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible.										
NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.										

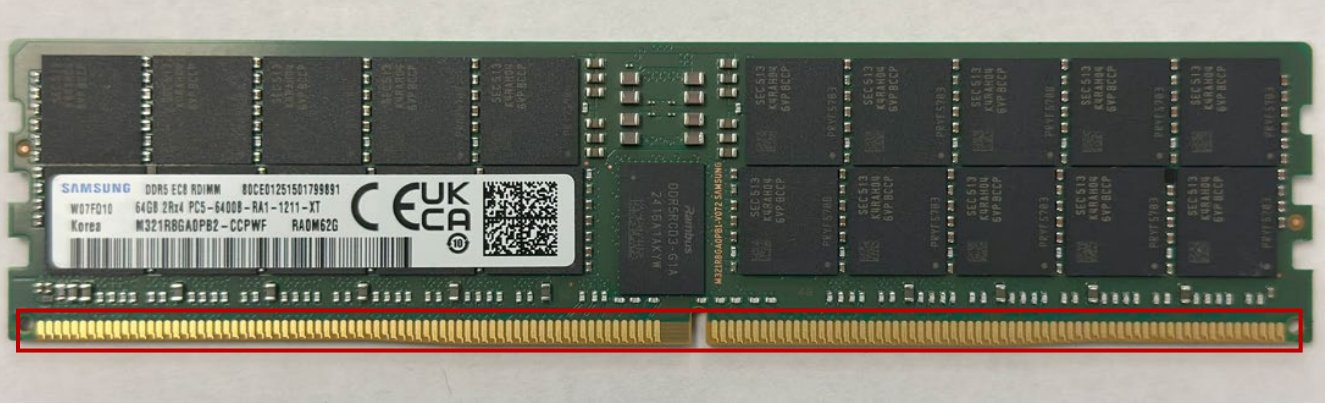
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-88 (JEDEC 305B) at 7 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)). These are the pins to be inserted and mated with the corresponding pins in the memory slot, thus indicating that the system bus includes one or more clock signal lines, control/address (C/A) signal lines and data signal lines.
[1.a] a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;	The Samsung Practicing '407 Products each comprise a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus. For example, as shown in the below images, each Samsung RDIMM has a green printed circuit board, at the bottom of which there are two rows of gold fingers (connectors, annotated in red boxes): Ex. ITC-1472 CC-116 (Samsung 4800MHz PC5-38400).

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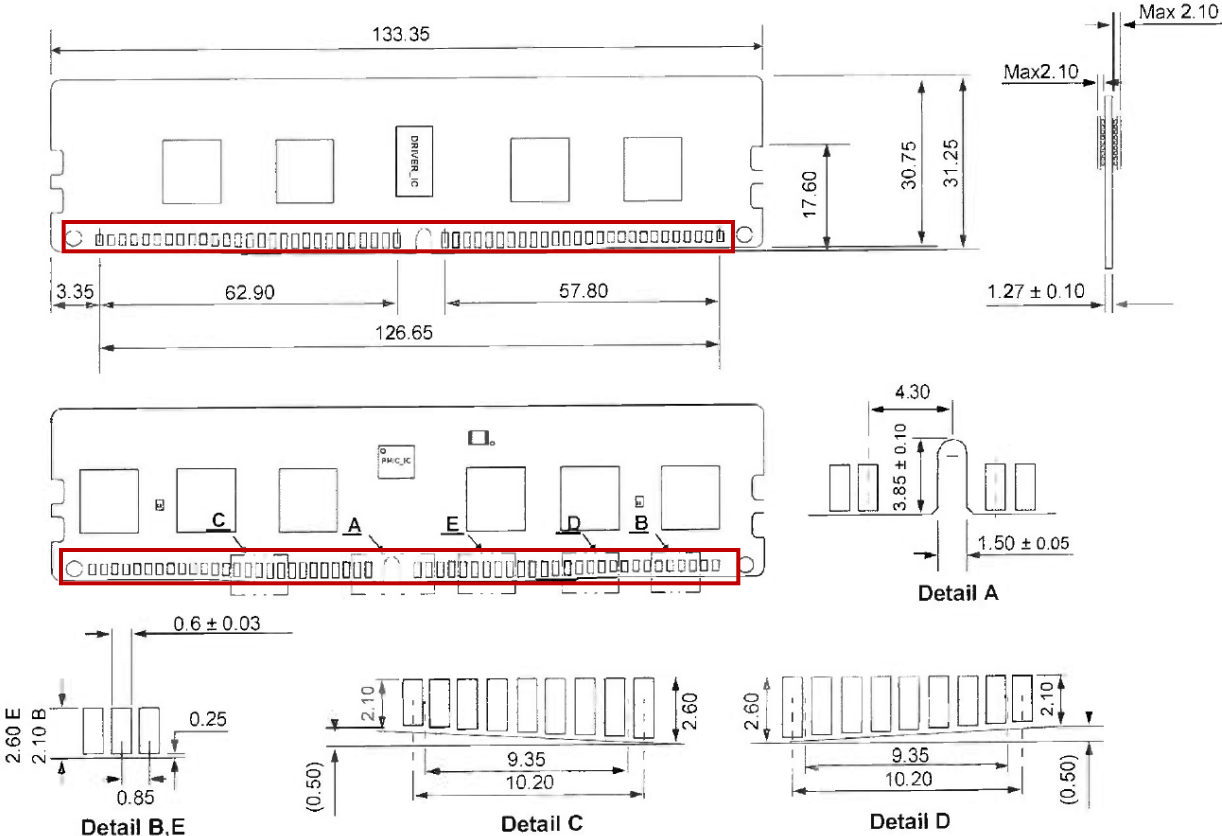
Claim 1	Evidence of Use
	A Samsung DDR5 RDIMM module with a green PCB and gold-plated pins. The module features a central white label with a barcode and technical specifications: SAMSUNG, DDR5 EC8 RDIMM, 80CE01234005418888, 64GB 2Rx4 PC5-5600B-RA0-1010-XT, W00J120, Korea, M321R8GA0PB0-CWMXJ, R80MR6R. A red box highlights the bottom edge of the module. Ex. ITC-1472 CC-117 (Samsung 5600MHz PC5-44800). A Samsung DDR5 RDIMM module with a green PCB and gold-plated pins. The module features a central white label with a barcode and technical specifications: SAMSUNG, DDR5 EC8 RDIMM, 80CE03251212251911, 64GB 2Rx4 PC5-4800B-RA0-1010-XT, Philippines, M321R8GA0BB0-CQKDS, MC3TD1G. A red box highlights the bottom edge of the module. Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module. The module is green with multiple black memory chips. A white label on the left side contains the following text: SAMSUNG, DDR5 E8 RDIMM, 80CE01251501799891, W07FD10, 64GB 2Rx4 PCS-64000-RA1-1211-XT, Korea, M321R8GA0PB2-CCPWF, RA0M62G. There is also a CE mark and a QR code. The gold-plated pins at the bottom are highlighted with a red rectangle. Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP). A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module, similar to the one above. The label text includes: SAMSUNG, DDR5 E8 RDIMM, 80CE01251101682065, W068040, 64GB 2Rx4 PCS-64000-RA1-1211-XT, Korea, M321R8GA0PB2-CLVX0, M01M110. The gold-plated pins at the bottom are highlighted with a red rectangle.

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Claim 1	Evidence of Use
	An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. <i>See</i> Exs. CC-111, CC-112. RDIMM Registered DIMM Include a register for enhancing the clock, command and control signals Error correction is available by adding 8-bit parity signals Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration Application : Server Ex. CC-95 (Samsung DDR DIMMs)

Claim 1	Evidence of Use
	19.0 PHYSICAL DIMENSIONS 19.1 2Gx8 based x80 Module (1 Rank) - M321R2GA3BB0 Units : mm  The technical drawings illustrate the physical dimensions of the M321R2GA3BB0 module. The top view shows a rectangular module with a total length of 133.35 mm and a total width of 31.25 mm. The bottom view shows the same module with a total length of 126.65 mm and a total width of 17.60 mm. The module features a central 'DRIVER IC' and several other components. The bottom edge of the module is highlighted with a red box, indicating the location of the contacts. The dimensions are provided in millimeters (mm). The bottom view includes labels A, B, C, D, and E, which correspond to the detail views. Detail A shows a cross-section of the module with dimensions 4.30, 3.85 ± 0.10, and 1.50 ± 0.05. Detail B and E show a cross-section of the module with dimensions 2.60 E, 2.10 B, 0.25, and 0.85. Detail C and D show a cross-section of the module with dimensions 2.10, 2.60, 9.35, 10.20, and (0.50). Ex. CC-89 at 93

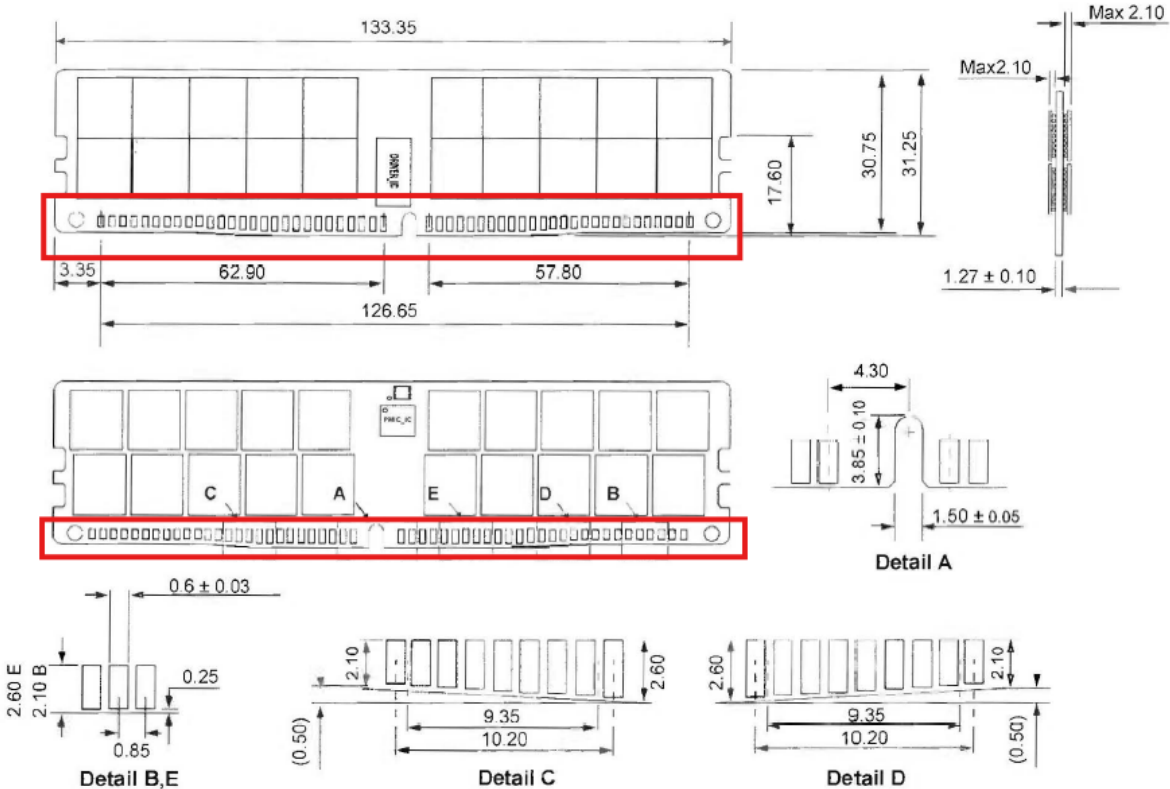
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	19.2 2Gx8 based x80 Module (2 Ranks) - M321R4GA3BB0 Units : mm The drawing shows two views of the module. The top view has a total length of 133.35 mm and a width of 31.25 mm. It features a central 'J1 REMBO' connector. The bottom view shows the same module with a 'PWC IC' label. Dimensions for the bottom view include a total length of 126.65 mm and a width of 17.60 mm. Callouts include 'Detail A' (showing a 4.30 mm dimension), 'Detail B,E' (showing 2.60 E, 2.10 B, and 0.85 mm dimensions), 'Detail C' (showing 9.35 mm and 10.20 mm dimensions), and 'Detail D' (showing 9.35 mm and 10.20 mm dimensions). Tolerances are specified as ± 0.15 unless otherwise specified. NOTE : Tolerances on all dimensions ± 0.15 unless otherwise specified Ex. CC-89 at 94

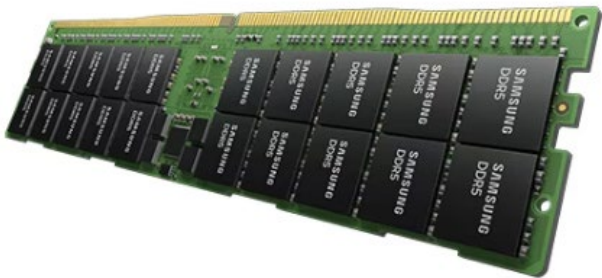
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	19.3 4Gx4 based x80 Module (1 Rank) - M321R4GA0BB0 Units : mm NOTE : Tolerances on all dimensions ± 0.15 unless otherwise specified Ex. CC-89 at 95.

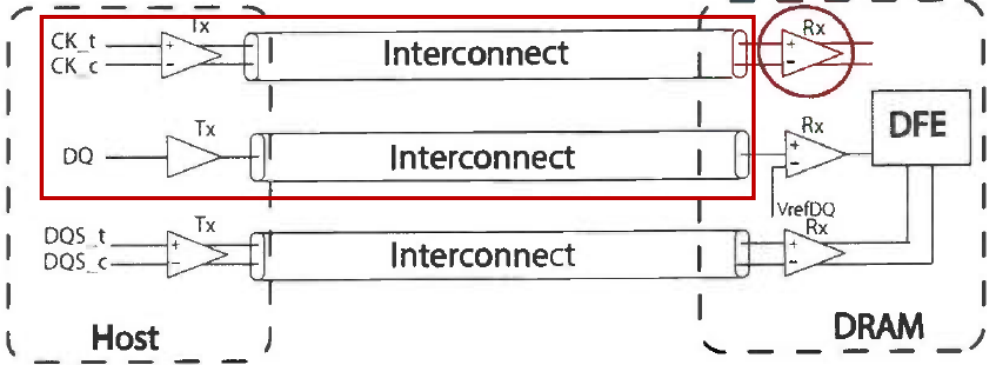
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	19.4 4Gx4 based x80 Module (2 Ranks) - M321R8GA0BB0 Units : mm  NOTE : Tolerances on all dimensions ± 0.15 unless otherwise specified Ex. CC-89 at 96.

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Claim 1	Evidence of Use
	Furthermore, the connectors are configured to provide electrical connections between the memory module and the system bus. For example, the connectors in the Samsung Practicing '407 Products are assigned to pin assignments for connecting with lines in the system bus, including clock signal lines, control/address (C/A) signal lines and data signal lines. <i>See, e.g.,</i> RDIMM Registered DIMM Include a register for enhancing the clock, command and control signals Error correction is available by adding 8-bit parity signals Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration Application : Server  Ex. CC-95 (Samsung DDR DIMMs)

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Claim 1	Evidence of Use
	 Figure 10. Example of DDR5 Memory Interconnect (Ex. CC-89 at 42, Fig. 10).

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Claim 1	Evidence of Use																																																		
	[Table 5] PIN Description <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>CA[6:0]_A CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </table> <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>SDRAM data strobes (positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>SDRAM data strobes (negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </table> Ex. CC-89 at 9.	Pin Name	Description	CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus	CS[1:0]_A CS[1:0]_B	SDRAM Chip Select	PAR_A PAR_B	SDRAM Parity input	CK_t	SDRAM Clocks (true/positive)	CK_c	SDRAM Clocks (complement/negative)	ALERT_n	SDRAM alert for CRC error	RESET_n	Set DRAM to known state	PCAMP	Control and Monitor Port	HSCL	I2C/I3C-Basic Host Sideband Bus Clock	HSDA	I2C/I3C-Basic Host Sideband Bus Data	HSA	I2C/I3C-Basic Host Sideband Bus Address	LBDQ	Loopback data output:	Pin Name	Description	DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B	CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B	DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)	DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.	VIN_BULK	DIMM Power Supply from system to PMIC	VIN_MGMT	DIMM Power Supply from system to PMIC	VSS	Power supply return (ground)	RFU	Reserved for future use	LBDQS	Loopback data strobe output
Pin Name	Description																																																		
CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus																																																		
CS[1:0]_A CS[1:0]_B	SDRAM Chip Select																																																		
PAR_A PAR_B	SDRAM Parity input																																																		
CK_t	SDRAM Clocks (true/positive)																																																		
CK_c	SDRAM Clocks (complement/negative)																																																		
ALERT_n	SDRAM alert for CRC error																																																		
RESET_n	Set DRAM to known state																																																		
PCAMP	Control and Monitor Port																																																		
HSCL	I2C/I3C-Basic Host Sideband Bus Clock																																																		
HSDA	I2C/I3C-Basic Host Sideband Bus Data																																																		
HSA	I2C/I3C-Basic Host Sideband Bus Address																																																		
LBDQ	Loopback data output:																																																		
Pin Name	Description																																																		
DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B																																																		
CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B																																																		
DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)																																																		
DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)																																																		
TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.																																																		
TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.																																																		
VIN_BULK	DIMM Power Supply from system to PMIC																																																		
VIN_MGMT	DIMM Power Supply from system to PMIC																																																		
VSS	Power supply return (ground)																																																		
RFU	Reserved for future use																																																		
LBDQS	Loopback data strobe output																																																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

[Table 4] PIN Configuration

Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side
1	VIN_BULK	145	VIN_BULK	50	VSS	194	DQ31_A	95	CB0_B	240	VSS
2	RFU	146	VIN_BULK	51	CB0_A	195	VSS	97	VSS	241	CB2_B
3	VIN_MGMT	147	PWR_GOOD(FAIL_n)	52	VSS	196	CB2_A	98	CB1_B	242	VSS
4	SCL	148	SAA	53	CB1_A	197	VSS	99	VSS	243	CB3_B
5	SDA	149	RFU	54	VSS	198	CB3_A	100	DQ0_B	244	VSS
6	VSS	150	RFU	55	DQ54_A_t	199	VSS	101	VSS	245	DQ2_B
7	DQ9_A	151	VSS	56	DQ54_A_c	200	DQ59_A_cTDQ59_A_c	102	DQ1_B	246	VSS
8	VSS	152	DQ2_A	57	VSS	201	DQ59_A_tTDQ59_A_t	103	VSS	247	DQ3_B
9	DQ1_A	153	VSS	58	CB4_A	202	VSS	104	DQ50_B_t	248	VSS
10	VSS	154	DQ3_A	59	VSS	203	CB6_A	105	DQ50_B_c	249	DQ55_B_cTDQ55_B_c
11	DQ50_A_t	155	VSS	60	CB5_A	204	VSS	106	VSS	250	DQ55_B_tTDQ55_B_t
12	DQ50_A_c	156	DQ55_A_cTDQ55_A_c	61	VSS	205	CB7_A	107	DQ4_B	251	VSS
13	VSS	157	DQ55_A_tTDQ55_A_t	62	ALERT_n	206	VSS	108	VSS	252	DQ6_B
14	DQ4_A	158	VSS	63	VSS	207	RESET_n	109	DQ6_B	253	VSS
15	VSS	159	DQ6_A	64	CS0_A_n	208	VSS	110	VSS	254	DQ7_B
16	DQ5_A	160	VSS	65	VSS	209	CS1_A_n	111	DQ8_B	255	VSS
17	VSS	161	DQ7_A	66	CA0_A	210	VSS	112	VSS	256	DQ10_B
18	DQ8_A	162	VSS	67	VSS	211	CA1_A	113	DQ9_B	257	VSS
19	VSS	163	DQ10_A	68	CA2_A	212	VSS	114	VSS	258	DQ11_B
20	DQ9_A	164	VSS	69	VSS	213	CA3_A	115	DQ51_B_t	259	VSS
21	VSS	165	DQ11_A	70	CA4_A	214	VSS	116	DQ51_B_c	260	DQ56_B_cTDQ56_B_c
22	DQ51_A_t	166	VSS	71	VSS	215	CA5_A	117	VSS	261	DQ56_B_tTDQ56_B_t
23	DQ51_A_c	167	DQ56_A_cTDQ56_A_c	72	CA6_A	216	VSS	118	DQ12_B	262	VSS
24	VSS	168	DQ56_A_tTDQ56_A_t	73	VSS	217	CK_t	119	VSS	263	DQ14_B
25	DQ12_A	169	VSS	74	PAR_A	218	CK_c	120	DQ13_B	264	VSS
26	VSS	170	DQ14_A	75	VSS	219	VSS	121	VSS	265	DQ15_B
27	DQ13_A	171	VSS	KEY				122	DQ16_B	266	VSS
28	VSS	172	DQ15_A					123	VSS	267	DQ18_B
29	DQ16_A	173	VSS					124	DQ17_B	268	VSS
30	VSS	174	DQ18_A					125	VSS	269	DQ19_B
31	DQ17_A	175	VSS	76	CA0_B	220	RFU	126	VSS	270	VSS
32	VSS	176	DQ19_A	77	VSS	221	CA1_B	126	DQ52_B_t	270	VSS
33	DQ52_A_t	177	VSS	78	CA2_B	222	VSS	127	DQ52_B_c	271	DQ57_B_cTDQ57_B_c
34	DQ52_A_c	178	DQ57_A_cTDQ57_A_c	79	VSS	223	CA3_B	128	VSS	272	DQ57_B_tTDQ57_B_t
35	VSS	179	DQ57_A_tTDQ57_A_t	80	CA4_B	224	VSS	129	DQ20_B	273	VSS
36	DQ20_A	180	VSS	81	VSS	225	CA5_B	130	VSS	274	DQ22_B
37	VSS	181	DQ22_A	82	CA6_B	226	VSS	131	DQ21_B	275	VSS
38	DQ21_A	182	VSS	83	VSS	227	PAR_B	132	VSS	276	DQ23_B
39	VSS	183	DQ23_A	84	CS0_B_n	228	VSS	133	DQ24_B	277	VSS
40	DQ24_A	184	VSS	85	VSS	229	CS1_B_n	134	VSS	278	DQ26_B
41	VSS	185	DQ26_A	86	L80_R5P_A_n	230	VSS	135	DQ25_B	279	VSS
42	DQ25_A	186	VSS	87	L85_R5P_B_n	231	RFU	136	VSS	280	DQ27_B
43	VSS	187	DQ27_A	88	VSS	232	RFU	137	DQ53_B_t	281	VSS
44	DQ53_A_t	188	VSS	89	CB4_B	233	VSS	138	DQ53_B_c	282	DQ58_B_cTDQ58_B_c
45	DQ53_A_c	189	DQ58_A_cTDQ58_A_c	90	VSS	234	CB6_B	139	VSS	283	DQ58_B_tTDQ58_B_t
46	VSS	190	DQ58_A_tTDQ58_A_t	91	CB5_B	235	VSS	140	DQ28_B	284	VSS
47	DQ28_A	191	VSS	92	VSS	236	CB7_B	141	VSS	285	DQ30_B
48	VSS	192	DQ30_A	93	DQ59_B_tTDQ59_B_t	237	VSS	142	DQ29_B	286	VSS
49	DQ29_A	193	VSS	94	DQ59_B_cTDQ59_B_c	238	DQ54_B_c	143	VSS	287	DQ31_B
				95	VSS	239	DQ54_B_t	144	RFU	288	VSS

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 6 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	V33	75	219	V33
VIN_MGMT	3	147	PWR_GOOD/FAIL_n	Key			
HSCL	4	148	HSA	CA0_B	76	220	RFU
HSDA	5	149	RFU	VSS	77	221	CA1_B
VSS	6	150	RFU	CA2_B	78	222	VSS
DQ0_A	7	151	VSS	VSS	79	223	CA3_B
VSS	8	152	DQ2_A	CA4_B	80	224	VSS
DQ1_A	9	153	VSS	VSS	81	225	CA5_B
VSS	10	154	DQ3_A	CA6_B	82	226	VSS
DQS0_A_t	11	155	VSS	VSS	83	227	PAR_B
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	CS0_B_n	84	228	VSS
VSS	13	157	DU55_A_t, DU55_A_t DBI0_A_n	VSS	85	229	CS1_B_n
DQ4_A	14	158	VSS	LBD, RSP_A_n	86	230	VSS
VSS	15	159	DQ6_A	LBS, RSP_B_n	87	231	RFU
DQ5_A	16	160	VSS	VSS	88	232	RFU
VSS	17	161	DQ7_A	CB4_B	89	233	VSS
DQ8_A	18	162	VSS	VSS	90	234	CB6_B
VSS	19	163	DQ10_A	CB5_B	91	235	VSS
DQ9_A	20	164	VSS	VSS	92	236	CB7_B
VSS	21	165	DQ11_A	DQS9_B_t, TDQS9_B_t DBI4_B_n	93	237	VSS
DQS1_A_t	22	166	VSS	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	VSS	95	239	DQS4_B_t
VSS	24	168	DQS6_A_t, TDQS6_A_t DBI1_A_n	CB0_B	96	240	VSS
DQ12_A	25	169	VSS	V33	97	241	OD2_D
VSS	26	170	DQ14_A	CB1_B	98	242	VSS
DQ13_A	27	171	VSS	VSS	99	243	CB3_B
VSS	28	172	DQ15_A	DQ0_B	100	244	VSS
DQ16_A	29	173	VSS	VSS	101	245	DQ2_B
VSS	30	174	DQ18_A	DQ1_B	102	246	VSS
DQ17_A	31	175	VSS	VSS	103	247	DQ3_BDQ3_B
VSS	32	176	DQ19_A	DQS0_B_t	104	248	VSS
DQS2_A_t	33	177	VSS	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	VSS	106	250	DQS5_B_t, TDQS5_B_t DBI0_B_n
VSS	35	179	DQS7_A_t, TDQS7_A_t DBI2_n	DQ4_B	107	251	VSS
DQ20_A	36	180	VSS	VSS	108	252	DQ6_B
VSS	37	181	DQ22_A	DQ5_B	109	253	VSS
DQ21_A	38	182	VSS				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use											
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B				
	DQ24_A	40	184	V33	DQ8_B	111	255	VSS				
	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B				
	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS				
	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B				
	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS				
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c				
	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t				
				DBI3_A_n				DBI1_B_n				
	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS				
	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B				
	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS				
	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B				
	CB0_A	51	195	VSS	DQ16_B	122	266	VSS				
	VSS	52	196	CB2_A	VSS	123	267	DQ18_B				
	CB1_A	53	197	VSS	DQ17_B	124	268	VSS				
	VSS	54	198	CB3_A	VSS	125	269	DQ19_B				
	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	V33				
	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c				
	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t				
				DBI4_A_n				DBI2_B_n				
	CB4_A	58	202	VSS	DQ20_B	129	273	VSS				
	VSS	59	203	CB6_A	VSS	130	274	DQ22_B				
	CB5_A	60	204	VSS	DQ21_B	131	275	VSS				
	VSS	61	205	CB7_A	VSS	132	276	DQ23_B				
	AI_FFT_n	62	206	VSS	DQ24_B	133	277	VSS				
	VSS	63	207	RESET_n	VSS	134	278	DQ26_B				
	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS				
	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B				
	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS				
	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c				
	CA2	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t				
	VSS	69	213	CA3_A				DBI3_B_n				
	CA4_A	70	214	VSS	DQ28_B	140	284	VSS				
	VSS	71	215	CA5_A	VSS	141	285	DQ30_B				
	CA6_A	72	216	VSS	DQ29_B	142	286	VSS				
	VSS	73	217	CK_t	VSS	143	287	DQ31_B				
					RFU	144	288	VSS				

Figure 1. Wiring Assignments

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7-8 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

DCA[6:0]_A	—	QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10]
DCS0_A_n	—	QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10]
DCS1_A_n	—	
DCA[6:0]_B	—	QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3]
DCS0_B_n	—	QACS0_B -> CS_n: SDRAMs D [7:3]
DCS1_B_n	—	
DCK_t	—	QACK_A_t/c -> CK_t/c: SDRAMs D [1,2,8,9,10]
DCK_c	—	QACK_A_t/c -> CK_t/c: SDRAMs D [7,3]
	REGISTER	
QLBD	—	DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7,3]
QLBS	—	DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7,3]
DRST_n	—	QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3]
ALERT_n	—	DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

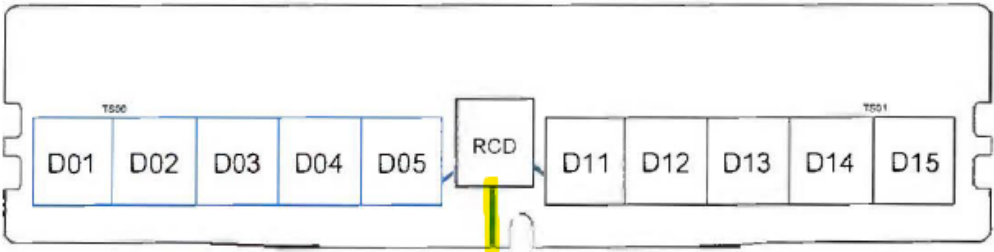
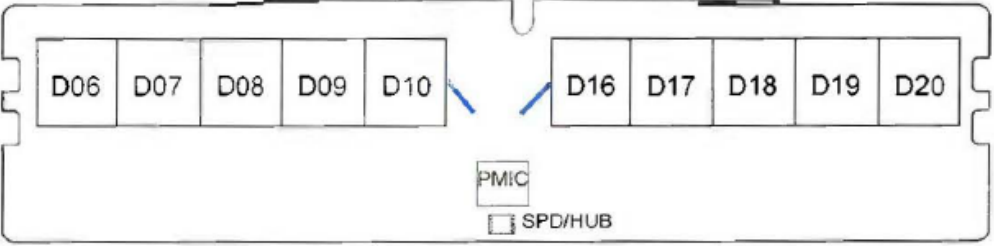
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates the internal architecture of a memory module, divided into Front and Back views. The Front view shows a sequence of components: D1, D2, RCD (Refresh Control Driver), D3, and D4, connected by blue lines representing Address, Control, and Clock lines. The Back view shows a sequence of components: D10, D9, D8, D7, D6, and D5, also connected by blue lines. Additional components include TS00 and TS01 (likely temperature sensors) connected to D9 and D6 respectively, and a PMIC (Power Management IC) and SPD/HUB (Serial Presence Detect/Hub) connected to the bottom of the module. A legend at the bottom indicates that blue lines represent Address, Control, and Clock lines. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21.

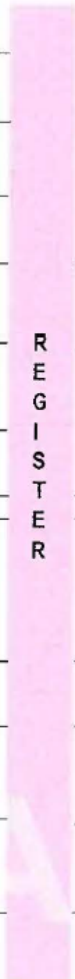
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
DCA[06:0]_A	QACA[13:00]_A -> CA [13:00] : SDRAMs D [10:01]
DCS0_A_n	QACS0_A -> CS_n : SDRAMs D [05:01]
DCS1_A_n	QACS1_A -> CS_n : SDRAMs D [10:06]
DCA[06:00]_B	QACA [13:00]_B -> CA [13:00]: SDRAMs D [20:11]
DCS0_B_n	QACS0_B -> CS_n : SDRAMs D [15:11]
DCS1_B_n	QACS1_B -> CS_n : SDRAMs D [20:16]
DCK_t/DCK_c	QACK_A_t -> CK_t: SDRAMs D [05:01] / QACK_A_c -> CK_c: SDRAMs D [05:01] QCK_A_t -> CK_t: SDRAMs D [10:06] / QCK_A_c -> CK_c: SDRAMs D [10:06] QACK_B_t -> CK_t: SDRAMs D [15:11] / QACK_B_c -> CK_c: SDRAMs D [15:11] QCK_B_t -> CK_t: SDRAMs D [20:16] / QCK_B_c -> CK_c: SDRAMs D [20:16]
QLBD	DLBD_A -> LBDQ: SDRAMs D [10:01] DLBD_B -> LBDQ: SDRAMs D [20:11]
QLBS	DLBS_A -> LBDQS: SDRAMs D [10:01] DLBS_B -> LBDQS: SDRAMs D [20:11]
DRST_n	QRT_A_n -> RESET_n: SDRAMs D [10:01] QRT_B_n -> RESET_n: SDRAMs D [20:11]
ALERT_n	DERROR_IN_A_n -> ALERT_n: SDRAMs D [10:01] DERROR_IN_B_n -> ALERT_n: SDRAMs D [20:11]

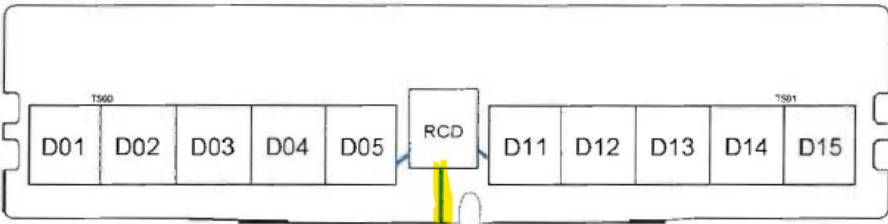
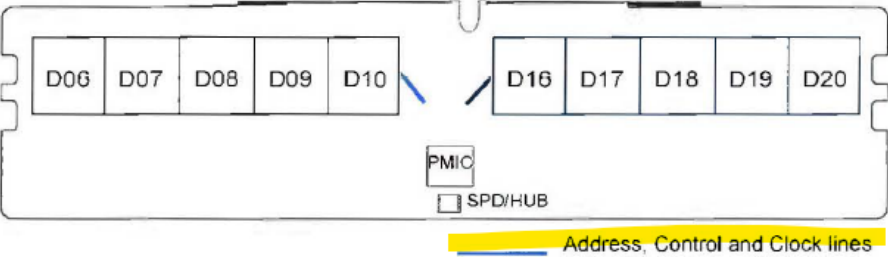
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:00]_A ——— QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] DCS0_A_n ——— QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n ——— DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMsD[15:11] QACS0_B -> CS_n: SDRAMsD[20:16] DCS1_B_n ——— DCK_t/DCK_c ——— QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMsD[05:01] QACK_A_t->CK_t: SDRAMsD[10:06] / QACK_A_c->CK_c: SDRAMsD[10:06] QACK_B_t->CK_t: SDRAMsD[15:11] / QACK_B_c->CK_c: SDRAMsD[15:11] QACK_B_t->CK_t: SDRAMsD[20:16] / QACK_B_c->CK_c: SDRAMsD[20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n->RESET_n: SDRAMsD[20:11] ALERT_n ——— DERROR_IN_A_n->ALERT_n: SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n: SDRAMsD[20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A — QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n — QACS0_A->CS_n: SDRAMs D[5:1] QBCS0_A -> CS_n: SDRAMs D [10:6] DCS1_A_n — QACS1_A->CS_n: SDRAMs D[15:11] QBCS1_A -> CS_n: SDRAMs D[20:16] DCA[6:0]_B — QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] DCS0_B_n — QACS0_B -> CS_n: SDRAMs D [25:21] QBCS0_B -> CS_n: SDRAMs D [30:26] DCS1_B_n — QACS1_B -> CS_n: SDRAMs D [35:31] QBCS1_B -> CS_n: SDRAMs D [40:36] DCK_t/DCK_c — QACK_A_t->CK_t: SDRAMs D[5:1] / QACK_A_c->CK_c: SDRAMs D[5:1] QBCK_A_t -> CK_t: SDRAMs D[10:6] / QBCK_A_c->CK_c: SDRAMs D[10:6] QCCK_A_t -> CK_t: SDRAMs D [15:11] / QCCK_A_c->CK_c: SDRAMs D [15:11] QDCK_A_t -> CK_t: SDRAMs D [20:16] / QDCK_A_c->CK_c: SDRAMs D [20:16] QACK_B_t -> CK_t: SDRAMs D [25:21] / QACK_B_c->CK_c: SDRAMs D [25:21] QBCK_B_t -> CK_t: SDRAMs D [30:26] / QBCK_B_c->CK_c: SDRAMs D [30:26] QCCK_B_t -> CK_t: SDRAMs D [35:31] / QCCK_B_c->CK_c: SDRAMs D [35:31] QDCK_B_t -> CK_t: SDRAMs D [40:36] / QDCK_B_c->CK_c: SDRAMs D [40:36] QLBD — DLBD_A->LBDQ : SDRAMs D[20:1] DLBD_B -> LBDQ : SDRAMs D [40:21] QLBS — DLBS_A -> LBDQS : SDRAMs D[20:1] DLBS_B -> LBDQS : SDRAMs D [40:21] DRST_n — QRT_A_n->RESET_n: SDRAMs D[20:1] QRT_B_n -> RESET_n: SDRAMs D [40:21] ALERT_n — DERROR_IN_A_n->ALERT_n: SDRAMs D[20:1] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

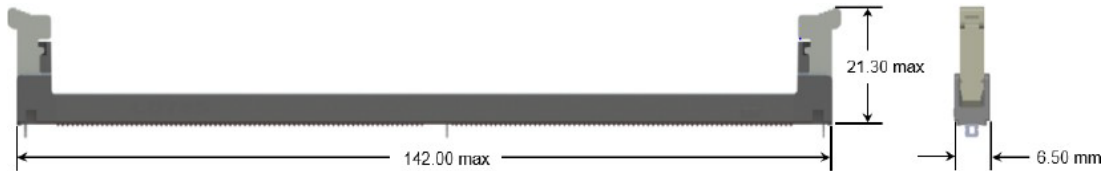
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33. As explained <i>supra</i>, Samsung's RDIMMs are JEDEC standard compliant and are therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below:

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Claim 1	Evidence of Use
	Ex. CC-70 (DDR5 Memory Channel Scaling Performance).

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Claim 1	Evidence of Use
	 Figure 1 DDR5 socket and module (SMT)  Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline Ex. CC-73 (JEDEC PS-005B) at 6.

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Claim 1	Evidence of Use
	Contacting the RDIMM's edge connectors with the corresponding pins in the memory slot forms electrical connections between the RDIMM and the system bus: Figure 6: DDR5 288-pin connector. Figure 7: Device under test (DUT) Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin on the memory slot). The JEDEC DDR5 RDIMM standard further demonstrates the Samsung Practicing '407 Products' edge connectors provide electrical connections between the memory module and the system bus because it includes connector pin assignments for clock signal lines, control/address (C/A) signal lines, and data signal lines included in the system bus. <i>See, e.g.,</i>

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Claim 1	Evidence of Use																																												
	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3. Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c.</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4.	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	Symbol	Type	I/O Levels	Function	CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating
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Claim 1	Evidence of Use																	
	Ex. CC-88 (JEDEC 305B) at 4. Table 4 — Input/Output Functional Description (cont'd) <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 5.	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
Symbol	Type	I/O Levels	Function															
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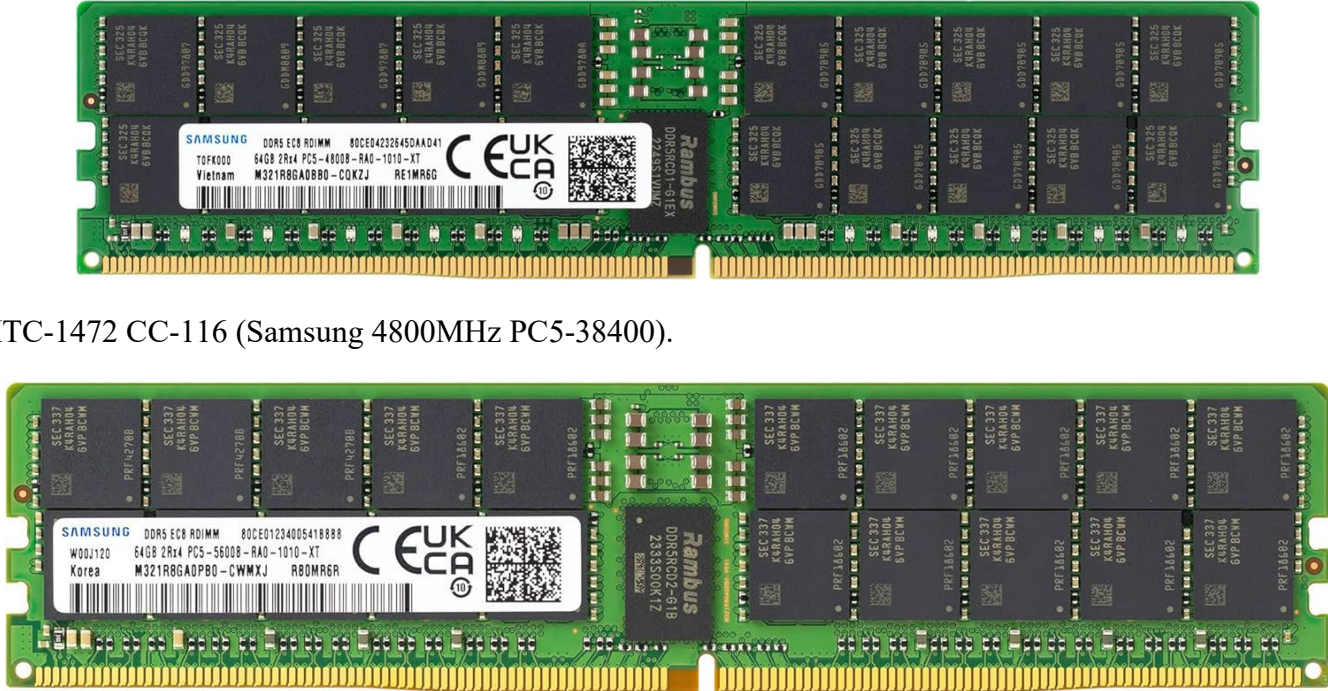
Claim 1	Evidence of Use									
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label		
	VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c		
	RFU	2	146	VIN_BULK	VSS	75	219	VSS		
	VIN_MGMT	3	147	PCAMP						
	HSCL	4	148	HSA						
	HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present		
	VSS	6	150	RFU	VSS	77	221	CA1_B		
	DQ0_A	7	151	VSS	CA2_B	78	222	VSS		
	VSS	8	152	DQ2_A	VSS	79	223	CA3_B		
	DQ1_A	9	153	VSS	CA4_B	80	224	VSS		
	VSS	10	154	DQ3_A	VSS	81	225	CA5_B		
	DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS		
	DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B		
	VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS		
	DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n		
	VSS	15	159	DQ6_A	DLBDQ	86	230	VSS		
	DQ5_A	16	160	VSS	DLBDQS	87	231	RFU		
	VSS	17	161	DQ7_A	VSS	88	232	RFU		
	DQ8_A	18	162	VSS	CB4_B	89	233	VSS		
	VSS	19	163	DQ10_A	VSS	90	234	CB6_B		
	DQ9_A	20	164	VSS	CB5_B	91	235	VSS		
	VSS	21	165	DQ11_A	VSS	92	236	CB7_B		
	DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS		
	DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c		
	VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t		
	DQ12_A	25	169	VSS	CB0_B	96	240	VSS		
	VSS	26	170	DQ14_A	VSS	97	241	CB2_B		
	DQ13_A	27	171	VSS	CB1_B	98	242	VSS		
	VSS	28	172	DQ15_A	VSS	99	243	CB3_B		
	DQ16_A	29	173	VSS	DQ0_B	100	244	VSS		
	VSS	30	174	DQ18_A	VSS	101	245	DQ2_B		
	DQ17_A	31	175	VSS	DQ1_B	102	246	VSS		
	VSS	32	176	DQ19_A	VSS	103	247	DQ3_B		
	DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS		
	DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c		
	VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t		
	DQ20_A	36	180	VSS	DQ4_B	107	251	VSS		
	VSS	37	181	DQ22_A	VSS	108	252	DQ6_B		
	DQ21_A	38	182	VSS	DQ5_B	109	253	VSS		

Ex. CC-88 (JEDEC 305B) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

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	<table><tr><th>Front Side</th><th>Pin</th><th>Pin</th><th>Back side</th><th>Front Side</th><th>Pin</th><th>Pin</th><th>Back side</th></tr><tr><th>Pin Label</th><th></th><th></th><th>Pin Label</th><th>Pin Label</th><th></th><th></th><th>Pin Label</th></tr><tr><td>VSS</td><td>39</td><td>183</td><td>DQ23_A</td><td>VSS</td><td>110</td><td>254</td><td>DQ7_B</td></tr><tr><td>DQ24_A</td><td>40</td><td>184</td><td>VSS</td><td>DQ8_B</td><td>111</td><td>255</td><td>VSS</td></tr><tr><td>VSS</td><td>41</td><td>185</td><td>DQ26_A</td><td>VSS</td><td>112</td><td>256</td><td>DQ10_B</td></tr><tr><td>DQ25_A</td><td>42</td><td>186</td><td>VSS</td><td>DQ9_B</td><td>113</td><td>257</td><td>VSS</td></tr><tr><td>VSS</td><td>43</td><td>187</td><td>DQ27_A</td><td>VSS</td><td>114</td><td>258</td><td>DQ11_B</td></tr><tr><td>DQS3_A_t</td><td>44</td><td>188</td><td>VSS</td><td>DQS1_B_t</td><td>115</td><td>259</td><td>VSS</td></tr><tr><td>DQS3_A_c</td><td>45</td><td>189</td><td>DQS8_A_c, TDQS8_A_c</td><td>DQS1_B_c</td><td>116</td><td>260</td><td>DQS6_B_c, TDQS6_B_c</td></tr><tr><td>VSS</td><td>46</td><td>190</td><td>DQS8_A_t, TDQS8_A_t</td><td>VSS</td><td>117</td><td>261</td><td>DQS6_B_t, TDQS6_B_t</td></tr><tr><td>DQ28_A</td><td>47</td><td>191</td><td>VSS</td><td>DQ12_B</td><td>118</td><td>262</td><td>VSS</td></tr><tr><td>VSS</td><td>48</td><td>192</td><td>DQ30_A</td><td>VSS</td><td>119</td><td>263</td><td>DQ14_B</td></tr><tr><td>DQ29_A</td><td>49</td><td>193</td><td>VSS</td><td>DQ13_B</td><td>120</td><td>264</td><td>VSS</td></tr><tr><td>VSS</td><td>50</td><td>194</td><td>DQ31_A</td><td>VSS</td><td>121</td><td>265</td><td>DQ15_B</td></tr><tr><td>CB0_A</td><td>51</td><td>195</td><td>VSS</td><td>DQ16_B</td><td>122</td><td>266</td><td>VSS</td></tr><tr><td>VSS</td><td>52</td><td>196</td><td>CB2_A</td><td>VSS</td><td>123</td><td>267</td><td>DQ18_B</td></tr><tr><td>CB1_A</td><td>53</td><td>197</td><td>VSS</td><td>DQ17_B</td><td>124</td><td>268</td><td>VSS</td></tr><tr><td>VSS</td><td>54</td><td>198</td><td>CB3_A</td><td>VSS</td><td>125</td><td>269</td><td>DQ19_B</td></tr><tr><td>DQS4_A_t</td><td>55</td><td>199</td><td>VSS</td><td>DQS2_B_t</td><td>126</td><td>270</td><td>VSS</td></tr><tr><td>DQS4_A_c</td><td>56</td><td>200</td><td>DQS9_A_c, TDQS9_A_c</td><td>DQS2_B_c</td><td>127</td><td>271</td><td>DQS7_B_c, TDQS7_B_c</td></tr><tr><td>VSS</td><td>57</td><td>201</td><td>DQS9_A_t, TDQS9_A_t</td><td>VSS</td><td>128</td><td>272</td><td>DQS7_B_t, TDQS7_B_t</td></tr><tr><td>CB4_A</td><td>58</td><td>202</td><td>VSS</td><td>DQ20_B</td><td>129</td><td>273</td><td>VSS</td></tr><tr><td>VSS</td><td>59</td><td>203</td><td>CB6_A</td><td>VSS</td><td>130</td><td>274</td><td>DQ22_B</td></tr><tr><td>CB5_A</td><td>60</td><td>204</td><td>VSS</td><td>DQ21_B</td><td>131</td><td>275</td><td>VSS</td></tr><tr><td>VSS</td><td>61</td><td>205</td><td>CB7_A</td><td>VSS</td><td>132</td><td>276</td><td>DQ23_B</td></tr><tr><td>ALERT_n</td><td>62</td><td>206</td><td>VSS</td><td>DQ24_B</td><td>133</td><td>277</td><td>VSS</td></tr><tr><td>VSS</td><td>63</td><td>207</td><td>RESET_n</td><td>VSS</td><td>134</td><td>278</td><td>DQ26_B</td></tr><tr><td>CS0_A_n</td><td>64</td><td>208</td><td>VSS</td><td>DQ25_B</td><td>135</td><td>279</td><td>VSS</td></tr><tr><td>VSS</td><td>65</td><td>209</td><td>CS1_A_n</td><td>VSS</td><td>136</td><td>280</td><td>DQ27_B</td></tr><tr><td>CA0_A</td><td>66</td><td>210</td><td>VSS</td><td>DQS3_B_t</td><td>137</td><td>281</td><td>VSS</td></tr><tr><td>VSS</td><td>67</td><td>211</td><td>CA1_A</td><td>DQS3_B_c</td><td>138</td><td>282</td><td>DQS8_B_c, TDQS8_B_c</td></tr><tr><td>CA2_A</td><td>68</td><td>212</td><td>VSS</td><td>VSS</td><td>139</td><td>283</td><td>DQS8_B_t, TDQS8_B_t</td></tr><tr><td>VSS</td><td>69</td><td>213</td><td>CA3_A</td><td>DQ28_B</td><td>140</td><td>284</td><td>VSS</td></tr><tr><td>CA4_A</td><td>70</td><td>214</td><td>VSS</td><td>VSS</td><td>141</td><td>285</td><td>DQ30_B</td></tr><tr><td>VSS</td><td>71</td><td>215</td><td>CA5_A</td><td>DQ29_B</td><td>142</td><td>286</td><td>VSS</td></tr><tr><td>CA6_A</td><td>72</td><td>216</td><td>VSS</td><td>VSS</td><td>143</td><td>287</td><td>DQ31_B</td></tr><tr><td>VSS</td><td>73</td><td>217</td><td>CK_t</td><td>RFU</td><td>144</td><td>288</td><td>VSS</td></tr></table> NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible. NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.										Front Side	Pin	Pin	Back side	Front Side	Pin	Pin	Back side	Pin Label			Pin Label	Pin Label			Pin Label	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B	DQ24_A	40	184	VSS	DQ8_B	111	255	VSS	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B	CB0_A	51	195	VSS	DQ16_B	122	266	VSS	VSS	52	196	CB2_A	VSS	123	267	DQ18_B	CB1_A	53	197	VSS	DQ17_B	124	268	VSS	VSS	54	198	CB3_A	VSS	125	269	DQ19_B	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	VSS	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t	CB4_A	58	202	VSS	DQ20_B	129	273	VSS	VSS	59	203	CB6_A	VSS	130	274	DQ22_B	CB5_A	60	204	VSS	DQ21_B	131	275	VSS	VSS	61	205	CB7_A	VSS	132	276	DQ23_B	ALERT_n	62	206	VSS	DQ24_B	133	277	VSS	VSS	63	207	RESET_n	VSS	134	278	DQ26_B	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c	CA2_A	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t	VSS	69	213	CA3_A	DQ28_B	140	284	VSS	CA4_A	70	214	VSS	VSS	141	285	DQ30_B	VSS	71	215	CA5_A	DQ29_B	142	286	VSS	CA6_A	72	216	VSS	VSS	143	287	DQ31_B	VSS	73	217	CK_t	RFU	144	288	VSS
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DQ24_A	40	184	VSS	DQ8_B	111	255	VSS																																																																																																																																																																																																																																																																																																											
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	Ex. CC-88 (JEDEC 305B) at 7.																																																																																																																																																																																																																																																																																																																	

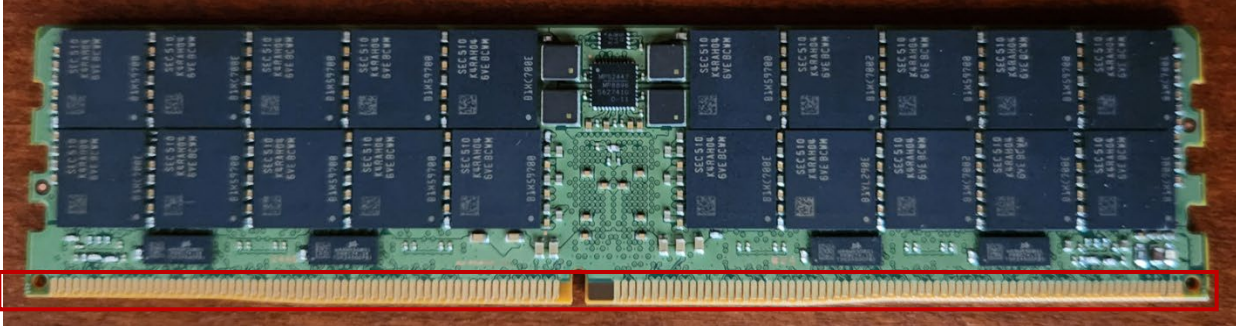
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
[1.b] memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines;	The Samsung Practicing '407 Products comprise memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines. For example, as seen below, each Samsung Practicing Product has multiple memory devices (there are one or more memory devices per DRAM package) mounted on the PCB and organized in a plurality of groups:  Ex. ITC-1472 CC-116 (Samsung 4800MHz PC5-38400). Ex. ITC-1472 CC-117 (Samsung 5600MHz PC5-44800).

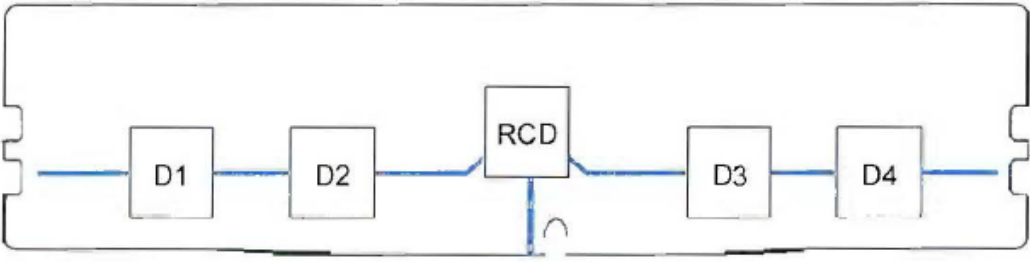
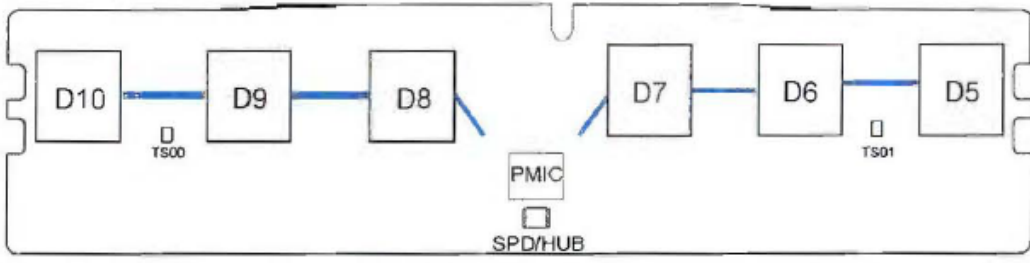
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Claim 1	Evidence of Use
	A photograph of a Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE03251213251911, and other specifications. The label also includes a barcode and a QR code. The module is populated with 16 black memory chips, each labeled with 'SEC510' and 'K4D8008'. Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK). A photograph of a Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number 80CE01251501799891, and other specifications. The label also includes a barcode and a QR code. The module is populated with 16 black memory chips, each labeled with 'SEC513' and 'K4D8008'. Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP).

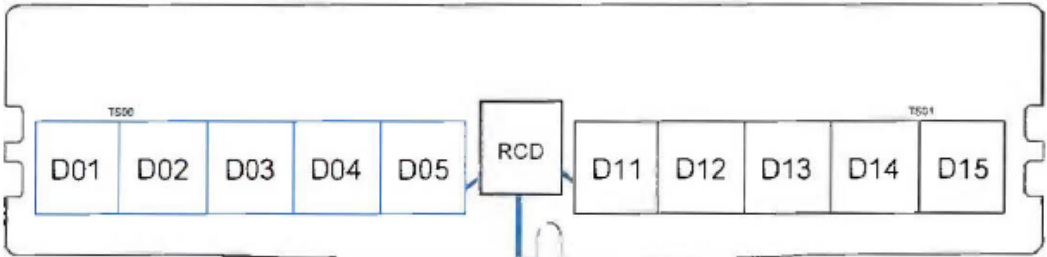
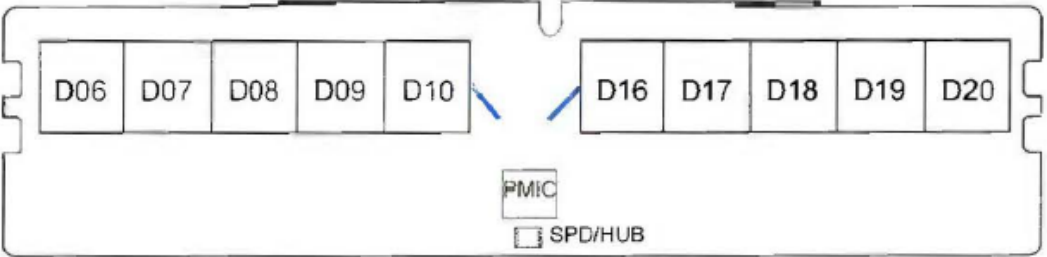
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	  An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. <i>See</i> Exs. CC-5, CC-6.

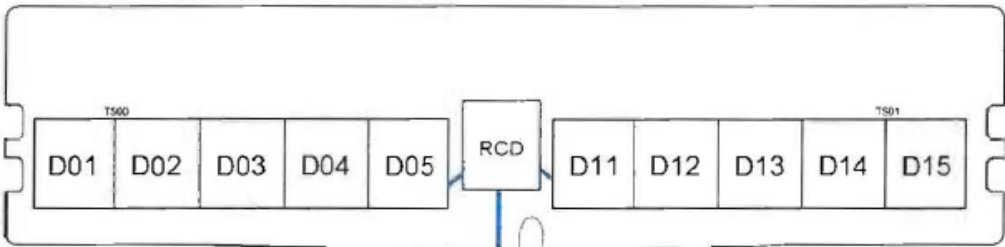
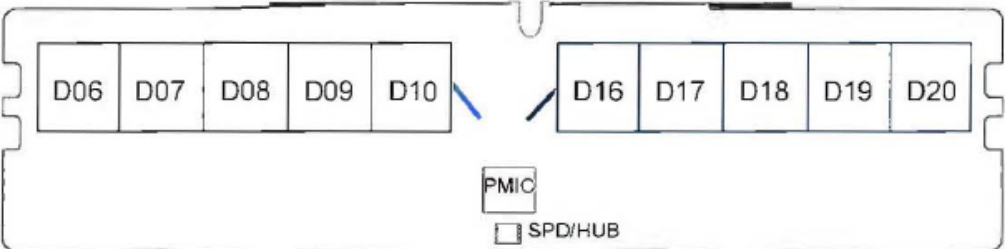
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21.

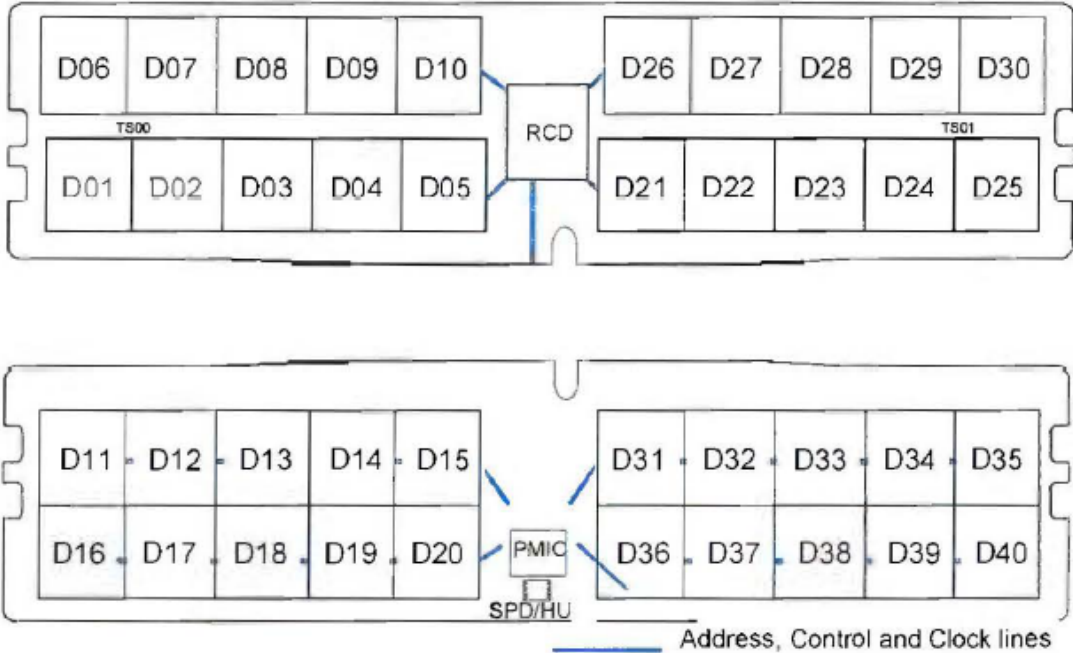
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  — Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29.

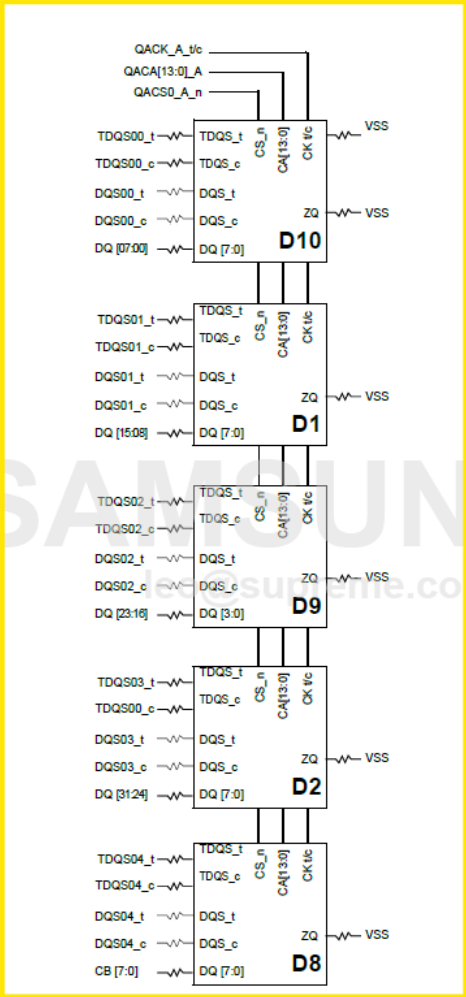
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Claim 1	Evidence of Use
	 The diagram illustrates the internal structure of a DDR5 RDIMM. It shows two main sections. The top section contains two banks of memory devices, labeled TS00 and TS01. Bank TS00 includes devices D06 through D10 in the top row and D01 through D05 in the bottom row. Bank TS01 includes devices D26 through D30 in the top row and D21 through D25 in the bottom row. These banks are connected to a central RCD (Refresh Control Driver). The bottom section contains two more banks of memory devices. The left bank includes devices D11 through D15 in the top row and D16 through D20 in the bottom row. The right bank includes devices D31 through D35 in the top row and D36 through D40 in the bottom row. These banks are connected to a PMIC (Power Management IC) and an SPD/HU (Serial Presence Detect/Hot Plug). A legend indicates that blue lines represent 'Address, Control and Clock lines'. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33. The memory devices are organized in a plurality of groups, and a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	REGISTER DCA[6:0]_A — QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10] DCS0_A_n — QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10] DCS1_A_n — DCA[6:0]_B — QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3] DCS0_B_n — QACS0_B -> CS_n: SDRAMs D [7:3] DCS1_B_n — DCK_t — QACK_A_t/c -> CK_t/c: SDRAMs D [1,2,8,9,10] DCK_c — QACK_A_t/c -> CK_t/c: SDRAMs D [7,3] QLBD — DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7,3] QLBS — DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7,3] DRST_n — QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3] ALERT_n — DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21] Ex. CC-89 at 21 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  The diagram illustrates the internal structure of Channel A, showing five memory modules (D10, D1, D9, D2, D8) connected to a common bus. Each module has inputs for TDQS_t, TDQS_e, DQS_t, DQS_e, and DQ. The modules are labeled D10, D1, D9, D2, and D8. A large 'SAMSUNG' watermark is visible across the diagram. Ex. CC-89 at 22 (Samsung DDR5 RDIMM Datasheet)

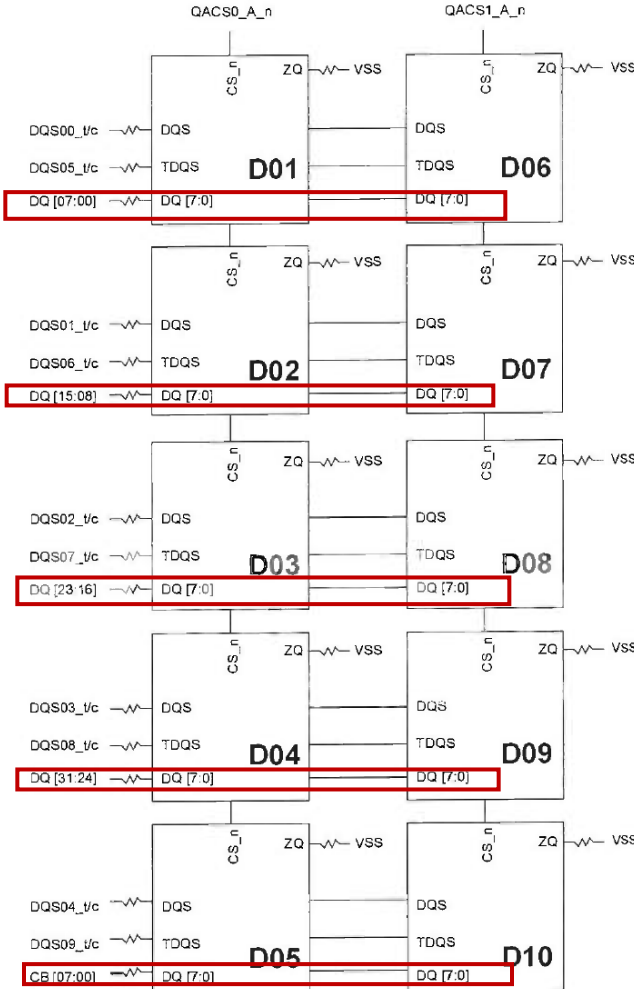
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B The diagram illustrates the Channel B configuration of a memory system. It shows five memory modules, labeled D7, D3, D6, D4, and D5, connected to a common bus. Each module has the following inputs/outputs: TDQS_t: Training Data Strobe (top) TDQS_c: Training Data Strobe (bottom) DQS_t: Data Strobe (top) DQS_c: Data Strobe (bottom) DQ: Data Bus (bottom) The modules are connected to a common bus, and the diagram is enclosed in a red box.
	Ex. CC-89 at 23 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[06:0]_A ——— QACA[13:00]_A -> CA[13:00]: SDRAMs D [10:01] DCS0_A_n ——— QACS0_A -> CS_n: SDRAMs D [05:01] DCS1_A_n ——— QACS1_A -> CS_n: SDRAMs D [10:06] DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D [20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMs D [15:11] DCS1_B_n ——— QACS1_B -> CS_n: SDRAMs D [20:16] DCK_t/DCK_c ——— QACK_A_t -> CK_t: SDRAMs D [05:01] / QACK_A_c -> CK_c: SDRAMs D [05:01] — QCCK_A_t -> CK_t: SDRAMs D [10:06] / QCCK_A_c -> CK_c: SDRAMs D [10:06] — QACK_B_t -> CK_t: SDRAMs D [15:11] / QACK_B_c -> CK_c: SDRAMs D [15:11] — QCCK_B_t -> CK_t: SDRAMs D [20:16] / QCCK_B_c -> CK_c: SDRAMs D [20:16] QLBD ——— DLBD_A -> LBDQ: SDRAMs D [10:01] — DLBD_B -> LBDQ: SDRAMs D [20:11] QLBS ——— DLBS_A -> LBDQS: SDRAMs D [10:01] — DLBS_B -> LBDQS: SDRAMs D [20:11] DRST_n ——— QRT_A_n -> RESET_n: SDRAMs D [10:01] — QRT_B_n -> RESET_n: SDRAMs D [20:11] ALERT_n ——— DERROR_IN_A_n -> ALERT_n: SDRAMs D [10:01] — DERROR_IN_B_n -> ALERT_n: SDRAMs D [20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet) Channel A  The diagram illustrates the internal structure of Channel A, showing ten DRAMs (D01 through D10) arranged in two columns. Each DRAM is connected to a common address bus (QACS0_A_n and QACS1_A_n) and a common data bus (DQ[7:0]). The DRAMs are organized into five pairs (D01-D02, D03-D04, D05-D06, D07-D08, D09-D10). Each pair is connected to a common data bus (DQ[7:0]). The diagram shows the internal structure of each DRAM, including the address bus (QACS0_A_n and QACS1_A_n), the data bus (DQ[7:0]), and the control signals (DQS, TDQS, and DQ[7:0]). Red boxes highlight the DQ[7:0] signal for each DRAM, indicating its connection to the common data bus.

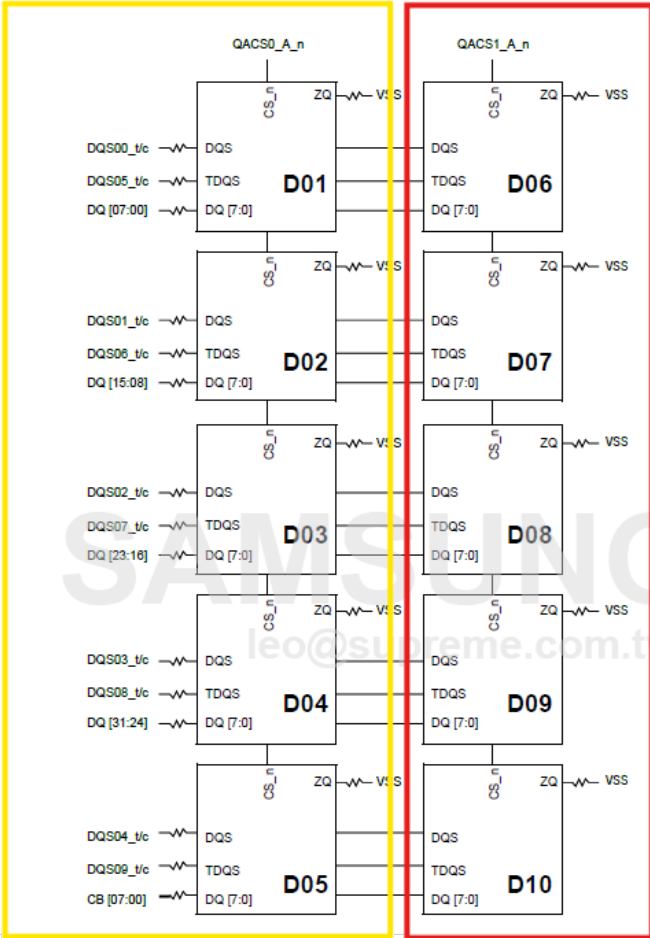
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use Ex. CC-89 at 26 (Samsung DDR5 RDIMM Datasheet) Channel B
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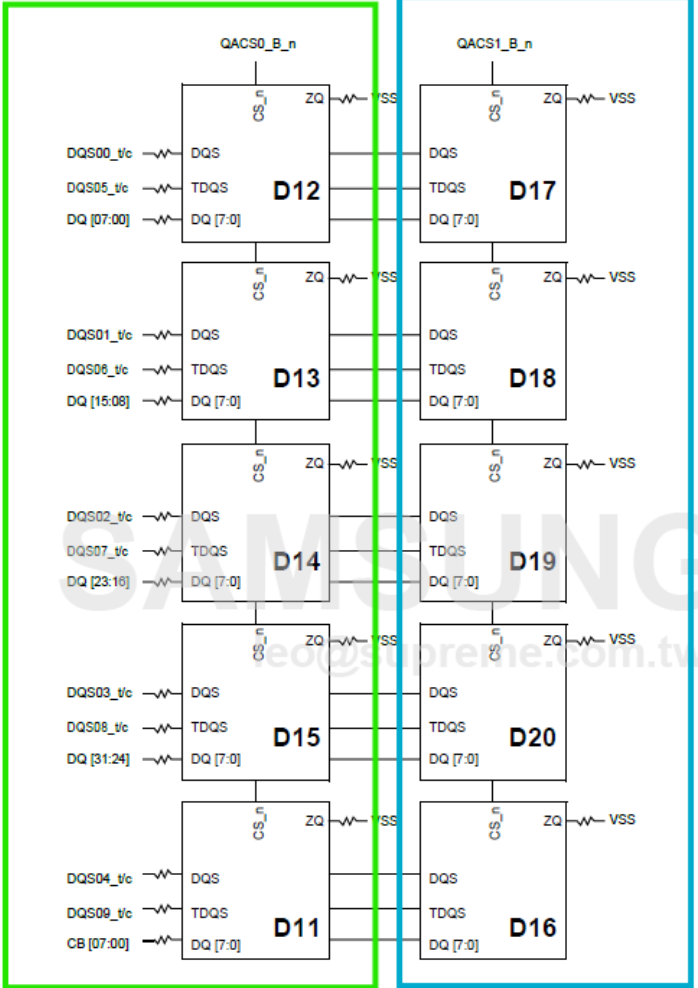
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 27 (Samsung DDR5 RDIMM Datasheet). The diagram illustrates the timing and signal connections for a DDR5 RDIMM. A central vertical pink bar is labeled REGISTER. Signals are connected to this register as follows: DCA[06:0]_A connects to QACA[13:00]_A, which maps to CA[13:00] : SDRAMs D [10:01]. DCS0_A_n connects to QACS0_A, which maps to CS_n : SDRAMs D [05:01]. DCS1_A_n connects to QACS1_A, which maps to CS_n : SDRAMs D [10:06]. DCA[06:00]_B connects to QACA[13:00]_B, which maps to CA[13:00] : SDRAMs D [20:11]. DCS0_B_n connects to QACS0_B, which maps to CS_n : SDRAMs D [15:11]. DCS1_B_n connects to QACS1_B, which maps to CS_n : SDRAMs D [20:16]. DCK_t/DCK_c connects to QACK_A_t, QACK_A_c, QACK_B_t, and QACK_B_c. These map to various SDRAM data bus connections: QACK_A_t and QACK_A_c map to CK_t : SDRAMs D [05:01] and CK_c : SDRAMs D [05:01]. QACK_B_t and QACK_B_c map to CK_t : SDRAMs D [15:11] and CK_c : SDRAMs D [15:11]. QACK_A_t and QACK_A_c also map to CK_t : SDRAMs D [10:06] and CK_c : SDRAMs D [10:06]. QACK_B_t and QACK_B_c also map to CK_t : SDRAMs D [20:16] and CK_c : SDRAMs D [20:16]. QLBD connects to DLBD_A and DLBD_B, which map to LBDQ : SDRAMs D [10:01] and LBDQ : SDRAMs D [20:11]. QLBS connects to DLBS_A and DLBS_B, which map to LBDQS : SDRAMs D [10:01] and LBDQS : SDRAMs D [20:11]. DRST_n connects to QRT_A_n and QRT_B_n, which map to RESET_n : SDRAMs D [10:01] and RESET_n : SDRAMs D [20:11]. ALERT_n connects to DERROR_IN_A_n and DERROR_IN_B_n, which map to ALERT_n : SDRAMs D [10:01] and ALERT_n : SDRAMs D [20:11]. Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  Ex. CC-89 at 26 (Samsung DDR5 RDIMM Datasheet).

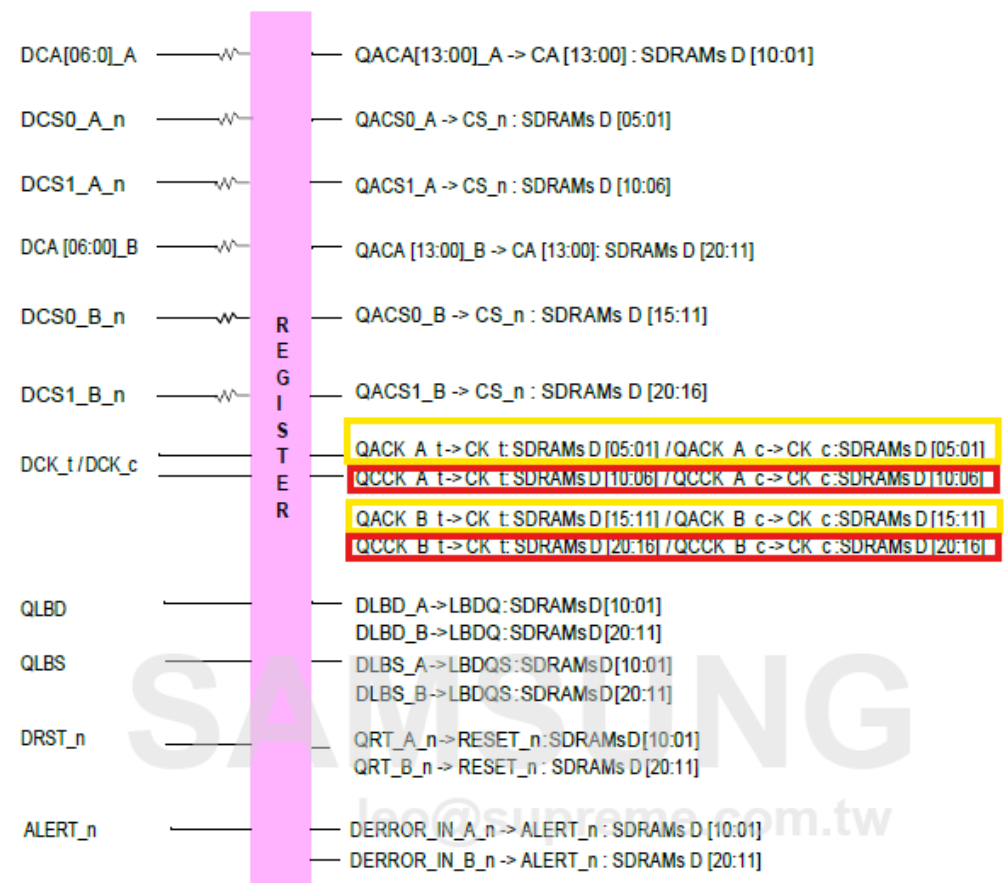
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 27 (Samsung DDR5 RDIMM Datasheet).

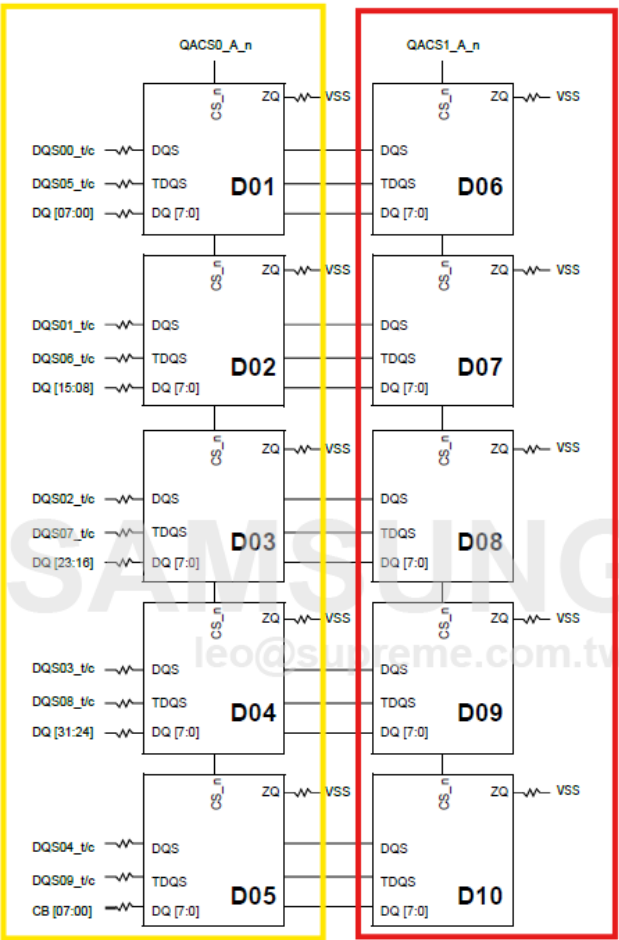
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A The diagram illustrates the internal structure of Channel A, which is divided into five horizontal sections, each containing two data blocks (D01-D10). Each block is connected to a common bus structure. The inputs to each block are DQS, TDQS, and DQ. The outputs are ZQ and VSS. The blocks are color-coded: D01-D02 (yellow), D03-D04 (green), D05-D06 (blue), D07-D08 (red), D09-D10 (orange). Block D01 (Yellow): Inputs: DQS00_t/c, DQS05_t/c, DQ [07:00]. Outputs: ZQ, VSS. Block D02 (Red): Inputs: DQS01_t/c, DQS06_t/c, DQ [15:08]. Outputs: ZQ, VSS. Block D03 (Green): Inputs: DQS02_t/c, DQS07_t/c, DQ [23:16]. Outputs: ZQ, VSS. Block D04 (Blue): Inputs: DQS03_t/c, DQS08_t/c, DQ [31:24]. Outputs: ZQ, VSS. Block D05 (Orange): Inputs: DQS04_t/c, DQS09_t/c, CB [07:00]. Outputs: ZQ, VSS. Block D06 (Yellow): Inputs: DQS, TDQS, DQ [7:0]. Outputs: ZQ, VSS. Block D07 (Red): Inputs: DQS, TDQS, DQ [7:0]. Outputs: ZQ, VSS. Block D08 (Green): Inputs: DQS, TDQS, DQ [7:0]. Outputs: ZQ, VSS. Block D09 (Blue): Inputs: DQS, TDQS, DQ [7:0]. Outputs: ZQ, VSS. Block D10 (Orange): Inputs: DQS, TDQS, DQ [7:0]. Outputs: ZQ, VSS.

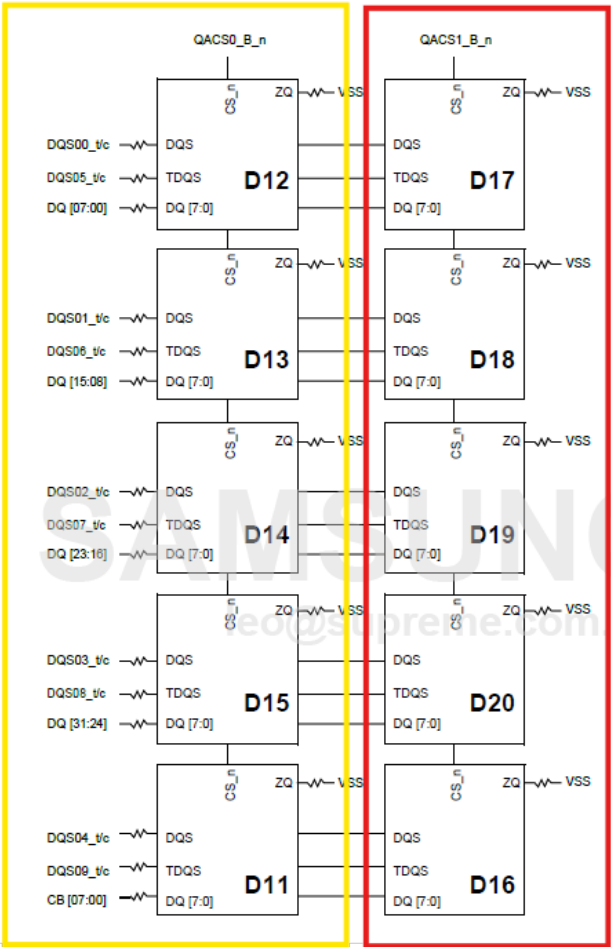
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 26 (Samsung DDR5 RDIMM Datasheet).  The diagram illustrates the internal register operations for various signals in a Samsung DDR5 RDIMM. A central vertical pink bar is labeled "REGISTER". Signals are connected to this register, and their internal operations are detailed on the right side of the diagram. DCA[06:00]_A connects to QACA[13:00]_A, which maps to CA[13:00] in SDRAMs D [10:01]. DCS0_A_n connects to QACS0_A, which maps to CS_n in SDRAMs D [05:01]. DCS1_A_n connects to QACS1_A, which maps to CS_n in SDRAMs D [10:06]. DCA [06:00]_B connects to QACA [13:00]_B, which maps to CA [13:00] in SDRAMs D [20:11]. DCS0_B_n connects to QACS0_B, which maps to CS_n in SDRAMs D [15:11]. DCS1_B_n connects to QACS1_B, which maps to CS_n in SDRAMs D [20:16]. DCK_t / DCK_c connects to REGISTER. This signal has four associated operations: QACK_A t->CK t:SDRAMs D [05:01] / QACK_A c->CK c:SDRAMs D [05:01] (highlighted in yellow) QCCK_A t->CK t:SDRAMs D [10:06] / QCCK_A c->CK c:SDRAMs D [10:06] (highlighted in red) QACK_B t->CK t:SDRAMs D [15:11] / QACK_B c->CK c:SDRAMs D [15:11] (highlighted in yellow) QCCK_B t->CK t:SDRAMs D [20:16] / QCCK_B c->CK c:SDRAMs D [20:16] (highlighted in red) QLBD connects to DLBD_A->LBDQ:SDRAMs D [10:01] and DLBD_B->LBDQ:SDRAMs D [20:11]. QLBS connects to DLBS_A->LBDQS:SDRAMs D [10:01] and DLBS_B->LBDQS:SDRAMs D [20:11]. DRST_n connects to QRT_A_n->RESET_n:SDRAMs D [10:01] and QRT_B_n->RESET_n:SDRAMs D [20:11]. ALERT_n connects to DERROR_IN_A_n->ALERT_n:SDRAMs D [10:01] and DERROR_IN_B_n->ALERT_n:SDRAMs D [20:11]. Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet).

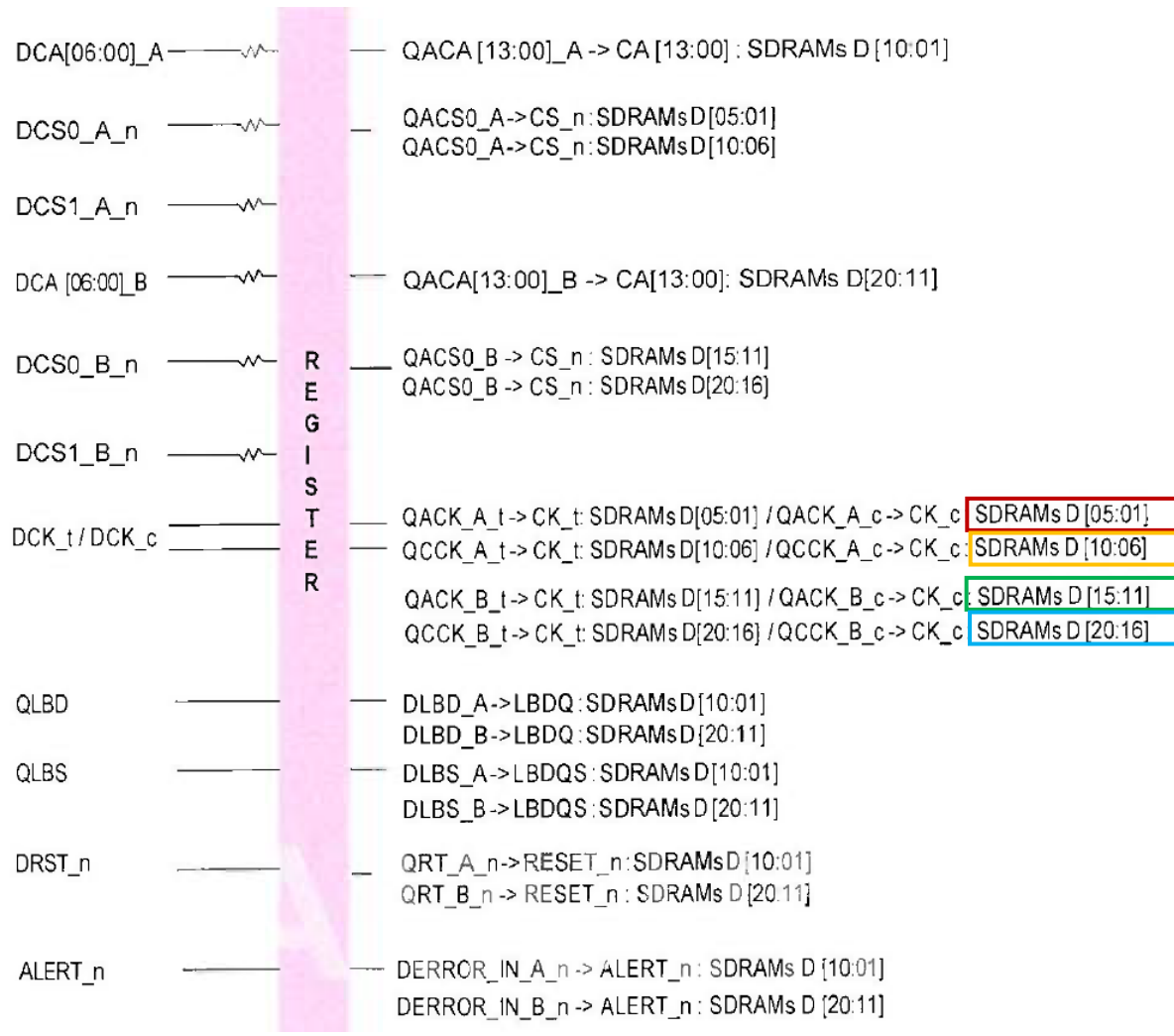
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  Ex. CC-89 at 26 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 27 (Samsung DDR5 RDIMM Datasheet).

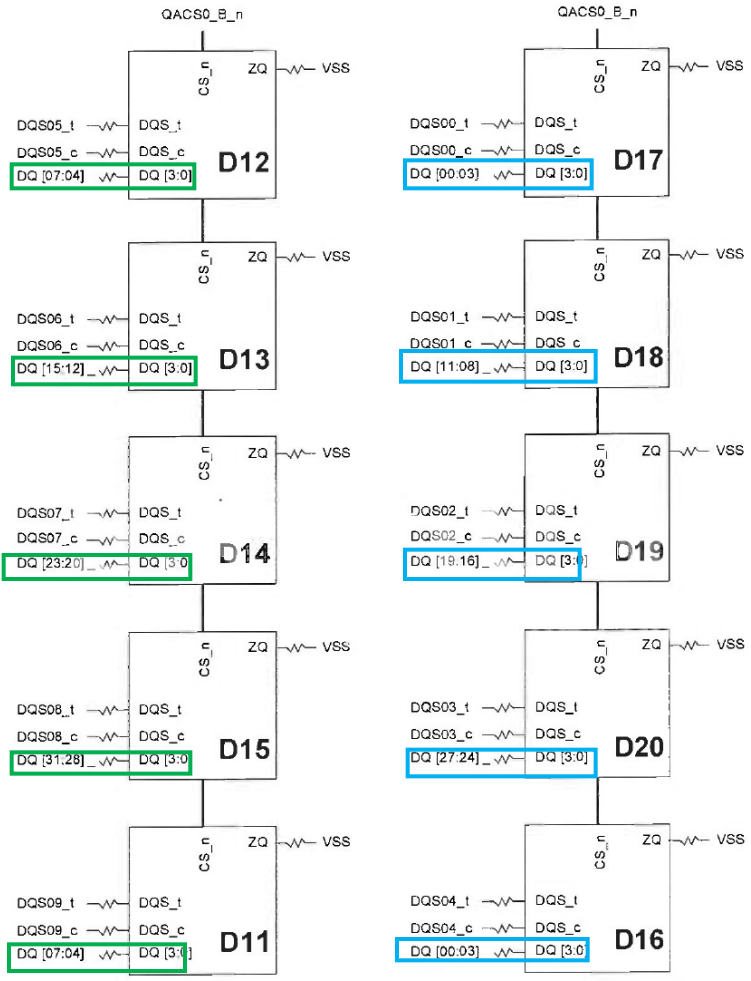
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:00]_A — QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n — QACS0_A->CS_n: SDRAMs D[05:01] QACS0_A->CS_n: SDRAMs D[10:06] DCS1_A_n — DCA [06:00]_B — QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n — QACS0_B -> CS_n: SDRAMs D[15:11] QACS0_B -> CS_n: SDRAMs D[20:16] DCS1_B_n — DCK_t/DCK_c — QACK_A_t->CK_t: SDRAMs D[05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] QCKK_A_t->CK_t: SDRAMs D[10:06] / QCKK_A_c->CK_c: SDRAMs D [10:06] QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QCKK_B_t->CK_t: SDRAMs D[20:16] / QCKK_B_c->CK_c: SDRAMs D [20:16] QLBD — DLBD_A->LBDQ: SDRAMs D[10:01] DLBD_B->LBDQ: SDRAMs D[20:11] QLBS — DLBS_A->LBDQS: SDRAMs D[10:01] DLBS_B->LBDQS: SDRAMs D[20:11] DRST_n — QRT_A_n->RESET_n: SDRAMs D [10:01] QRT_B_n-> RESET_n: SDRAMs D [20:11] ALERT_n — DERROR_IN_A_n-> ALERT_n: SDRAMs D [10:01] DERROR_IN_B_n-> ALERT_n: SDRAMs D [20:11] REGISTER
	Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A Ex. CC-89 at 30 (Samsung DDR5 RDIMM Datasheet)

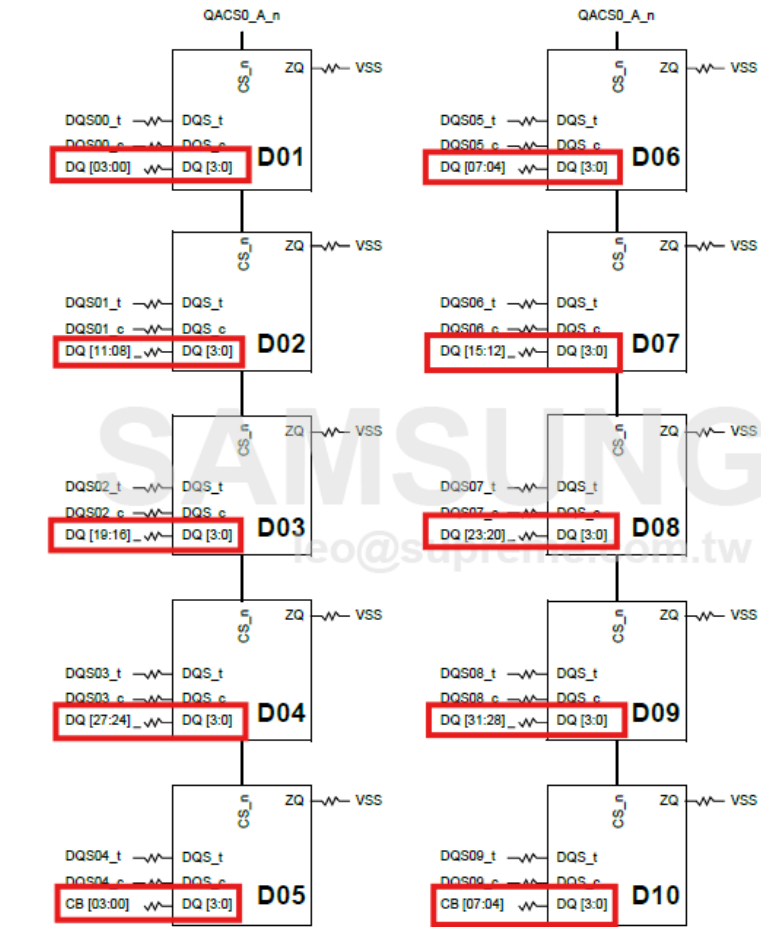
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 31 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram shows a central vertical pink bar labeled "REGISTER". To its left are signal names, and to its right are their functions. Signals with a triangle symbol are inputs, while others are outputs. - DCA[06:00]_A — QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] - DCS0_A_n — QACS0_A->CS_n: SDRAMs D[05:01] QACS0_A->CS_n: SDRAMs D[10:06] - DCS1_A_n - DCA[06:00]_B — QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] - DCS0_B_n — QACS0_B -> CS_n: SDRAMs D[15:11] QACS0_B -> CS_n: SDRAMs D[20:16] - DCS1_B_n - DCK_t / DCK_c — QACK_A_t->CK_t: SDRAMs D[05:01] / QACK_A_c->CK_c: SDRAMs D[05:01] QCK_A_t->CK_t: SDRAMs D[10:06] / QCK_A_c->CK_c: SDRAMs D[10:06] QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D[15:11] QCK_B_t->CK_t: SDRAMs D[20:16] / QCK_B_c->CK_c: SDRAMs D[20:16] - QLBD — DLBD_A->LBDQ: SDRAMs D[10:01] DLBD_B->LBDQ: SDRAMs D[20:11] - QLBS — DLBS_A->LBDQS: SDRAMs D[10:01] DLBS_B->LBDQS: SDRAMs D[20:11] - DRST_n — QRT_A_n->RESET_n: SDRAMs D[10:01] QRT_B_n->RESET_n: SDRAMs D[20:11] - ALERT_n — DERROR_IN_A_n->ALERT_n: SDRAMs D[10:01] DERROR_IN_B_n->ALERT_n: SDRAMs D[20:11] REGISTER Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  Ex. CC-89 at 30 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

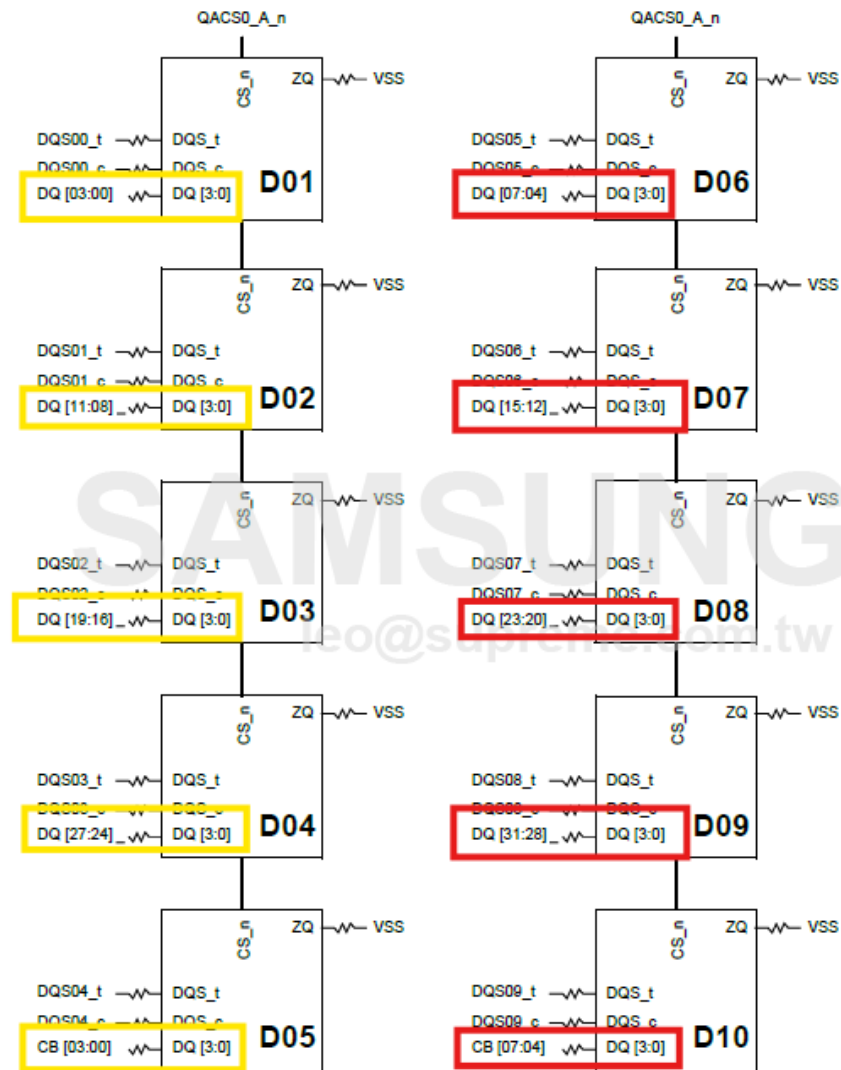
Claim 1	Evidence of Use
	Channel B Ex. CC-89 at 31 (Samsung DDR5 RDIMM Datasheet).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

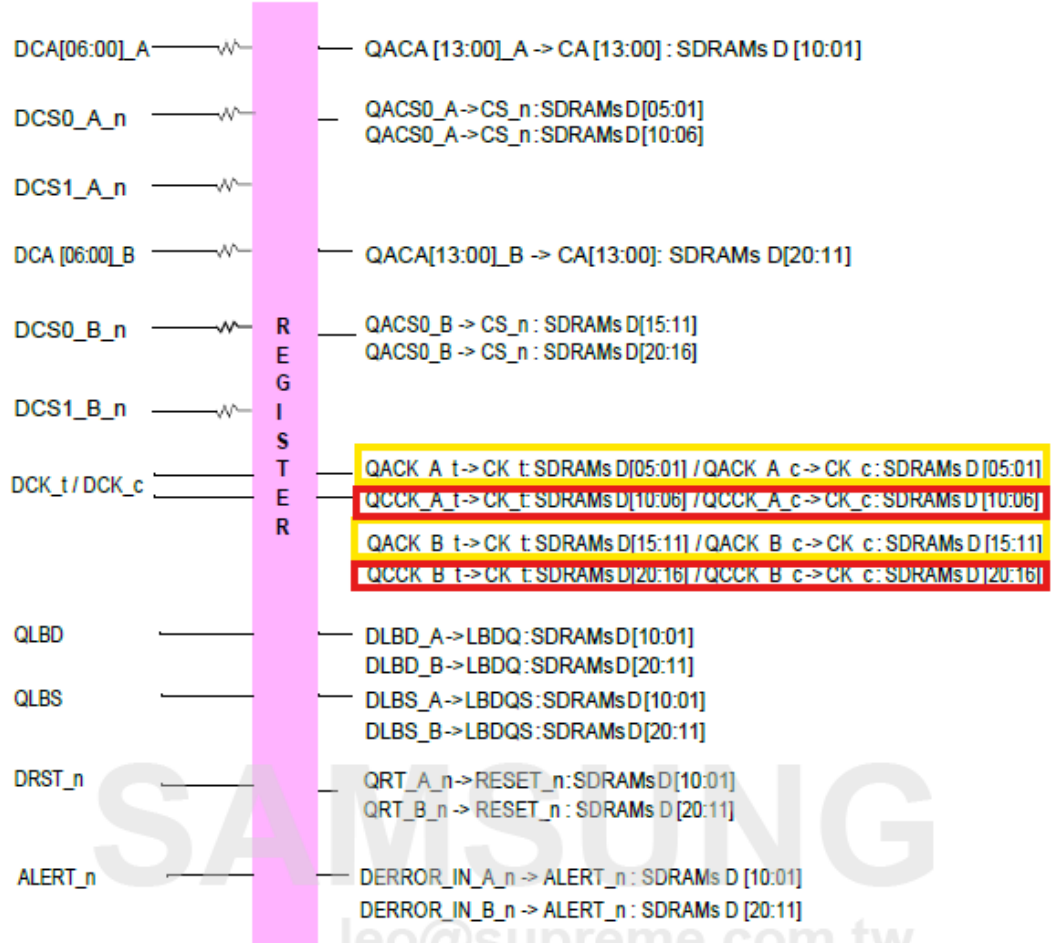
Claim 1	Evidence of Use
DCA[06:00]_A	QACA [13:00]_A -> CA [13:00]: SDRAMs D [10:01]
DCS0_A_n	QACS0_A->CS_n:SDRAMsD[05:01] QACS0_A->CS_n:SDRAMsD[10:06]
DCS1_A_n	
DCA [06:00]_B	QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11]
DCS0_B_n	QACS0_B -> CS_n : SDRAMs D[15:11] QACS0_B -> CS_n : SDRAMs D[20:16]
DCS1_B_n	
DCK_t / DCK_c	QACK_A_t->CK_t:SDRAMs D[05:01] / QACK_A_c->CK_c:SDRAMs D [05:01] QCKK_A_t->CK_t:SDRAMs D[10:06] / QCKK_A_c->CK_c:SDRAMs D [10:06] QACK_B_t->CK_t:SDRAMs D[15:11] / QACK_B_c->CK_c:SDRAMs D [15:11] QCKK_B_t->CK_t:SDRAMs D[20:16] / QCKK_B_c->CK_c:SDRAMs D [20:16]
QLBD	DLBD_A->LBDQ:SDRAMsD[10:01] DLBD_B->LBDQ:SDRAMsD[20:11]
QLBS	DLBS_A->LBDQS:SDRAMsD[10:01] DLBS_B->LBDQS:SDRAMsD[20:11]
DRST_n	QRT_A_n->RESET_n:SDRAMsD[10:01] QRT_B_n->RESET_n:SDRAMs D [20:11]
ALERT_n	DERROR_IN_A_n->ALERT_n: SDRAMs D [10:01] DERROR_IN_B_n->ALERT_n: SDRAMs D [20:11]

Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet).

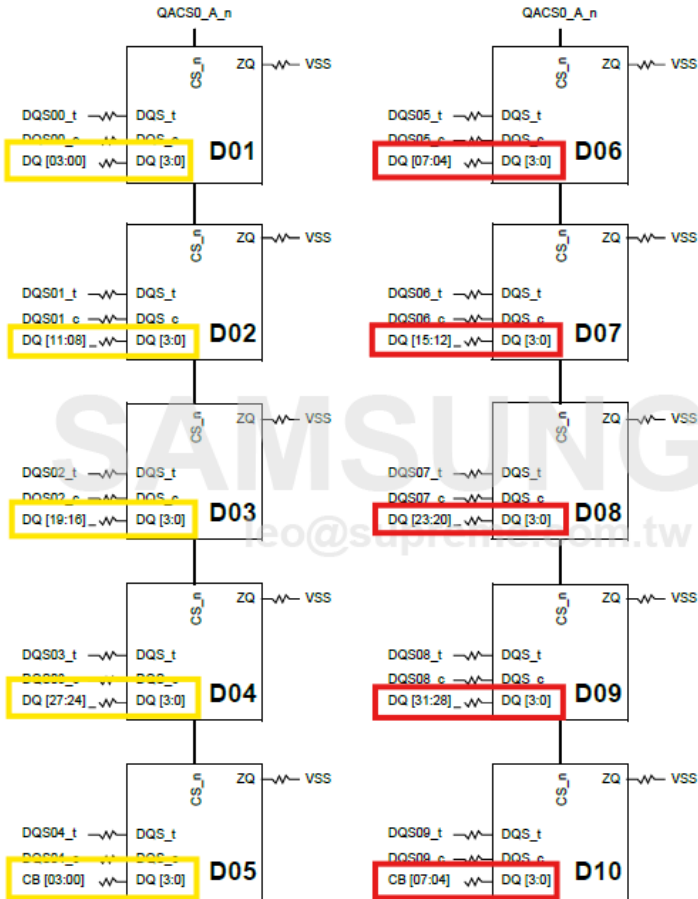
Channel A



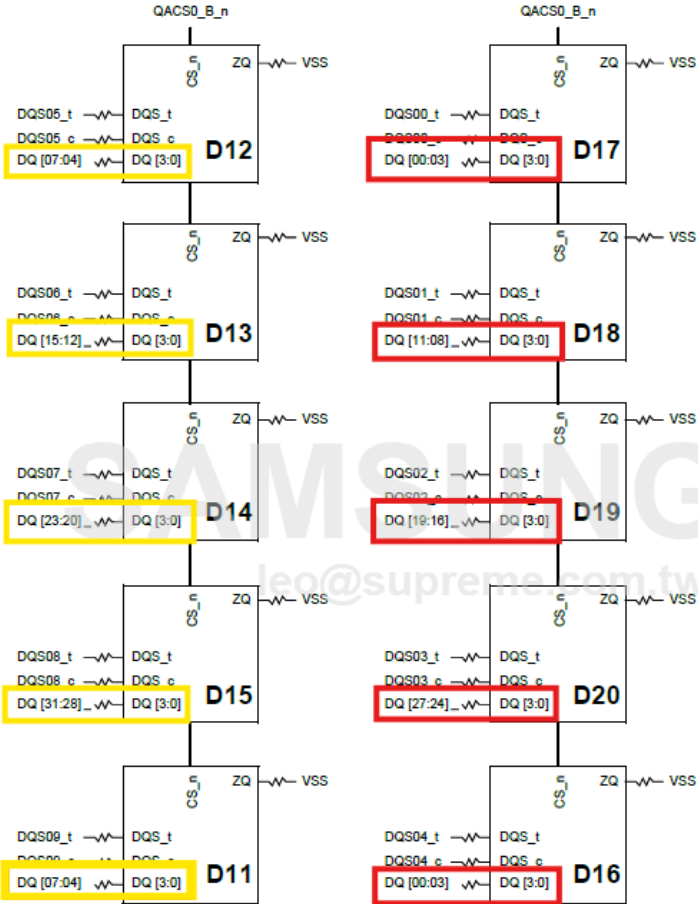
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 30 (Samsung DDR5 RDIMM Datasheet).
	 DCA[06:00]_A — QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n — QACS0_A->CS_n:SDRAMsD[05:01] QACS0_A->CS_n:SDRAMsD[10:06] DCS1_A_n — DCA [06:00]_B — QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D[15:11] QACS0_B -> CS_n : SDRAMs D[20:16] DCS1_B_n — DCK_t / DCK_c — QACK_A t->CK t:SDRAMsD[05:01] / QACK_A c->CK c:SDRAMsD[05:01] QACK_A t->CK t:SDRAMsD[10:06] / QACK_A c->CK c:SDRAMsD[10:06] QACK_B t->CK t:SDRAMsD[15:11] / QACK_B c->CK c:SDRAMsD[15:11] QACK_B t->CK t:SDRAMsD[20:16] / QACK_B c->CK c:SDRAMsD[20:16] QLBD — DLBD_A->LBDQ:SDRAMsD[10:01] DLBD_B->LBDQ:SDRAMsD[20:11] QLBS — DLBS_A->LBDQS:SDRAMsD[10:01] DLBS_B->LBDQS:SDRAMsD[20:11] DRST_n — QRT_A_n->RESET_n:SDRAMsD[10:01] QRT_B_n->RESET_n:SDRAMsD[20:11] ALERT_n — DERROR_IN_A_n->ALERT_n:SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n:SDRAMsD[20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet). Channel A  The diagram illustrates the internal structure of Channel A, showing two columns of DRAMs (D01-D05 on the left, D06-D10 on the right) connected to a common QACS0_A_n signal line. Each DRAM is connected to a ZQ signal and a VSS ground. The DRAMs are organized into two columns, each with five DRAMs. The DRAMs are labeled D01 through D10. Each DRAM has a DQS signal (DQS00_t to DQS04_t on the left, DQS05_t to DQS09_t on the right) and a DQ signal (DQ [3:0] to DQ [31:28] on the left, DQ [3:0] to DQ [31:28] on the right). The DQ signals are highlighted in yellow for D01-D05 and red for D06-D10. A large 'SAMSUNG' watermark is visible across the center of the diagram.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 30 (Samsung DDR5 RDIMM Datasheet). Channel B  Ex. CC-89 at 31 (Samsung DDR5 RDIMM Datasheet).

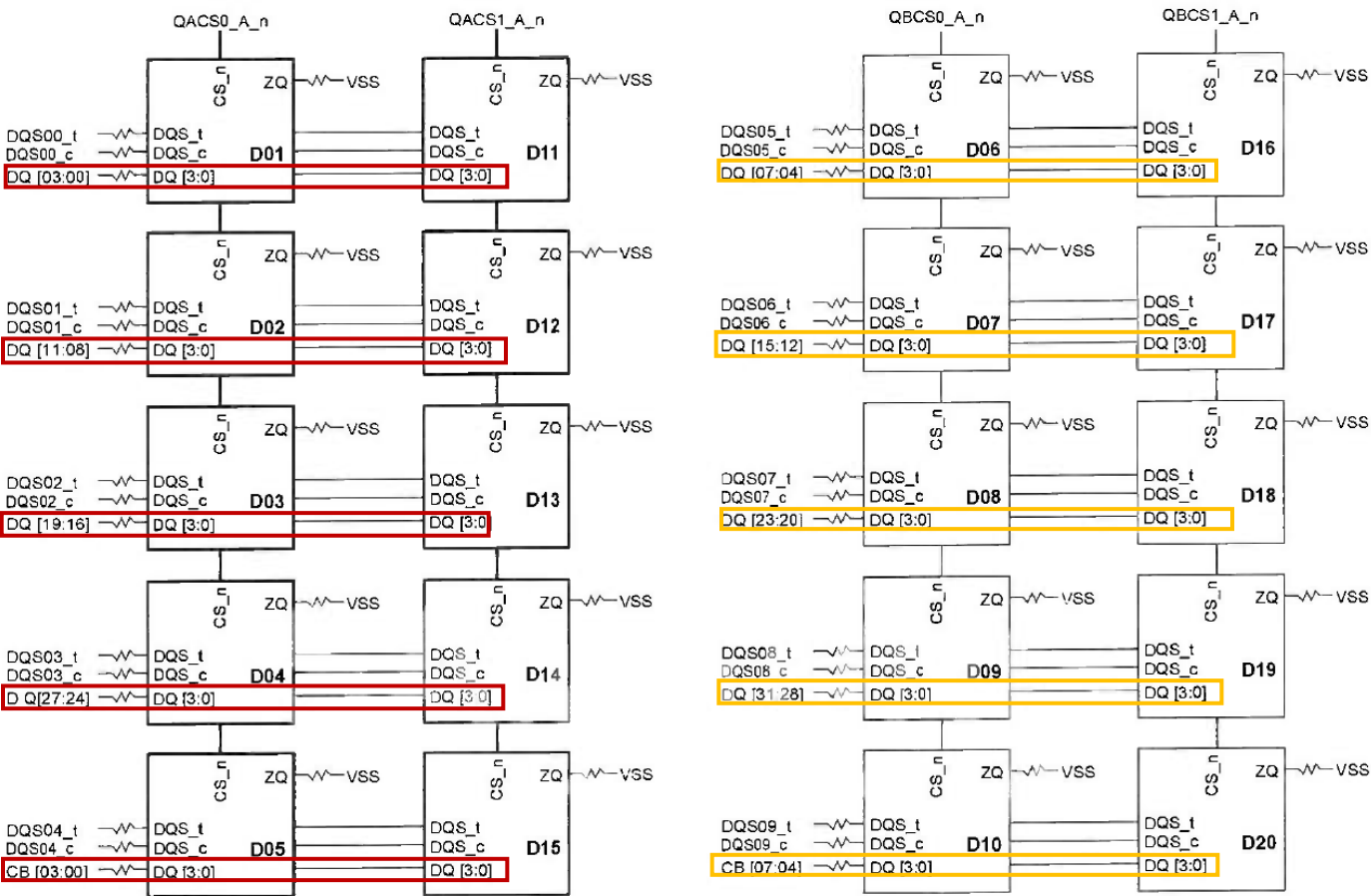
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates a memory array architecture with two memory banks, TS00 and TS01, connected to a central RCD (Row Control Driver) and a PMIC/SPD/HU (Power Management IC/Serial Presence Detect/Host Interface) block. The memory banks are organized into two rows of five columns each. The top row contains data lines D06, D07, D08, D09, and D10 for TS00, and D26, D27, D28, D29, and D30 for TS01. The bottom row contains data lines D01, D02, D03, D04, and D05 for TS00, and D21, D22, D23, D24, and D25 for TS01. The RCD block is connected to the data lines of both banks. The PMIC/SPD/HU block is connected to the data lines of both banks. A legend indicates that the blue lines represent Address, Control and Clock lines.

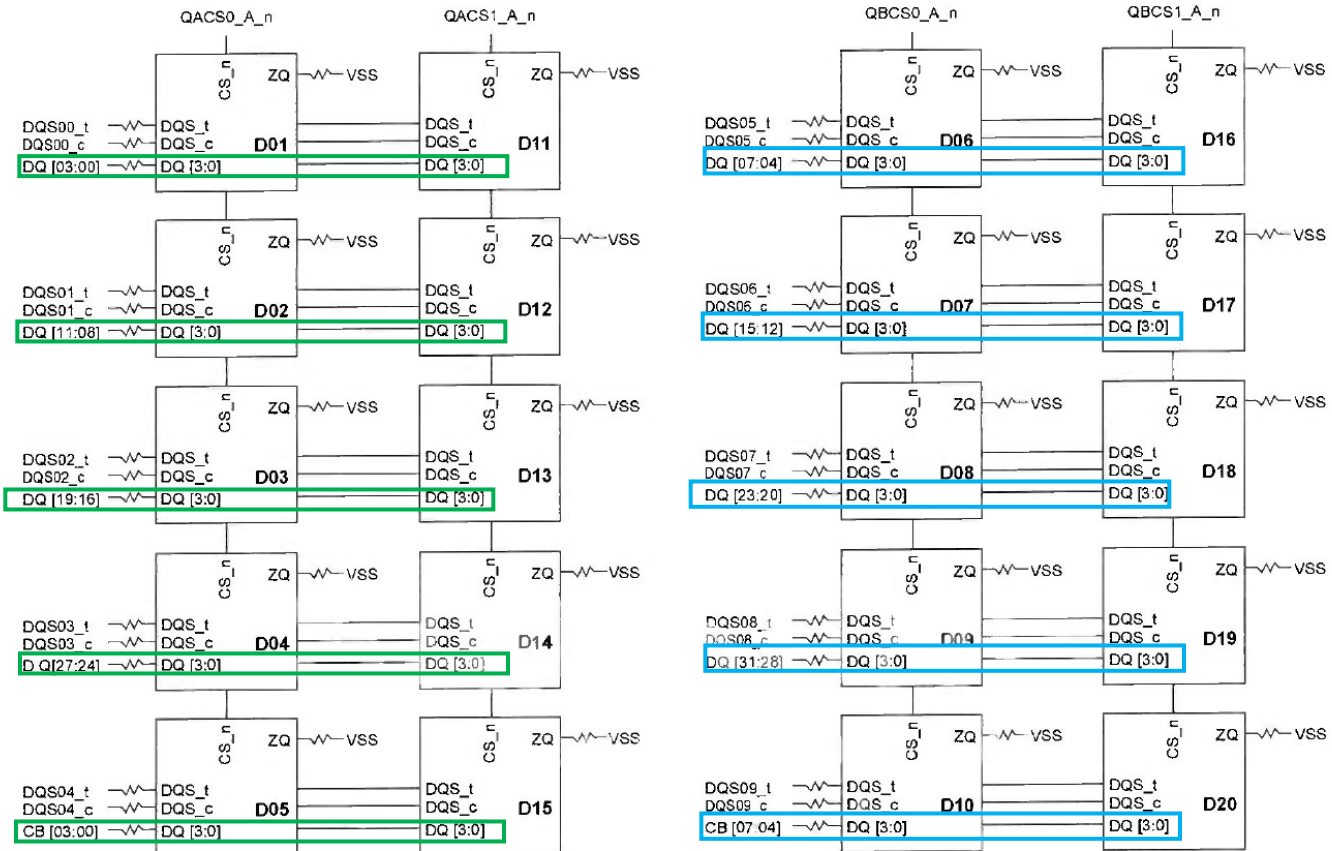
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

DCA[6:0]_A	—	QACA [13:0]_A → CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A → CA [13:0] : SDRAMs D [10:6, 20:16]
DCS0_A_n	—	QACS0_A → CS_n: SDRAMs D [5:1] QBCS0_A → CS_n: SDRAMs D [10:6]
DCS1_A_n	—	QACS1_A → CS_n: SDRAMs D [15:11] QBCS1_A → CS_n: SDRAMs D [20:16]
DCA[6:0]_B	—	QACA [13:0]_B → CA [13:0]: SDRAMs D [25:21, 35:31] QBCA [13:0]_B → CA [13:0]: SDRAMs D [30:26, 40:36]
DCS0_B_n	—	QACS0_B → CS_n: SDRAMs D [25:21] QBCS0_B → CS_n: SDRAMs D [30:26]
DCS1_B_n	—	QACS1_B → CS_n: SDRAMs D [35:31] QBCS1_B → CS_n: SDRAMs D [40:36]
DCK_t/DCK_c	—	QACK_A_t → CK_t: SDRAMs D [5:1] / QACK_A_c → CK_c: SDRAMs D [5:1] QBCK_A_t → CK_t: SDRAMs D [10:6] / QBCK_A_c → CK_c: SDRAMs D [10:6] QCCK_A_t → CK_t: SDRAMs D [15:11] / QCCK_A_c → CK_c: SDRAMs D [15:11] QDCK_A_t → CK_t: SDRAMs D [20:16] / QDCK_A_c → CK_c: SDRAMs D [20:16] QACK_B_t → CK_t: SDRAMs D [25:21] / QACK_B_c → CK_c: SDRAMs D [25:21] QBCK_B_t → CK_t: SDRAMs D [30:26] / QBCK_B_c → CK_c: SDRAMs D [30:26] QCCK_B_t → CK_t: SDRAMs D [35:31] / QCCK_B_c → CK_c: SDRAMs D [35:31] QDCK_B_t → CK_t: SDRAMs D [40:36] / QDCK_B_c → CK_c: SDRAMs D [40:36]
QLBD	—	DLBD_A → LBDQ: SDRAMs D [20:1] DLBD_B → LBDQ: SDRAMs D [40:21]
QLBS	—	DLBS_A → LBDQS: SDRAMs D [20:1] DLBS_B → LBDQS: SDRAMs D [40:21]
DRST_n	—	QRT_A_n → RESET_n: SDRAMs D [20:1] QRT_B_n → RESET_n: SDRAMs D [40:21]
ALERT_n	—	DERROR_IN_A_n → ALERT_n: SDRAMs D [20:1] DERROR_IN_B_n → ALERT_n: SDRAMs D [40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet) Channel A  The diagram illustrates the internal structure of Channel A, showing two columns of memory modules. Each module is connected to a common bus (QACS0_A_n, QACS1_A_n, QBCS0_A_n, QBCS1_A_n) and has its own ZQ and VSS connections. Data signals are shown as DQS_t, DQS_c, and DQ [3:0]. Red boxes highlight DQ [3:0] signals for modules D01-D05 and D11-D15. Yellow boxes highlight DQ [3:0] signals for modules D06-D10 and D16-D20.

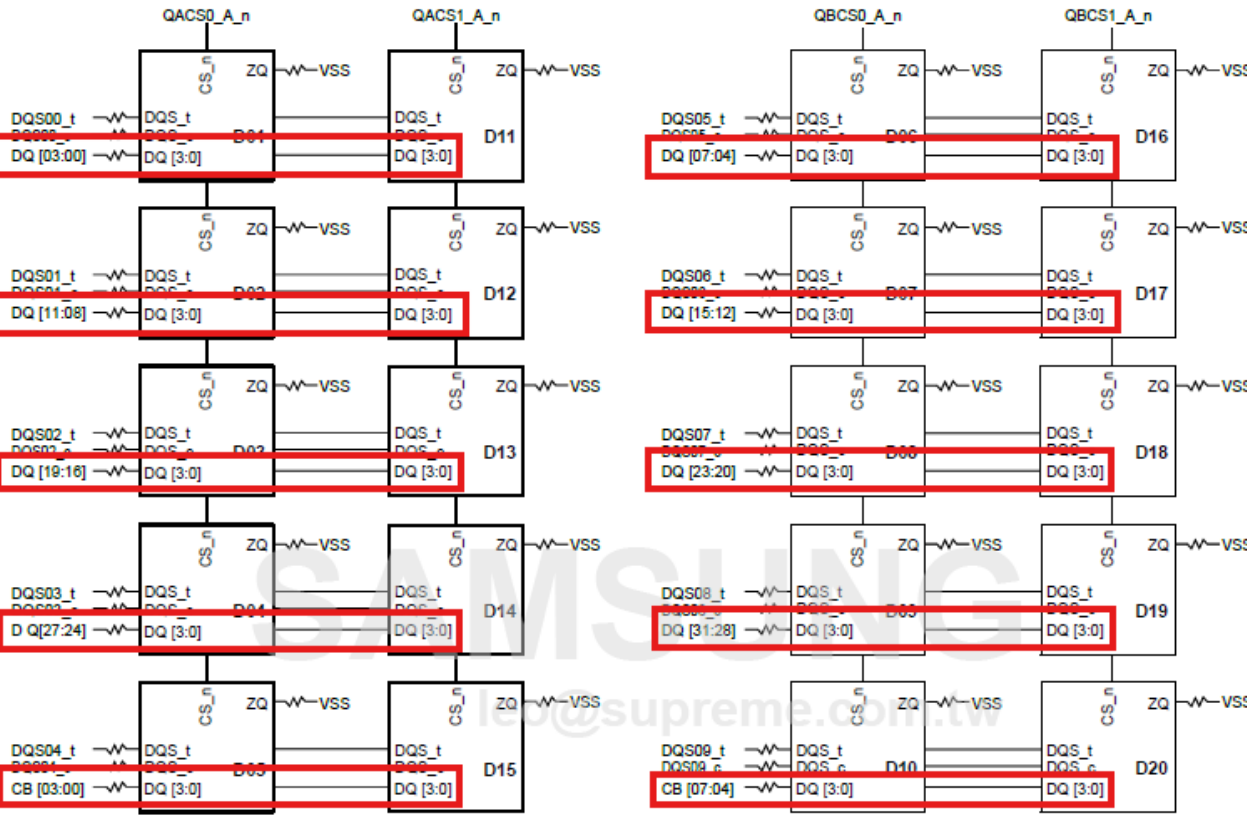
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 34 (Samsung DDR5 RDIMM Datasheet) Channel B  Ex. CC-89 at 35 (Samsung DDR5 RDIMM Datasheet)

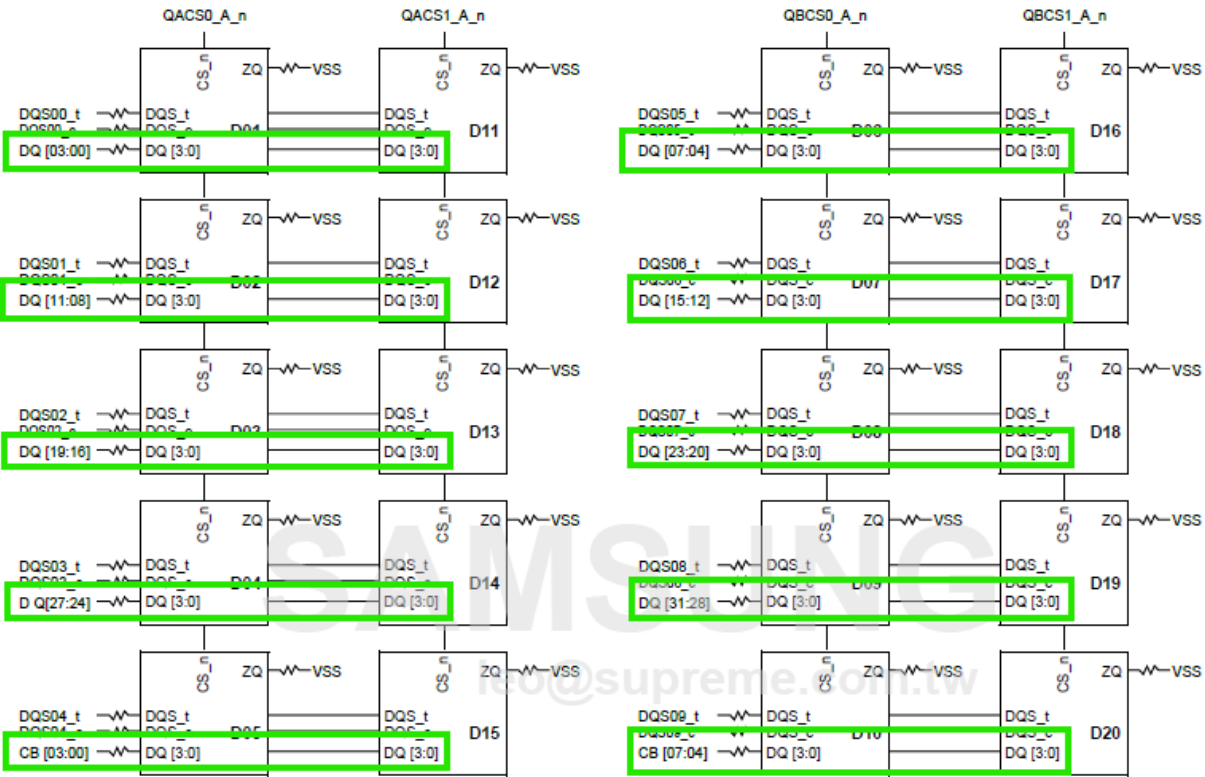
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A ——— QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] DCS1_A_n ——— QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B ——— QACA [13:0]_B -> CA [13:0]: SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0]: SDRAMs D [30:26, 40:36] DCS0_B_n ——— QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] DCS1_B_n ——— QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] DCK_t/DCK_c ——— QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCK_A_t->CK_t:SDRAMsD[15:11]/QCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCK_B_t->CK_t:SDRAMsD[35:31]/QCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD ——— DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS ——— DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n ——— QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n ——— DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21] REGISTER Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet)

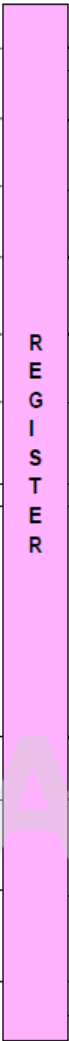
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  The diagram illustrates the internal structure of Channel A, showing two columns of memory banks. Each bank (D11-D15 and D16-D20) contains a CS_n input, a ZQ input, and a VSS input. The data paths are labeled with DQS_t, DQS_c, and DQ. Red boxes highlight specific data paths and control signals, including DQ [03:00], DQ [07:04], DQ [11:08], DQ [15:12], DQ [19:16], DQ [23:20], DQ [27:24], DQ [31:28], DQ [3:0], and CB [03:00]. Ex. CC-89 at 34 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 35 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[6:0]_A ——— QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n: SDRAMs D[10:6] DCS1_A_n ——— QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B ——— QACA[13:0]_B->CA[13:0]:SDRAMsD[25:21, 35:31] QBCA[13:0]_B->CA[13:0]:SDRAMsD[30:26, 40:36] DCS0_B_n ——— QACS0_B->CS_n: SDRAMs D[25:21] QBCS0_B->CS_n: SDRAMs D[30:26] DCS1_B_n ——— QACS1_B->CS_n: SDRAMs D[35:31] QBCS1_B->CS_n: SDRAMs D[40:36] DCK_t/DCK_c ——— QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD ——— DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS ——— DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n ——— QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n ——— DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21] Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A Ex. CC-89 at 34 (Samsung DDR5 RDIMM Datasheet)

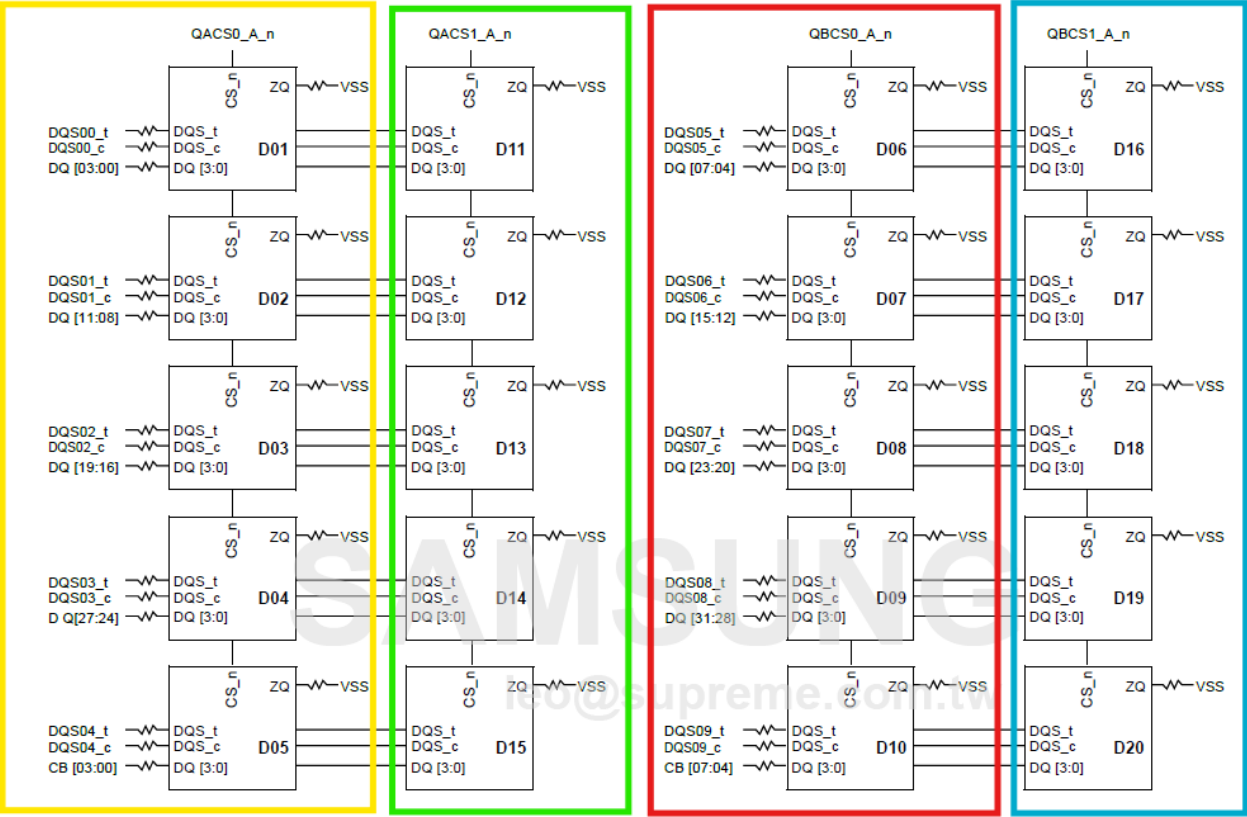
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A — QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n — QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] DCS1_A_n — QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B — QACA [13:0]_B->CA [13:0]:SDRAMsD [25:21, 35:31] QBCA [13:0]_B->CA [13:0]:SDRAMsD [30:26, 40:36] DCS0_B_n — QACS0_B->CS_n:SDRAMsD [25:21] QBCS0_B->CS_n:SDRAMsD [30:26] DCS1_B_n — QACS1_B->CS_n:SDRAMsD [35:31] QBCS1_B->CS_n:SDRAMsD [40:36] DCK_t/DCK_c — QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCK_A_t->CK_t:SDRAMsD [15:11] /QCK_A_c->CK_c:SDRAMsD [15:11] QDCK_A_t->CK_t:SDRAMsD [20:16] /QDCK_A_c->CK_c:SDRAMsD [20:16] QACK_B_t->CK_t:SDRAMsD [25:21] /QACK_B_c->CK_c:SDRAMsD [25:21] QBCK_B_t->CK_t:SDRAMsD [30:26] /QBCK_B_c->CK_c:SDRAMsD [30:26] QCK_B_t->CK_t:SDRAMsD [35:31] /QCK_B_c->CK_c:SDRAMsD [35:31] QDCK_B_t->CK_t:SDRAMsD [40:36] /QDCK_B_c->CK_c:SDRAMsD [40:36] QLBD — DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD [40:21] QLBS — DLBS_A->LBDQS:SDRAMsD [20:1] DLBS_B->LBDQS:SDRAMsD [40:21] DRST_n — QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD [40:21] ALERT_n — ERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] ERROR_IN_B_n->ALERT_n:SDRAMsD [40:21]
	Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet)

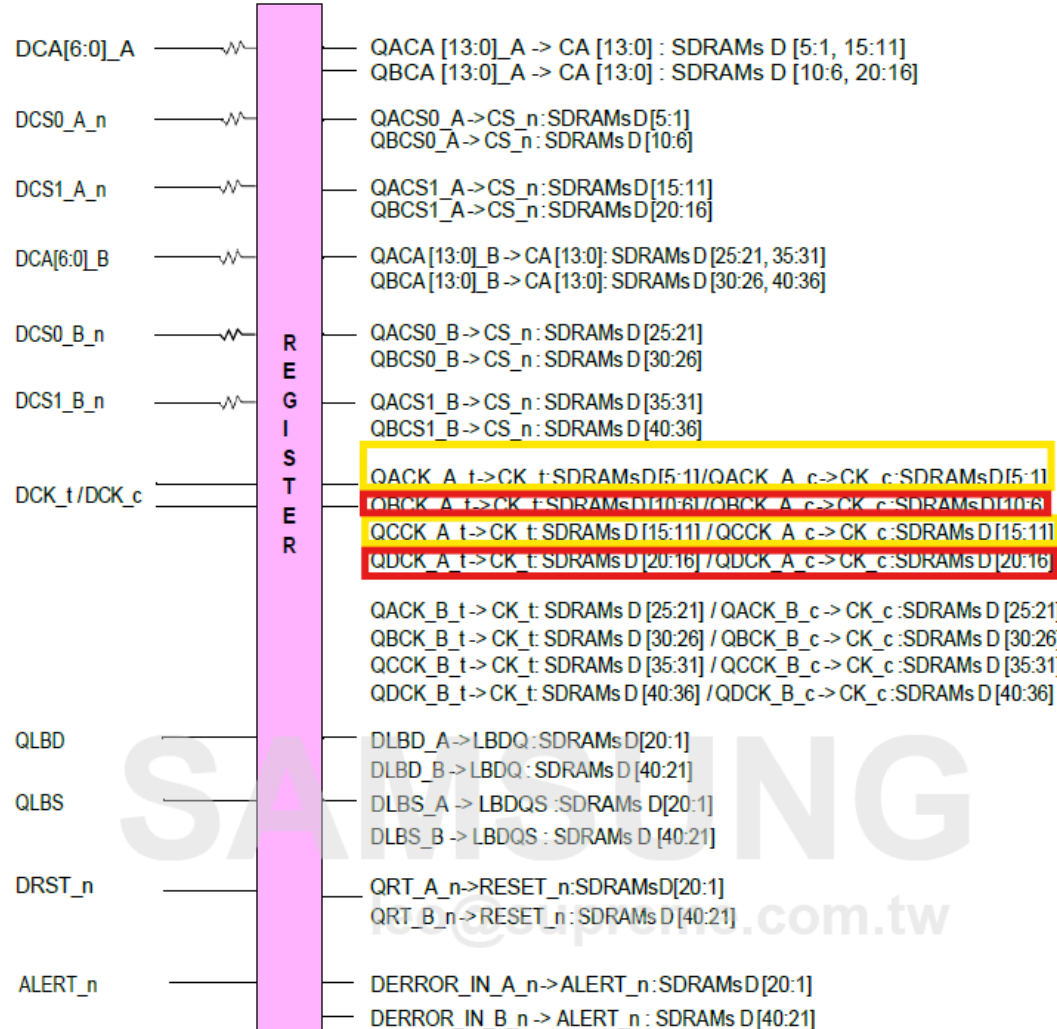
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A Ex. CC-89 at 34 (Samsung DDR5 RDIMM Datasheet)

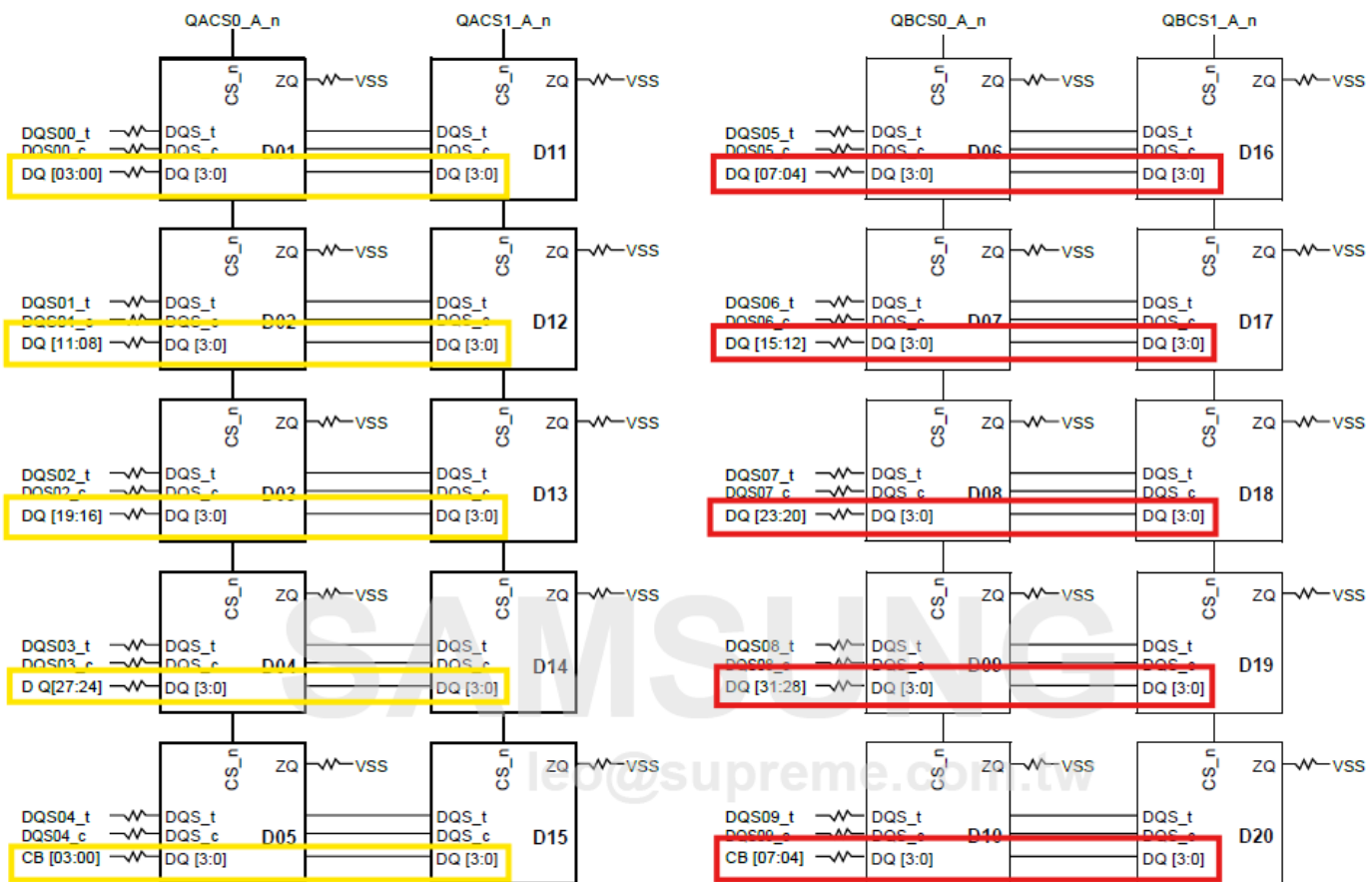
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 35 (Samsung DDR5 RDIMM Datasheet)

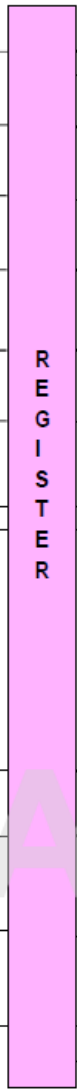
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[6:0]_A ——— QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] DCS1_A_n ——— QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B ——— QACA [13:0]_B->CA [13:0]:SDRAMsD[25:21, 35:31] QBCA [13:0]_B->CA [13:0]:SDRAMsD[30:26, 40:36] DCS0_B_n ——— QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] DCS1_B_n ——— QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] DCK_t/DCK_c ——— QACK_A_t->CK_t:SDRAMsD[5:11]/QACK_A_c->CK_c:SDRAMsD[15:11] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD ——— DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS ——— DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n ——— QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n ——— DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21] REGISTER Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A  The diagram illustrates the internal structure of memory banks in Channel A. It is organized into two columns of banks. The left column contains banks D00 through D05, and the right column contains banks D16 through D20. Each bank is represented by a block diagram showing its internal components and connections. The components include QACS0_A_n and QACS1_A_n for the left column, and QBCS0_A_n and QBCS1_A_n for the right column. Each bank also shows connections to VSS and ZQ. The data and control signals for each bank are listed on the left and right sides of the bank block. The signals for banks D00-D05 are highlighted in yellow, and the signals for banks D16-D20 are highlighted in red. The signals include DQS_t, DQS_c, DQ [3:0], and CB [03:00]. Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet)

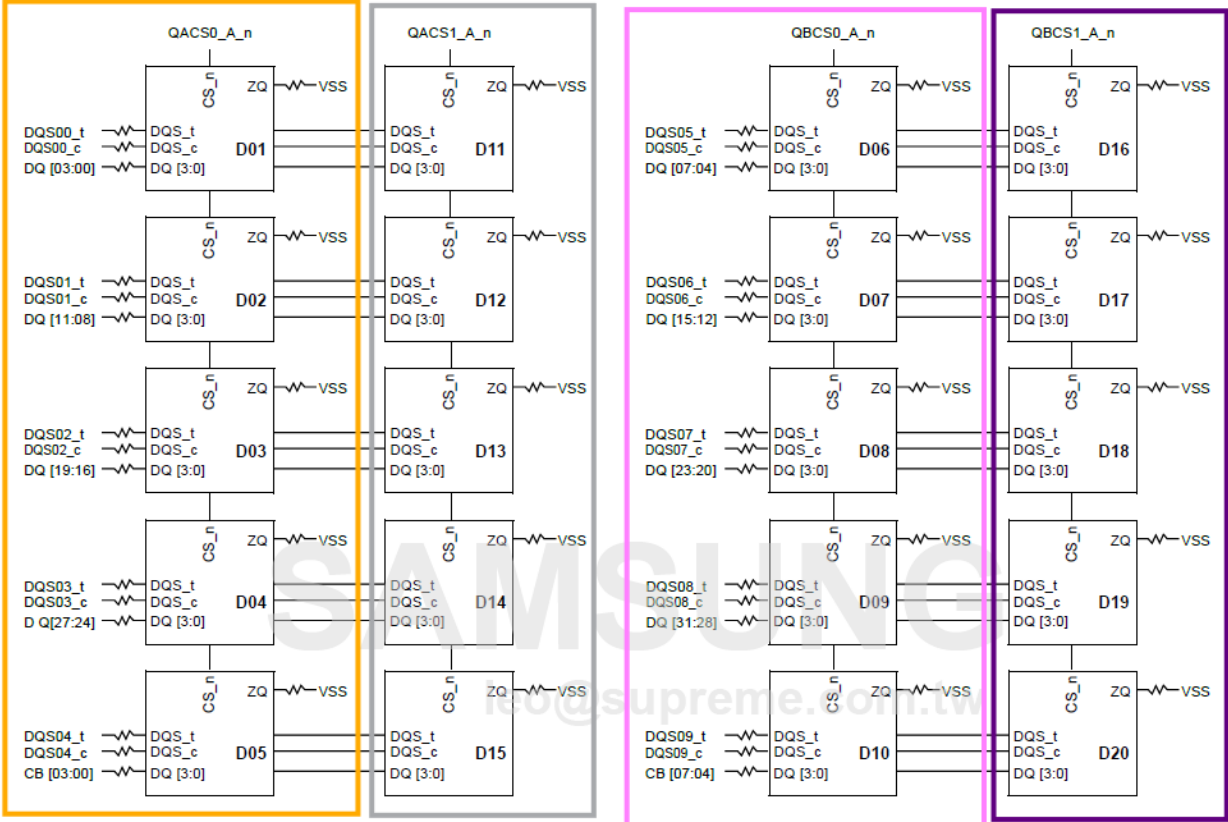
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[6:0]_A ——— QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n: SDRAMs D[10:6] DCS1_A_n ——— QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B ——— QACA[13:0]_B->CA[13:0]:SDRAMsD[25:21, 35:31] QBCA[13:0]_B->CA[13:0]:SDRAMsD[30:26, 40:36] DCS0_B_n ——— QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] DCS1_B_n ——— QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] DCK_t/DCK_c ——— QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD ——— DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS ——— DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n ——— QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n ——— DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 at 31 (Samsung DDR5 RDIMM Datasheet) Channel A Ex. CC-89 at 32 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet) As explained <i>supra</i>, the Samsung Practicing '407 Products use a JEDEC compliant 288-pin edge connector. The JEDEC DDR5 RDIMM standard further demonstrates the Samsung Practicing '407 Products' edge connectors

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																
	allow each respective group of the memory devices to be configured to communicate data with the memory controller via a corresponding subset of the data signal lines. <i>See, e.g.,</i> Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (showing each channel has separate CA Bus and CS commands). <table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (showing each channel has separate data lines).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.
Pin Name	Description	Pin Name	Description														
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B														
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B														
DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.														

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

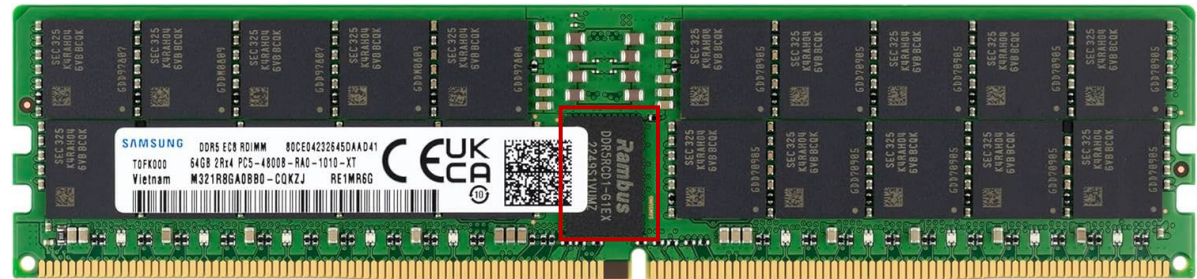
1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

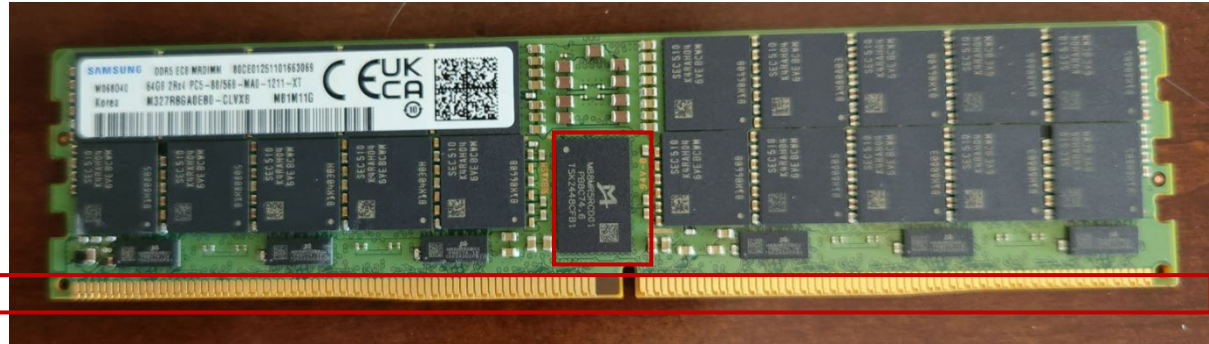
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																																																																																												
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="10">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td>CB[7:0]</td><td></td></tr><tr><th colspan="10">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td>[31:24]</td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td></td><td></td><td>CB[7:0]</td><td></td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive CA, CS, and data signals via different subsets of CA, CS, and data signal lines).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)										Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top			[15:8]				[31:24]				Bottom	[7:0]				[23:16]				CB[7:0]		1Rx8 Sub Channel 1 (Channel B)										Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top	[7:0]				[23:16]		[31:24]				Bottom			[15:8]						CB[7:0]	
1Rx8 (less than 8000 Mbps)																																																																																																																																																													
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Top	Lower	QCCK	QACS0	QACA	0																																																																																																																																																								
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1Rx8 Sub Channel 0 (Channel A)																																																																																																																																																													
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																			
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																			
Top			[15:8]				[31:24]																																																																																																																																																						
Bottom	[7:0]				[23:16]				CB[7:0]																																																																																																																																																				
1Rx8 Sub Channel 1 (Channel B)																																																																																																																																																													
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																			
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																			
Top	[7:0]				[23:16]		[31:24]																																																																																																																																																						
Bottom			[15:8]						CB[7:0]																																																																																																																																																				
[1.c] circuitry mounted on the PCB and configurable to:	The Samsung Practicing '407 Products each comprise circuitry, such as an RCD, mounted on the PCB. Some examples are shown below, with the RCDs annotated with red boxes:  Ex. ITC-1472 CC-116 (Samsung 4800MHz PC5-38400).																																																																																																																																																												

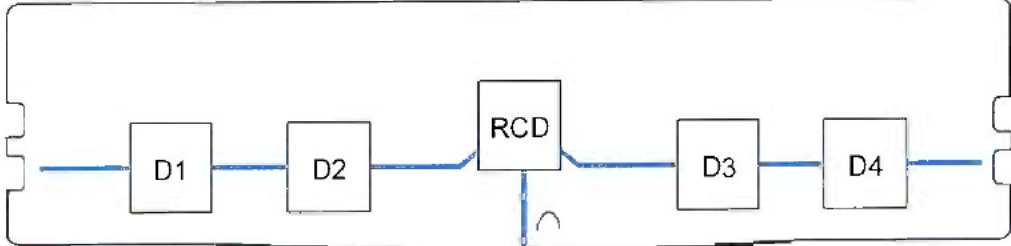
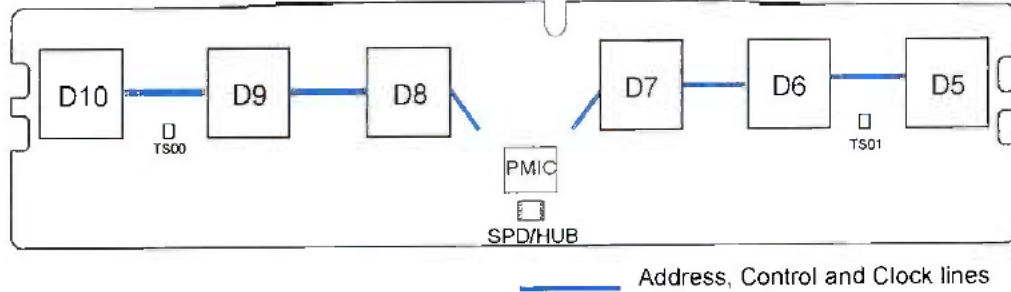
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	A photograph of a Samsung DDR5 EC8 RDIMM memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number W00J120, and other specifications. A red box highlights the RAMBUS logo and the part number DDR5NCD2-618 2388S00K17. Ex. ITC-1472 CC-117 (Samsung 5600MHz PC5-44800). A photograph of a Samsung DDR5 EC8 RDIMM memory module. The module is green with gold-plated pins. It features a white label with the Samsung logo, part number W00J120, and other specifications. A red box highlights the RAMBUS logo and the part number DDR5NCD2-618 2388S00K17. Ex. CC-118 (Samsung 64GB DDR5 PC5-4800 EC8 RDIMM 2Rx4 M321R8GA0BB0-CQK).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-119 (Samsung 64GB DDR5 PC5-6400 EC8 RDIMM 2Rx4 M321R8GA0PB2-CCP).  An image of a Samsung MRDIMM M327R8GA0EB0-CLVXB. <i>See</i> Ex. CC-111. This is confirmed by a representative datasheet for a Samsung RDIMM:

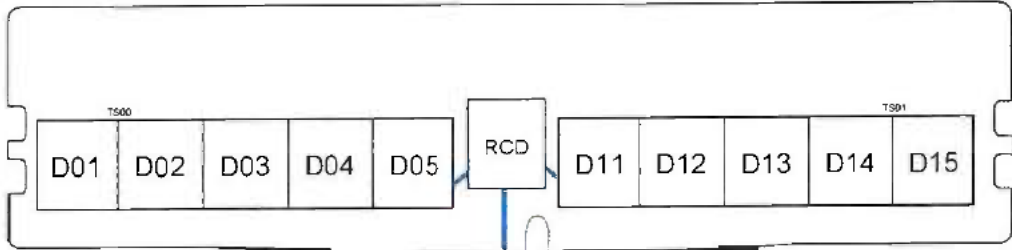
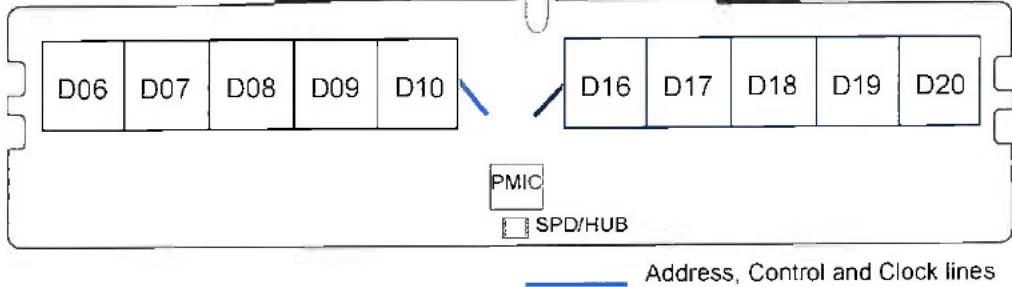
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 at 21 (Samsung DDR5 RDIMM Datasheet)

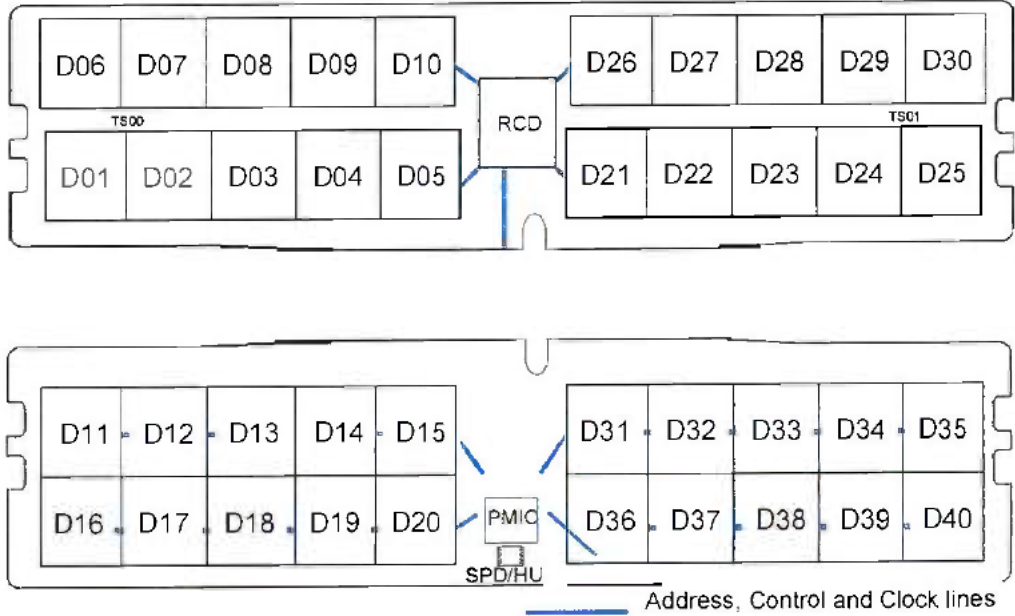
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front Back Address, Control and Clock lines Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  The diagram shows the front of a DDR5 RDIMM. It features two data buffers: the first buffer contains data elements D01, D02, D03, D04, and D05, with a label 'T500' above it; the second buffer contains data elements D11, D12, D13, D14, and D15, with a label 'T501' above it. A central control element labeled 'RCD' is connected to both buffers. A blue line representing an address, control, and clock line runs vertically through the center of the RCD. Back  The diagram shows the back of a DDR5 RDIMM. It features two data buffers: the first buffer contains data elements D06, D07, D08, D09, and D10; the second buffer contains data elements D16, D17, D18, D19, and D20. Below the buffers are two control elements: 'PMIC' and 'SPD/HUB'. Blue lines representing address, control, and clock lines connect the buffers to these control elements. A legend at the bottom right indicates that a blue line represents 'Address, Control and Clock lines'. Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet)
[1.c.1] receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;	The Samsung Practicing '407 Products comprise circuitry mounted on the PCB and configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines. For example, Samsung described its RDIMMs as including “a register [RCD] for enhancing the clock, command and control signals.” Ex. CC-95 (Samsung DRAM Modules). This indicates that the register can be configured to receive input clock, control and command signals:

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use	
	RDIMM Registered DIMM Include a register for enhancing the clock, command and control signals Error correction is available by adding 8-bit parity signals Supports x4 / x8 Organization / up to 2 ranks per DIMM and 3DPC configuration Application : Server	
	The clock signals are received from the system as a system clock via the one or more clock signals, and the control and address (C/A) signals received by the RCD at the host-facing input side are received via the C/A signal lines:	

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																		
	[Table 5] PIN Description <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>CA[6:0]_A CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </table> <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>SDRAM data strobes (positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>SDRAM data strobes (negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </table> Ex. CC-89 at 9.	Pin Name	Description	CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus	CS[1:0]_A CS[1:0]_B	SDRAM Chip Select	PAR_A PAR_B	SDRAM Parity input	CK_t	SDRAM Clocks (true/positive)	CK_c	SDRAM Clocks (complement/negative)	ALERT_n	SDRAM alert for CRC error	RESET_n	Set DRAM to known state	PCAMP	Control and Monitor Port	HSCL	I2C/I3C-Basic Host Sideband Bus Clock	HSDA	I2C/I3C-Basic Host Sideband Bus Data	HSA	I2C/I3C-Basic Host Sideband Bus Address	LBDQ	Loopback data output:	Pin Name	Description	DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B	CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B	DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)	DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.	VIN_BULK	DIMM Power Supply from system to PMIC	VIN_MGMT	DIMM Power Supply from system to PMIC	VSS	Power supply return (ground)	RFU	Reserved for future use	LBDQS	Loopback data strobe output
Pin Name	Description																																																		
CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus																																																		
CS[1:0]_A CS[1:0]_B	SDRAM Chip Select																																																		
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HSCL	I2C/I3C-Basic Host Sideband Bus Clock																																																		
HSDA	I2C/I3C-Basic Host Sideband Bus Data																																																		
HSA	I2C/I3C-Basic Host Sideband Bus Address																																																		
LBDQ	Loopback data output:																																																		
Pin Name	Description																																																		
DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B																																																		
CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B																																																		
DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)																																																		
DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)																																																		
TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.																																																		
TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.																																																		
VIN_BULK	DIMM Power Supply from system to PMIC																																																		
VIN_MGMT	DIMM Power Supply from system to PMIC																																																		
VSS	Power supply return (ground)																																																		
RFU	Reserved for future use																																																		
LBDQS	Loopback data strobe output																																																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

[Table 4] PIN Configuration

Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side
1	VIN_BULK	145	VIN_BULK	50	VSS	194	DQ31_A	95	CB0_B	240	VSS
2	RFU	146	VIN_BULK	51	CB0_A	195	VSS	97	VSS	241	CB2_B
3	VIN_MGMT	147	PWR_GOODIFAIL_n	52	VSS	196	CB2_A	98	CB1_B	242	VSS
4	SCL	148	SAA	53	CB1_A	197	VSS	99	VSS	243	CB3_B
5	SDA	149	RFU	54	VSS	198	CB3_A	100	DQ0_B	244	VSS
6	VSS	150	RFU	55	DQ54_A_1	199	VSS	101	VSS	245	DQ2_B
7	DQ0_A	151	VSS	56	DQ54_A_c	200	DQ59_A_c,TDQ59_A_c	102	DQ1_B	246	VSS
8	VSS	152	DQ2_A	57	VSS	201	DQ59_A_1,TDQ59_A_1	103	VSS	247	DQ3_B
9	DQ1_A	153	VSS	58	CB4_A	202	VSS	104	DQ50_B_1	248	VSS
10	VSS	154	DQ3_A	59	VSS	203	CB6_A	105	DQ50_B_c	249	DQ55_B_c,TDQ55_B_c
11	DQ50_A_1	155	VSS	60	CB5_A	204	VSS	106	VSS	250	DQ55_B_1,TDQ55_B_1
12	DQ50_A_c	156	DQ55_A_c,TDQ55_A_c	61	VSS	205	CB7_A	107	DQ4_B	251	VSS
13	VSS	157	DQ55_A_1,TDQ55_A_1	62	ALERT_n	206	VSS	108	VSS	252	DQ6_B
14	DQ4_A	158	VSS	63	VSS	207	RESET_n	109	DQ6_B	253	VSS
15	VSS	159	DQ6_A	64	CS0_A_n	208	VSS	110	VSS	254	DQ7_B
16	DQ5_A	160	VSS	65	VSS	209	CS1_A_n	111	DQ8_B	255	VSS
17	VSS	161	DQ7_A	66	CA0_A	210	VSS	112	VSS	256	DQ10_B
18	DQ8_A	162	VSS	67	VSS	211	CA1_A	113	DQ9_B	257	VSS
19	VSS	163	DQ10_A	68	CA2_A	212	VSS	114	VSS	258	DQ11_B
20	DQ9_A	164	VSS	69	VSS	213	CA3_A	115	DQ51_B_1	259	VSS
21	VSS	165	DQ11_A	70	CA4_A	214	VSS	116	DQ51_B_c	260	DQ56_B_c,TDQ56_B_c
22	DQ51_A_1	166	VSS	71	VSS	215	CA5_A	117	VSS	261	DQ56_B_1,TDQ56_B_1
23	DQ51_A_c	167	DQ56_A_c,TDQ56_A_c	72	CA6_A	216	VSS	118	DQ12_B	262	VSS
24	VSS	168	DQ56_A_1,TDQ56_A_1	73	VSS	217	CK_1	119	VSS	263	DQ14_B
25	DQ12_A	169	VSS	74	PAR_A	218	CK_c	120	DQ13_B	264	VSS
26	VSS	170	DQ14_A	75	VSS	219	VSS	121	VSS	265	DQ15_B
27	DQ13_A	171	VSS	KEY				122	DQ16_B	266	VSS
28	VSS	172	DQ15_A					123	VSS	267	DQ18_B
29	DQ16_A	173	VSS					124	DQ17_B	268	VSS
30	VSS	174	DQ18_A					125	VSS	269	DQ19_B
31	DQ17_A	175	VSS	76	CA0_B	220	RFU	126	DQ52_B_1	270	VSS
32	VSS	176	DQ19_A	77	VSS	221	CA1_B	127	DQ52_B_c	271	DQ57_B_c,TDQ57_B_c
33	DQ52_A_1	177	VSS	78	CA2_B	222	VSS	128	VSS	272	DQ57_B_1,TDQ57_B_1
34	DQ52_A_c	178	DQ57_A_c,TDQ57_A_c	79	VSS	223	CA3_B	129	DQ20_B	273	VSS
35	VSS	179	DQ57_A_1,TDQ57_A_1	80	CA4_B	224	VSS	130	VSS	274	DQ22_B
36	DQ20_A	180	VSS	81	VSS	225	CA5_B	131	DQ21_B	275	VSS
37	VSS	181	DQ22_A	82	CA6_B	226	VSS	132	VSS	276	DQ23_B
38	DQ21_A	182	VSS	83	VSS	227	PAR_B	133	DQ24_B	277	VSS
39	VSS	183	DQ23_A	84	CS0_B_n	228	VSS	134	VSS	278	DQ26_B
40	DQ24_A	184	VSS	85	VSS	229	CS1_B_n	135	DQ25_B	279	VSS
41	VSS	185	DQ26_A	86	LBS, RSP_A_n	230	VSS	136	VSS	280	DQ27_B
42	DQ25_A	186	VSS	87	LBS, RSP_B_n	231	RFU	137	DQ53_B_1	281	VSS
43	VSS	187	DQ27_A	88	VSS	232	RFU	138	DQ53_B_c	282	DQ58_B_c,TDQ58_B_c
44	DQ53_A_1	188	VSS	89	CB4_B	233	VSS	139	VSS	283	DQ58_B_1,TDQ58_B_1
45	DQ53_A_c	189	DQ58_A_c,TDQ58_A_c	90	VSS	234	CB5_B	140	DQ28_B	284	VSS
46	VSS	190	DQ58_A_1,TDQ58_A_1	91	CB5_B	235	VSS	141	VSS	285	DQ30_B
47	DQ28_A	191	VSS	92	VSS	236	CB7_B	142	DQ29_B	286	VSS
48	VSS	192	DQ30_A	93	DQ59_B_1,TDQ59_B_1	237	VSS	143	VSS	287	DQ31_B
49	DQ29_A	193	VSS	94	DQ59_B_c,TDQ59_B_c	238	DQ54_B_c	144	RFU	288	VSS
				95	VSS	239	DQ54_B_1				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 6 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	V33	75	219	V33
VIN_MGMT	3	147	PWR_GOOD/FAIL_n	Key			
HSCL	4	148	HSA	CA0_B	76	220	RFU
HSDA	5	149	RFU	VSS	77	221	CA1_B
VSS	6	150	RFU	CA2_B	78	222	VSS
DQ0_A	7	151	VSS	VSS	79	223	CA3_B
VSS	8	152	DQ2_A	CA4_B	80	224	VSS
DQ1_A	9	153	VSS	VSS	81	225	CA5_B
VSS	10	154	DQ3_A	CA6_B	82	226	VSS
DQS0_A_t	11	155	VSS	VSS	83	227	PAR_B
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	CS0_B_n	84	228	VSS
VSS	13	157	DU55_A_t, DU55_A_t DBI0_A_n	VSS	85	229	CS1_B_n
DQ4_A	14	158	VSS	LBD, RSP_A_n	86	230	VSS
VSS	15	159	DQ6_A	LBS, RSP_B_n	87	231	RFU
DQ5_A	16	160	VSS	VSS	88	232	RFU
VSS	17	161	DQ7_A	CB4_B	89	233	VSS
DQ8_A	18	162	VSS	VSS	90	234	CB6_B
VSS	19	163	DQ10_A	CB5_B	91	235	VSS
DQ9_A	20	164	VSS	VSS	92	236	CB7_B
VSS	21	165	DQ11_A	DQS9_B_t, TDQS9_B_t DBI4_B_n	93	237	VSS
DQS1_A_t	22	166	VSS	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	VSS	95	239	DQS4_B_t
VSS	24	168	DQS6_A_t, TDQS6_A_t DBI1_A_n	CB0_B	96	240	VSS
DQ12_A	25	169	VSS	V33	97	241	CD2_D
VSS	26	170	UQ14_A	CB1_B	98	242	VSS
DQ13_A	27	171	VSS	VSS	99	243	CB3_B
VSS	28	172	DQ15_A	DQ0_B	100	244	VSS
DQ16_A	29	173	VSS	VSS	101	245	DQ2_B
VSS	30	174	DQ18_A	DQ1_B	102	246	VSS
DQ17_A	31	175	VSS	VSS	103	247	DQ3_BDQ3_B
VSS	32	176	DQ19_A	DQS0_B_t	104	248	VSS
DQS2_A_t	33	177	VSS	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	VSS	106	250	DQS5_B_t, TDQS5_B_t DBI0_B_n
VSS	35	179	DQS7_A_t, TDQS7_A_t DBI2_n	DQ4_B	107	251	VSS
DQ20_A	36	180	VSS	VSS	108	252	DQ6_B
VSS	37	181	DQ22_A	DQ5_B	109	253	VSS
DQ21_A	38	182	VSS				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

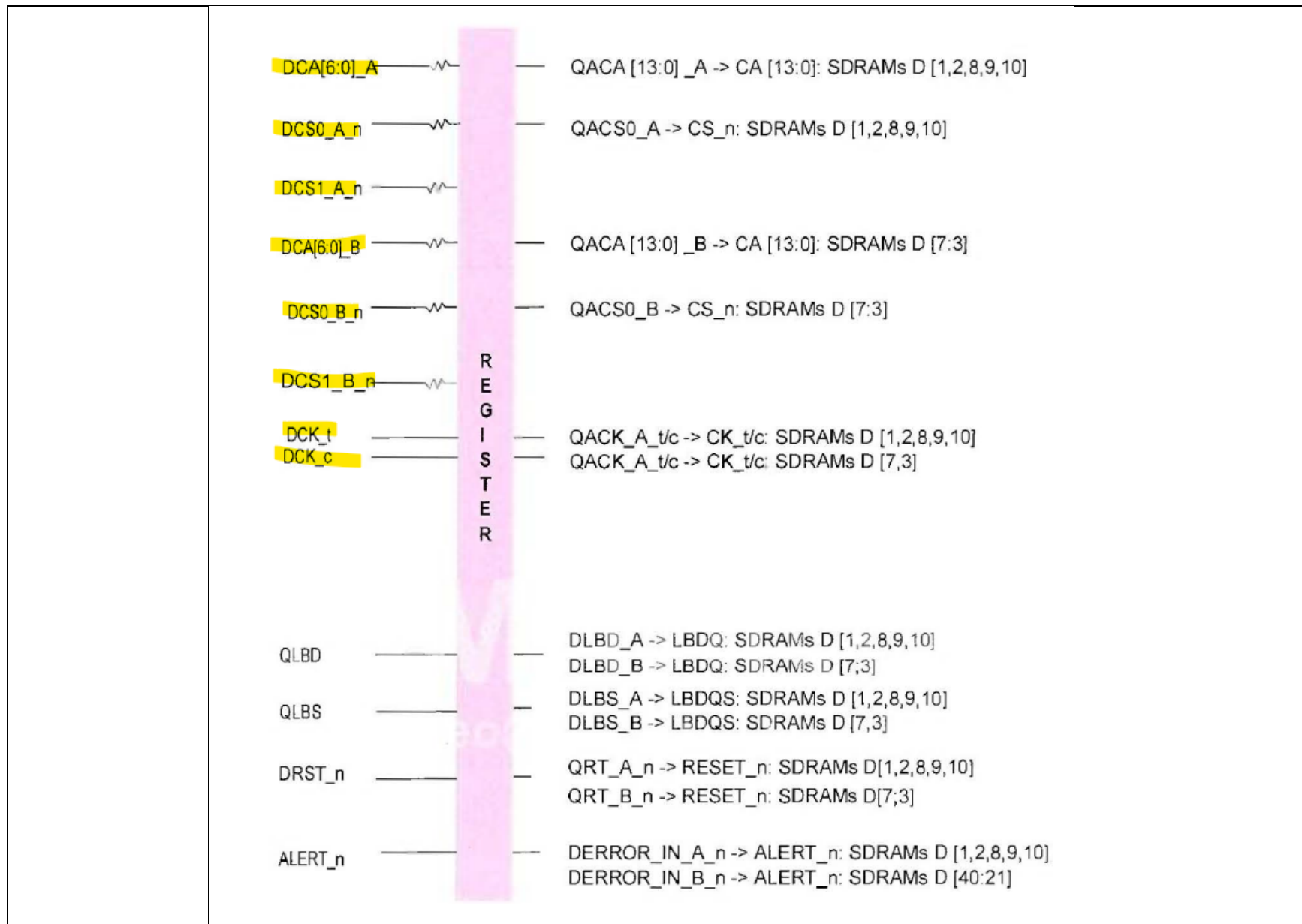
Claim 1	Evidence of Use											
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label				
	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B				
	DQ24_A	40	184	V33	DQ8_B	111	255	VSS				
	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B				
	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS				
	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B				
	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS				
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c				
	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t				
				DBI3_A_n				DBI1_B_n				
	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS				
	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B				
	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS				
	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B				
	CB0_A	51	195	VSS	DQ16_B	122	266	VSS				
	VSS	52	196	CB2_A	VSS	123	267	DQ18_B				
	CB1_A	53	197	VSS	DQ17_B	124	268	VSS				
	VSS	54	198	CB3_A	VSS	125	269	DQ19_B				
	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	V33				
	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c				
	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t				
				DBI4_A_n				DBI2_B_n				
	CB4_A	58	202	VSS	DQ20_B	129	273	VSS				
	VSS	59	203	CB6_A	VSS	130	274	DQ22_B				
	CB5_A	60	204	VSS	DQ21_B	131	275	VSS				
	VSS	61	205	CB7_A	VSS	132	276	DQ23_B				
	AI FFT_n	62	206	VSS	DQ24_B	133	277	VSS				
	VSS	63	207	RESET_n	VSS	134	278	DQ26_B				
	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS				
	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B				
	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS				
	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c				
	CA2	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t				
	VSS	69	213	CA3_A				DBI3_B_n				
	CA4_A	70	214	VSS	DQ28_B	140	284	VSS				
	VSS	71	215	CA5_A	VSS	141	285	DQ30_B				
	CA6_A	72	216	VSS	DQ29_B	142	286	VSS				
	VSS	73	217	CK_t	VSS	143	287	DQ31_B				
					RFU	144	288	VSS				

Figure 1. Wiring Assignments

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7-8 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)



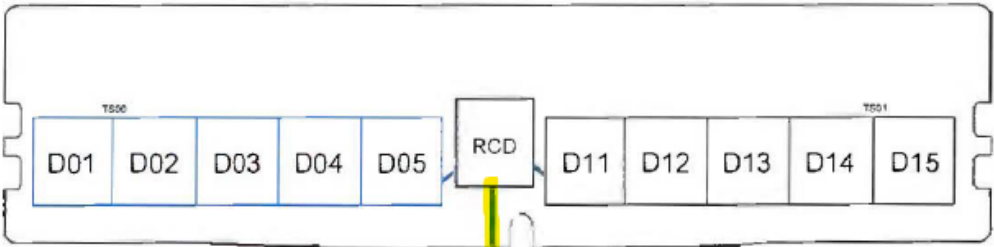
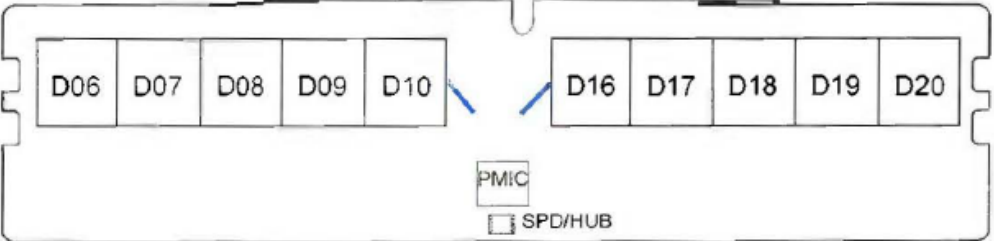
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front Back TS00 TS01 PMIC SPD/HUB Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21.

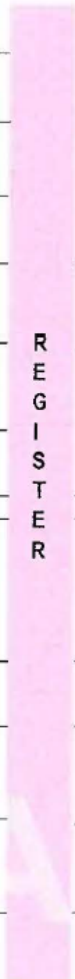
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
DCA[06:0]_A	QACA[13:00]_A -> CA [13:00] : SDRAMs D [10:01]
DCS0_A_n	QACS0_A -> CS_n : SDRAMs D [05:01]
DCS1_A_n	QACS1_A -> CS_n : SDRAMs D [10:06]
DCA[06:00]_B	QACA [13:00]_B -> CA [13:00]: SDRAMs D [20:11]
DCS0_B_n	QACS0_B -> CS_n : SDRAMs D [15:11]
DCS1_B_n	QACS1_B -> CS_n : SDRAMs D [20:16]
DCK_t/DCK_c	QACK_A_t -> CK_t: SDRAMs D [05:01] / QACK_A_c -> CK_c: SDRAMs D [05:01] QCK_A_t -> CK_t: SDRAMs D [10:06] / QCK_A_c -> CK_c: SDRAMs D [10:06] QACK_B_t -> CK_t: SDRAMs D [15:11] / QACK_B_c -> CK_c: SDRAMs D [15:11] QCK_B_t -> CK_t: SDRAMs D [20:16] / QCK_B_c -> CK_c: SDRAMs D [20:16]
QLBD	DLBD_A -> LBDQ: SDRAMs D [10:01] DLBD_B -> LBDQ: SDRAMs D [20:11]
QLBS	DLBS_A -> LBDQS: SDRAMs D [10:01] DLBS_B -> LBDQS: SDRAMs D [20:11]
DRST_n	QRT_A_n -> RESET_n: SDRAMs D [10:01] QRT_B_n -> RESET_n: SDRAMs D [20:11]
ALERT_n	DERROR_IN_A_n -> ALERT_n: SDRAMs D [10:01] DERROR_IN_B_n -> ALERT_n: SDRAMs D [20:11]

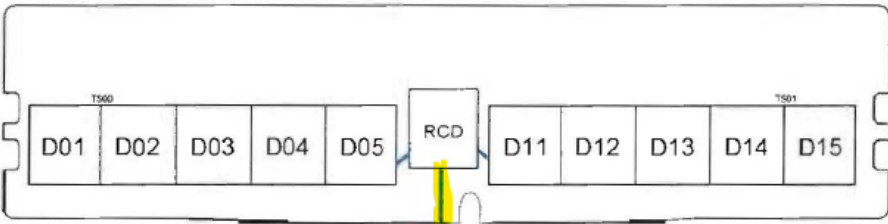
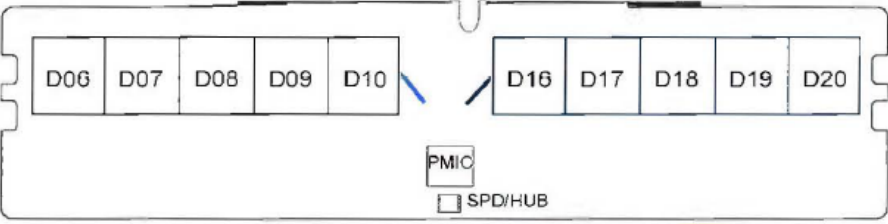
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use	
		DCA[06:00]_A — QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] DCS0_A_n — QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n — DCA[06:00]_B — QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n — QACS0_B -> CS_n: SDRAMsD[15:11] QACS0_B -> CS_n: SDRAMsD[20:16] DCS1_B_n — DCK_t/DCK_c — QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMsD[05:01] QACK_A_t->CK_t: SDRAMsD[10:06] / QACK_A_c->CK_c: SDRAMsD[10:06] QACK_B_t->CK_t: SDRAMsD[15:11] / QACK_B_c->CK_c: SDRAMsD[15:11] QACK_B_t->CK_t: SDRAMsD[20:16] / QACK_B_c->CK_c: SDRAMsD[20:16] QLBD — DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS — DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] DRST_n — QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n->RESET_n: SDRAMsD[20:11] ALERT_n — DERROR_IN_A_n->ALERT_n: SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n: SDRAMsD[20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A — QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n — QACS0_A -> CS_n : SDRAMs D [5:1] QBCS0_A -> CS_n : SDRAMs D [10:6] DCS1_A_n — QACS1_A -> CS_n : SDRAMs D [15:11] QBCS1_A -> CS_n : SDRAMs D [20:16] DCA[6:0]_B — QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D [25:21] QBCS0_B -> CS_n : SDRAMs D [30:26] DCS1_B_n — QACS1_B -> CS_n : SDRAMs D [35:31] QBCS1_B -> CS_n : SDRAMs D [40:36] DCK_t/DCK_c — QACK_A_t -> CK_t : SDRAMs D [5:1] / QACK_A_c -> CK_c : SDRAMs D [5:1] QBCK_A_t -> CK_t : SDRAMs D [10:6] / QBCK_A_c -> CK_c : SDRAMs D [10:6] QCCK_A_t -> CK_t : SDRAMs D [15:11] / QCCK_A_c -> CK_c : SDRAMs D [15:11] QDCK_A_t -> CK_t : SDRAMs D [20:16] / QDCK_A_c -> CK_c : SDRAMs D [20:16] QACK_B_t -> CK_t : SDRAMs D [25:21] / QACK_B_c -> CK_c : SDRAMs D [25:21] QBCK_B_t -> CK_t : SDRAMs D [30:26] / QBCK_B_c -> CK_c : SDRAMs D [30:26] QCCK_B_t -> CK_t : SDRAMs D [35:31] / QCCK_B_c -> CK_c : SDRAMs D [35:31] QDCK_B_t -> CK_t : SDRAMs D [40:36] / QDCK_B_c -> CK_c : SDRAMs D [40:36] QLBD — DLBD_A -> LBDQ : SDRAMs D [20:1] DLBD_B -> LBDQ : SDRAMs D [40:21] QLBS — DLBS_A -> LBDQS : SDRAMs D [20:1] DLBS_B -> LBDQS : SDRAMs D [40:21] DRST_n — QRT_A_n -> RESET_n : SDRAMs D [20:1] QRT_B_n -> RESET_n : SDRAMs D [40:21] ALERT_n — DERROR_IN_A_n -> ALERT_n : SDRAMs D [20:1] DERROR_IN_B_n -> ALERT_n : SDRAMs D [40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 33. As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The DCA, DCS and DCK signals are therefore received via the corresponding C/A, C/S and CK signals lines in the system bus because the DCA, DCS and DCK pins of an RDIMM each mate with a corresponding connector in the memory slot to form electrical connection to the corresponding slot pin and signal traces, as illustrated below:

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Figure 6: DDR5 288-pin connector. Figure 7: Device under test (DUT) Labels in Figure 6: DIMM-PCB, Memory-Slot-Connector-with-its-pins-soldered-to-the-PCB-of-the-System-Board, Conduit-is-a-Pin-of-the-Memory-Slot-Connector-along-with-pad-and-traces-on-the-system-board, System-Board-(PCB). Labels in Figure 7: 0.635 mm, Gold finger (on module), Edge-Connection-of-the-DIMM (Gold-finger), Pin, SMT Pad (on baseboard), 0.635 mm. Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin in the memory slot and the corresponding signal trace/line on the system board). The JEDEC DDR5 RDIMM standard further demonstrates the RCD circuitry is configurable to receive from the memory controller a system clock via the one or more clock signal lines (e.g., “CK_t” and “CK_c” / “DCK_t” and “DCK_c”) and input control and address (C/A) signals via the C/A signal lines (e.g., “CA[]_A” and “CA[]_B” / “DCA[]_A” and “DCA[]_B” / “CS[]_A_n” and “CS[]_B_n” / “DCS[]_A_n” and “DCS[]_B_n”). <i>See, e.g.,</i>

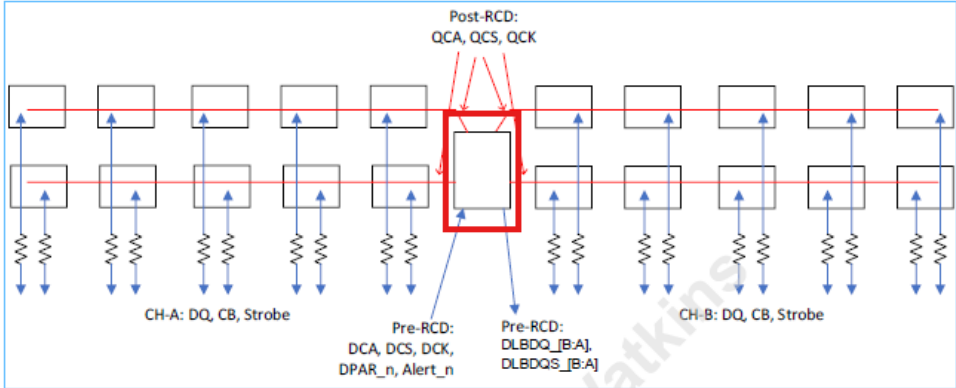
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																				
	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (illustrating circuitry inputs for receiving clock signal lines (<i>e.g.</i>, “CK_t” and “CK_c”) and control/address (C/A) signal lines (<i>e.g.</i>, “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”) from the memory controller). Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c,</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating circuitry inputs for receiving clock signal lines (<i>e.g.</i>, “CK_t” and “CK_c”) and control/address (C/A) signal lines (<i>e.g.</i>, “CA[]_A” and “CA[]_B”) from the memory controller).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	Symbol	Type	I/O Levels	Function	CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.
Pin Name	Description	Pin Name	Description																																		
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Claim 1	Evidence of Use
	6.1 Signal Groups This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are: Pre-RCD - Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A] - Pre-RCD Clock group includes: DCK_t and DCK_c - Pre-RCD Other group includes: ALERT_n, DRST_n - Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A] Post-RCD to SDRAM - Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A] - Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n - Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c - Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n - Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A] DQ and DQS - edge connector pin to SDRAM with series resistor SideBand - Host to Hub group: HSDA, HSCL, HSA - Hub local bus group: LSDA, LSCL (star topology from Hub) Ex. CC-88 (JEDEC 305B) at 12 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller).

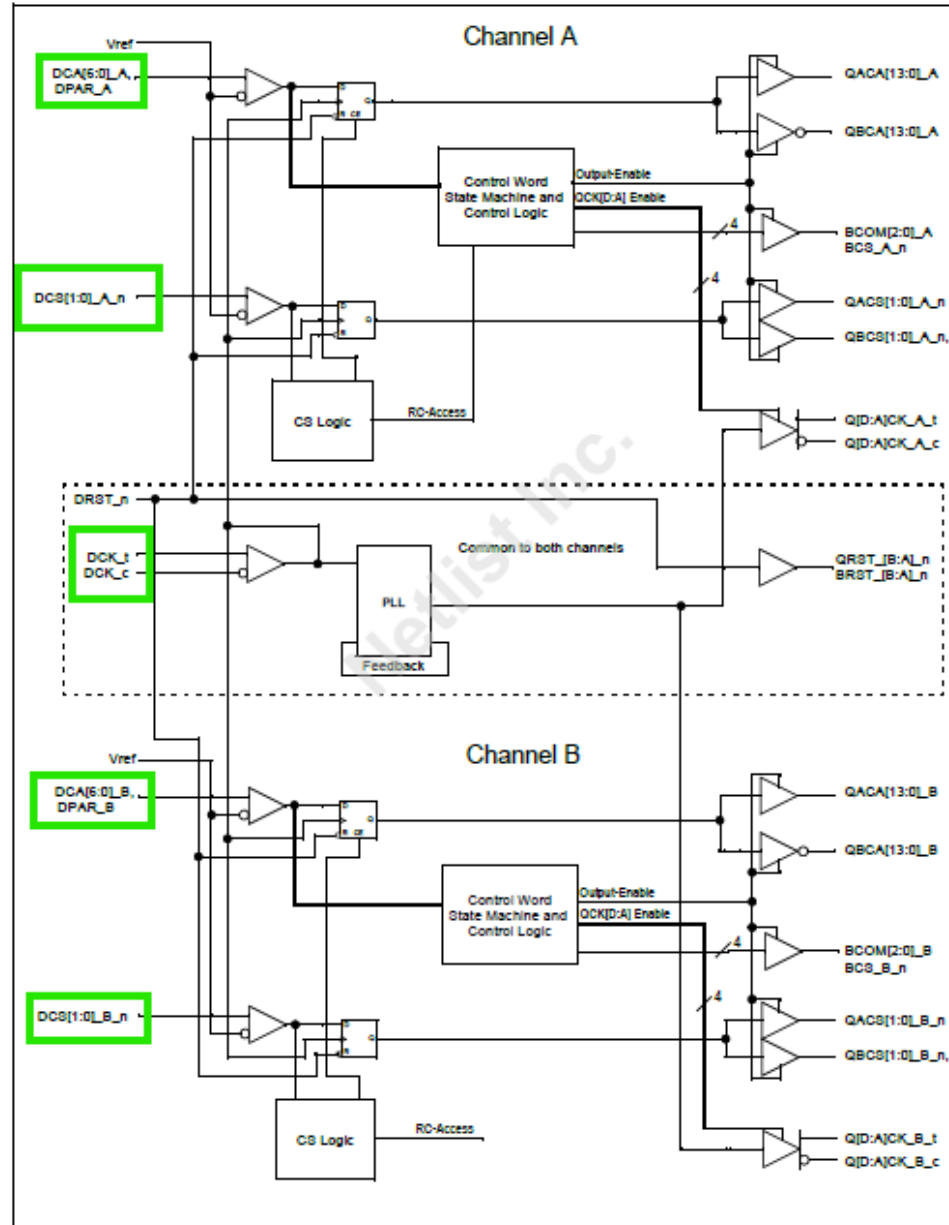
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” control and address signals (“DCA” and “DCS”) and clock signals (“DCK”) from the memory controller). Furthermore, according to the JEDEC RCD specification, the RCD receives on its input side DCA, DCS and DCK signals.

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Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15 (illustrating circuitry inputs for receiving control/address (C/A) signal lines (e.g., “DCA[]_A” and “DCA[]_B”) from the memory controller).	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

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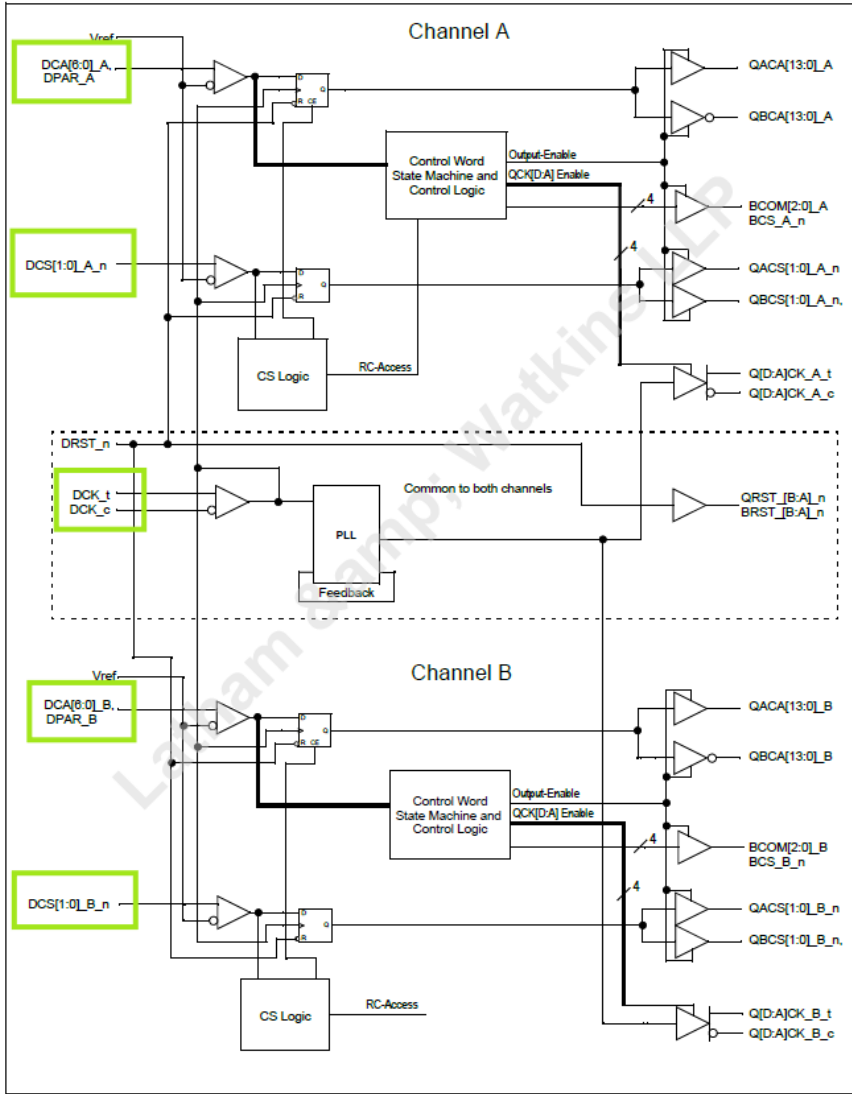
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	Ex. CC-32 (JEDEC 82-512) at 7 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller). <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller). 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. Ex. CC-32 (JEDEC 82-512) at 6.	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Signal Group	Signal Name	Type	Description																		
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Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-30 (JEDEC 82-513) at 15.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

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Claim 1	Evidence of Use
	 The diagram illustrates a dual-channel ADC system, specifically Channel A and Channel B. Each channel has an input stage with a differential pair of transistors (n and p) and a current source (CS). The input signals are DCA[6:0] and DCS[1:0]. The output of the input stage is connected to a control word state machine and control logic, which also receives a feedback signal from the output. The control logic outputs enable signals to the output registers. The output registers are QACA[13:0], QBCA[13:0], BCOM[2:0], QACS[1:0], QBCS[1:0], and Q[DA]CK. The diagram also shows a common PLL block for both channels, which receives a reference signal (Vref) and a clock signal (DCK_t, DCK_c) and outputs a common clock signal (QRST, BRST). The diagram is labeled 'Ex. CC-30 (JEDEC 82-513) at 7.'

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Claim 1	Evidence of Use																				
	2.2 Terminal Functions Table 2 — Terminal Functions <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr></table> Ex. CC-30 (JEDEC 82-513) at 4. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. Ex. CC-30 (JEDEC 82-513) at 6.	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Signal Group	Signal Name	Type	Description																		
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Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.																		
[1.c.2] generate module C/A signals in response to the input C/A signals;	The Samsung Practicing '407 Products each comprise circuitry mounted on the PCB and configurable to generate module C/A signals in response to the input C/A signals, as illustrated below:																				

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Claim 1	Evidence of Use
	DCA[6:0]_A DCS0_A_n DCS1_A_n DCA[6:0]_B DCS0_B_n DCS1_B_n DCK_t DCK_c QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10] QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10] QACS1_A -> CS_n: SDRAMs D [1,2,8,9,10] QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3] QACS0_B -> CS_n: SDRAMs D [7:3] QACS1_B -> CS_n: SDRAMs D [7:3] QACK_A_t/c -> CK_t/c: SDRAMs D [1,2,8,9,10] QACK_A_c -> CK_c: SDRAMs D [7,3]
	QLBD QLBS DRST_n ALERT_n DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7:3] DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7:3] QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3] DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

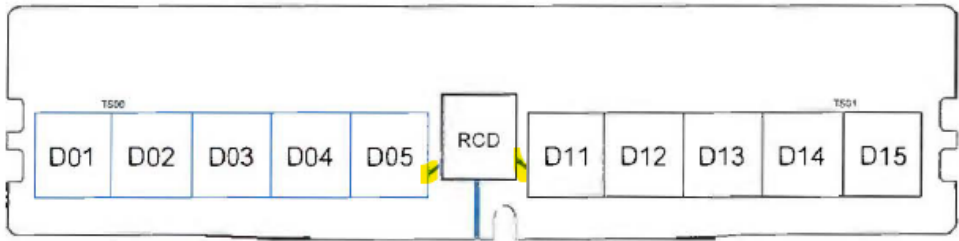
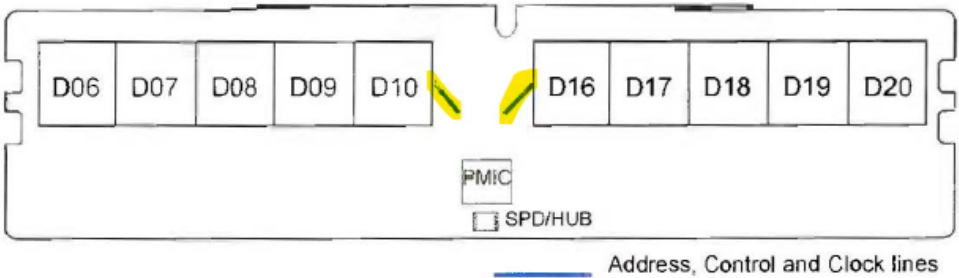
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Claim 1	Evidence of Use
	Front Back Address, Control and Clock lines Ex. CC-89 at 21 (Samsung DDR5 RDIMM Datasheet)

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Claim 1	Evidence of Use
	DCA[06:0]_A — QACA[13:00]_A -> CA[13:00] : SDRAMs D [10:01] DCS0_A_n — QACS0_A -> CS_n : SDRAMs D [05:01] DCS1_A_n — QACS1_A -> CS_n : SDRAMs D [10:06] DCA[06:00]_B — QACA[13:00]_B -> CA[13:00] : SDRAMs D [20:11] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D [15:11] DCS1_B_n — QACS1_B -> CS_n : SDRAMs D [20:16] DCK_t/DCK_c — QACK_A_t->CK_t: SDRAMs D [05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] — QCCK_A_t->CK_t: SDRAMs D [10:06] / QCCK_A_c->CK_c: SDRAMs D [10:06] — QACK_B_t->CK_t: SDRAMs D [15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] — QCCK_B_t->CK_t: SDRAMs D [20:16] / QCCK_B_c->CK_c: SDRAMs D [20:16] QLBD — DLBD_A->LBDQ: SDRAMs D [10:01] — DLBD_B->LBDQ: SDRAMs D [20:11] QLBS — DLBS_A->LBDQS: SDRAMs D [10:01] — DLBS_B->LBDQS: SDRAMs D [20:11] DRST_n — QRT_A_n->RESET_n: SDRAMs D [10:01] — QRT_B_n->RESET_n: SDRAMs D [20:11] ALERT_n — DERROR_IN_A_n -> ALERT_n : SDRAMs D [10:01] — DERROR_IN_B_n -> ALERT_n : SDRAMs D [20:11]

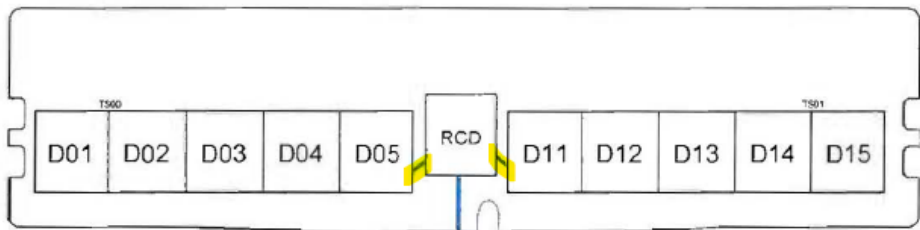
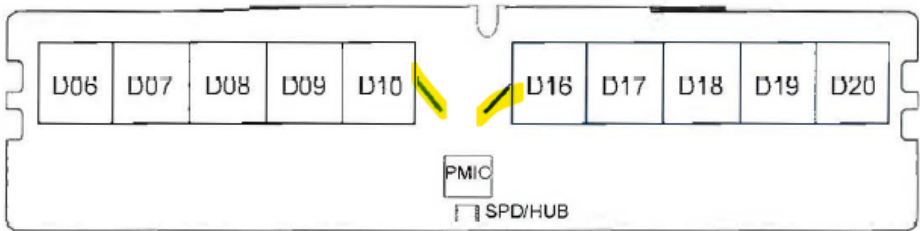
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Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[06:00]_A ——— QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n ——— QACS0_A->CS_n: SDRAMsD[05:01] DCS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n ——— DCA [06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n : SDRAMs D[15:11] DCS0_B -> CS_n : SDRAMs D[20:16] DCS1_B_n ——— DCK_t / DCK_c ——— QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] QCCCK_A_t->CK_t: SDRAMsD[10:06] / QCCCK_A_c->CK_c: SDRAMs D [10:06] QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QCCCK_B_t->CK_t: SDRAMsD[20:16] / QCCCK_B_c->CK_c: SDRAMs D [20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD [10:01] DLBD_B->LBDQ: SDRAMsD [20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD [10:01] DLBS_B->LBDQS: SDRAMsD [20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD [10:01] QRT_B_n-> RESET_n : SDRAMs D [20:11] ALERT_n ——— DERROR_IN_A_n-> ALERT_n : SDRAMs D [10:01] DERROR_IN_B_n-> ALERT_n : SDRAMs D [20:11]

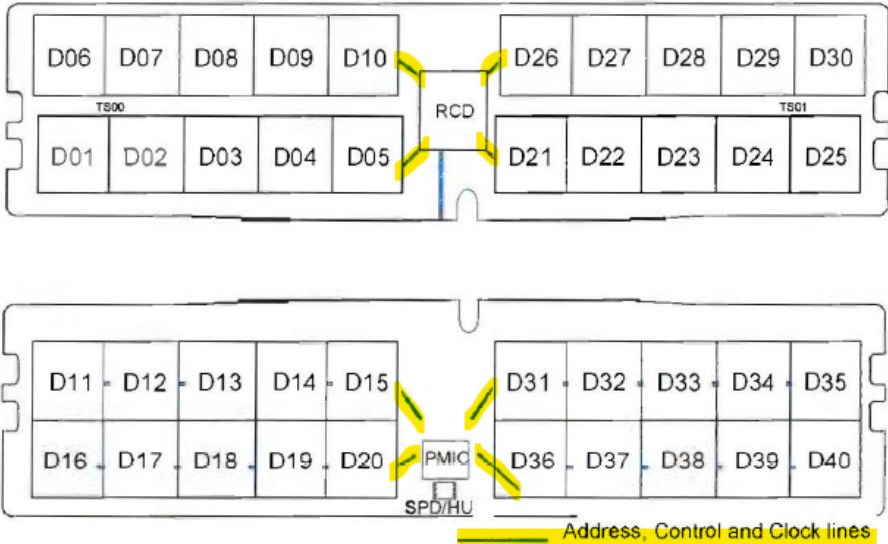
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A ——— QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n ——— QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] DCS1_A_n ——— QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B ——— QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] DCS0_B_n ——— QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] DCS1_B_n ——— QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] DCK_t/DCK_c ——— QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD ——— DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS ——— DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n ——— QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n ——— DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet) As explained supra, the Samsung Practicing '407 Products are JEDEC compliant. According to the JEDEC RCD specifications, RCD outputs module C/A signals (such as QACA, QBCA, QACS and QBCS signals) in response to the input C/A signals (DCA and DCS) received from the host. 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

Ex. CC-32 (JEDEC 82-512) at 6.

2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input Control bus	DCS[1:0]_[B:A]_n	POD V_{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Input Address and Command bus	DCA[6:0]_[B:A]	POD V_{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Parity input	DPAR_[B:A]	POD V_{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V_{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs	Q[D:A]CK_[B:A]_t	POD differential	Clock outputs to the DRAMs. Four copies per channel.

Ex. CC-32 (JEDEC 82-512) at 4.

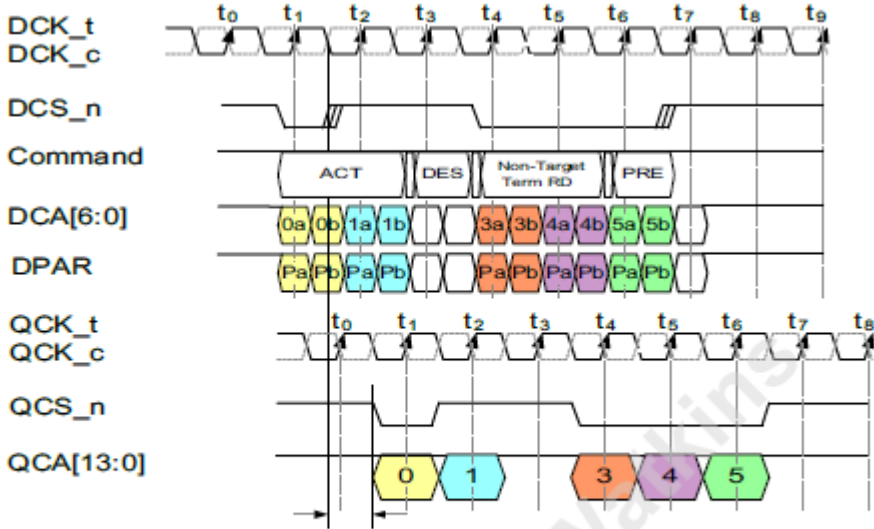
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-32 (JEDEC 82-512) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15. 3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled RW01[1] = 0. • When CA Pass-through mode is enabled RW00[2] = 1.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-32 (JEDEC 82-512) at 9.  NOTE Timing Diagram for 0 nCK command latency adder. Figure 9 — DDR CA Multi-Command Timing Diagram Example Ex. CC-32 (JEDEC 82-512) at 17 (timing diagram showing the generation of module C/A signals (QCA[13:0]) in response to the input C/A signals (DCA[6:0]).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Control bus			
Input	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Address and Command bus			
Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V _{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Control bus			
Output	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Address and Command bus			

Ex. CC-30 (JEDEC 82-513) at 4.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-30 (JEDEC 82-513) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-30 (JEDEC 82-513) at 15.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled <code>RW01[1] = 0</code>. • When CA Pass-through mode is enabled <code>RW00[2] = 1</code>. Ex. CC-30 (JEDEC 82-513) at 9.

Claim 1	Evidence of Use
	3.4.1 Double Data Rate DCA (cont'd) The diagram illustrates the timing for a Double Data Rate (DDR) Command Address (CA) Multi-Command sequence. It shows the relationship between clock signals (DCK_t, DCK_c, QCK_t, QCK_c), command strobes (DCS_n, QCS_n), and data bursts (DCA[6:0], DPAR, QCA[13:0]) across time steps t₀ to t₉. Command: The sequence includes ACT (Active), DES (Decoding), Non-Target Term RD (Non-Target Terminate Read), and PRE (Precharge). DCA[6:0]: Data bursts are shown as 0a, 0b, 1a, 1b, 3a, 3b, 4a, 4b, 5a, 5b. DPAR: Data bursts are shown as Pa, Pb, Pa, Pb, Pa, Pb, Pa, Pb, Pa, Pb. QCA[13:0]: Data bursts are shown as 0, 1, 3, 4, 5. NOTE Timing Diagram for 0 nCK command latency adder. Figure 9 — DDR CA Multi-Command Timing Diagram Example Ex. CC-30 (JEDEC 82-513) at 17.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The JEDEC DDR5 RDIMM standard further demonstrates that the Samsung Practicing '407 Products generate module C/A signals (e.g., “QACA[]_A,” “QBCA[]_A,” “QACA[]_B,” and “QBCA[]_B”) in response to the input C/A signals (e.g., “DCA[]_A” and “DCA[]_B”). <i>See, e.g.,</i> 6.1 Signal Groups This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are: Pre-RCD - Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A] - Pre-RCD Clock group includes: DCK_t and DCK_c - Pre-RCD Other group includes: ALERT_n, DRST_n - Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A] Post-RCD to SDRAM - Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A] - Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n - Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c - Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n - Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A] DQ and DQS - edge connector pin to SDRAM with series resistor SideBand - Host to Hub group: HSDA, HSCL, HSA - Hub local bus group: LSDA, LSCL (star topology from Hub) Ex. CC-88 (JEDEC 305B) at 12 (that RCD receives from the memory controller Pre-RCD command and address signals and generate Post-RCD command and address signals).

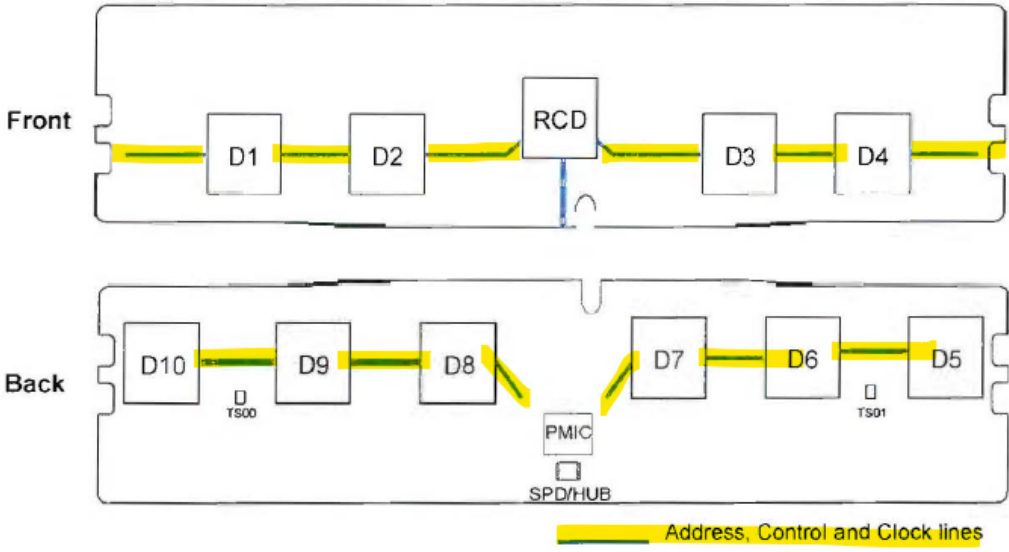
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
[1.c.3] generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices,	The Samsung Practicing '407 Products comprise circuitry mounted on the PCB and configurable to generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices. For example, the RCD circuitry is configurable to generate local clock outputs corresponding to the groups of memory devices associated with Channels A and B in different ranks and different locations on the PCB. <i>See, e.g.,</i>

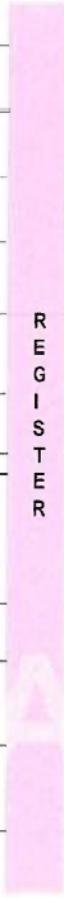
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

DCA[6:0]_A	QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10]
DCS0_A_n	QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10]
DCS1_A_n	
DCA[6:0]_B	QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3]
DCS0_B_n	QACS0_B -> CS_n: SDRAMs D [7:3]
DCS1_B_n	
DCK_t	QACK_A t/c -> CK t/c: SDRAMs D [1,2,8,9,10]
DCK_c	QACK_A t/c -> CK t/c: SDRAMs D [7,3]
QLBD	DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7,3]
QLBS	DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7,3]
DRST_n	QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3]
ALERT_n	DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21 (RCD generates two sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c and QACK_B_t/c).

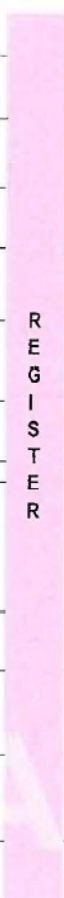
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:0]_A ——— QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] DCS0_A_n ——— QACS0_A -> CS_n: SDRAMs D[05:01] DCS1_A_n ——— QACS1_A -> CS_n: SDRAMs D[10:06] DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMs D[15:11] DCS1_B_n ——— QACS1_B -> CS_n: SDRAMs D[20:16] DCK_t/DCK_c ——— QACK_A_t -> CK_t: SDRAMs D[05:01] / QACK_A_c -> CK_c: SDRAMs D[05:01] — QACK_B_t -> CK_t: SDRAMs D[15:11] / QACK_B_c -> CK_c: SDRAMs D[15:11] — QCCK_A_t -> CK_t: SDRAMs D[10:06] / QCCK_A_c -> CK_c: SDRAMs D[10:06] — QCCK_B_t -> CK_t: SDRAMs D[20:16] / QCCK_B_c -> CK_c: SDRAMs D[20:16] QLBD ——— DLBD_A -> LBDQ: SDRAMs D[10:01] — DLBD_B -> LBDQ: SDRAMs D[20:11] QLBS ——— DLBS_A -> LBDQS: SDRAMs D[10:01] — DLBS_B -> LBDQS: SDRAMs D[20:11] DRST_n ——— QRT_A_n -> RESET_n: SDRAMs D[10:01] — QRT_B_n -> RESET_n: SDRAMs D[20:11] ALERT_n ——— DERROR_IN_A_n -> ALERT_n: SDRAMs D[10:01] — DERROR_IN_B_n -> ALERT_n: SDRAMs D[20:11]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates a memory module with two views: Front and Back. The Front view shows a row of memory devices labeled D01 through D15. Devices D01-D05 are grouped under the label TS00, and devices D11-D15 are grouped under TS01. A central component labeled RCD is positioned between the two groups. The Back view shows a row of memory devices labeled D06 through D20. Devices D06-D10 are on the left, and D16-D20 are on the right. A central component labeled PMIC is positioned below the devices, and a component labeled SPD/HUB is located below the PMIC. A legend at the bottom indicates that blue lines represent Address, Control and Clock lines. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25 (RCD generates four sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QCKK_A_t/c, QACK_B_t/c, and QCKK_B_t/c).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:00]_A ——— QACA[13:00]_A -> CA[13:00] : SDRAMs D[10:01] DCS0_A_n ——— QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n ——— DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00] : SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMsD[15:11] QACS0_B -> CS_n: SDRAMsD[20:16] DCS1_B_n ——— DCK_t / DCK_c ——— QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMsD[05:01] QACK_A_t->CK_t: SDRAMsD[10:06] / QACK_A_c->CK_c: SDRAMsD[10:06] QACK_B_t->CK_t: SDRAMsD[15:11] / QACK_B_c->CK_c: SDRAMsD[15:11] QACK_B_t->CK_t: SDRAMsD[20:16] / QACK_B_c->CK_c: SDRAMsD[20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n->RESET_n: SDRAMsD[20:11] ALERT_n ——— DERROR_IN_A_n->ALERT_n: SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n: SDRAMsD[20:11]

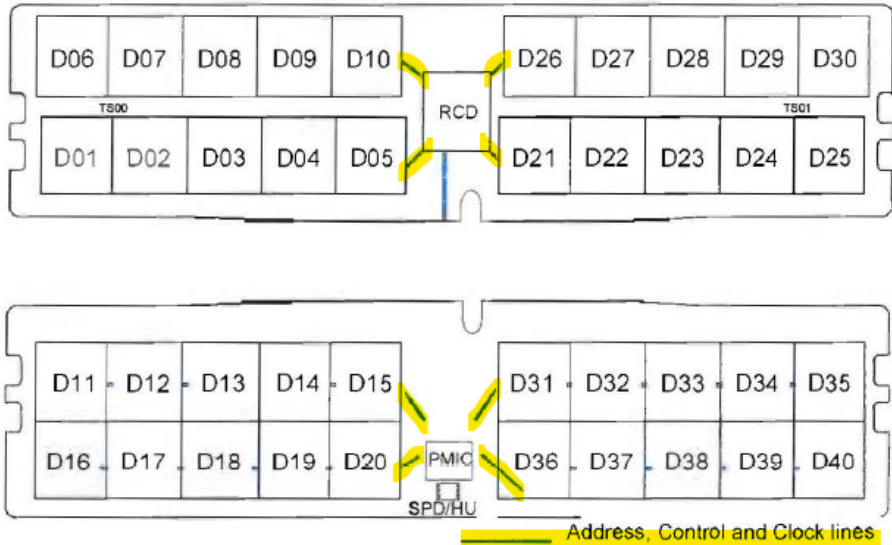
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates the internal architecture of a DDR5 RDIMM. It is divided into two main sections: 'Front' and 'Back'. Front View: Shows a central RCD (Refresh Control Driver) connected to two groups of memory devices. On the left, devices D01 through D05 are connected to the RCD. On the right, devices D11 through D15 are connected. The RCD is also connected to a PMIC (Power Management IC) and SPD/HUB (Serial Presence Detect/Hub) located on the back of the module. Back View: Shows the remaining memory devices, D06 through D10 on the left and D16 through D20 on the right. These devices are connected to the RCD and the PMIC/SPD/HUB. Legend: A yellow line indicates 'Address, Control and Clock lines'. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29 (RCD generates four sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QCKK_A_t/c, QACK_B_t/c, QCKK_B_t/c).

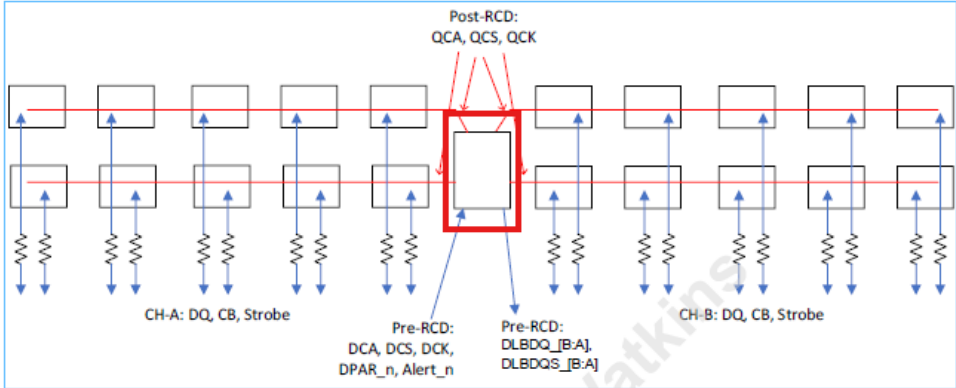
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A — QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n — QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] DCS1_A_n — QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] DCA[6:0]_B — QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] DCS0_B_n — QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] DCS1_B_n — QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] DCK_t/DCK_c — QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] QLBD — DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] QLBS — DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] DRST_n — QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] ALERT_n — DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 The diagram illustrates the internal structure of a Samsung DDR5 RDIMM. It shows two main sections of memory devices, each organized into two rows. The top section contains devices D06-D10 and D26-D30 in the top row, and D01-D05 and D21-D25 in the bottom row. The bottom section contains devices D11-D15 and D31-D35 in the top row, and D16-D20 and D36-D40 in the bottom row. A central RCD (Row Address Decoder) is connected to the top section, and a PMIC (Power Management IC) is connected to the bottom section. Yellow arrows indicate the distribution of address, control, and clock signals from the RCD and PMIC to the memory devices. A legend at the bottom of the diagram identifies the yellow arrows as 'Address, Control and Clock lines'. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29 (RCD generates eight sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QBCK_A_t/c, QCCCK_A_t/c, QDCK_A_t/c, QACK_B_t/c, QBCK_B_t/c, QCCCK_B_t/c, and QDCK_B_t/c). As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates each Samsung Practicing '407 Products' RCD generates multiple local clock signals corresponding, respectively, to the plurality of groups of the memory devices. For example, the RCD generates multiple QCK clock signals.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” clock (“DCK”) and generating “Post-RCD” control and address signals (“QCK”).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

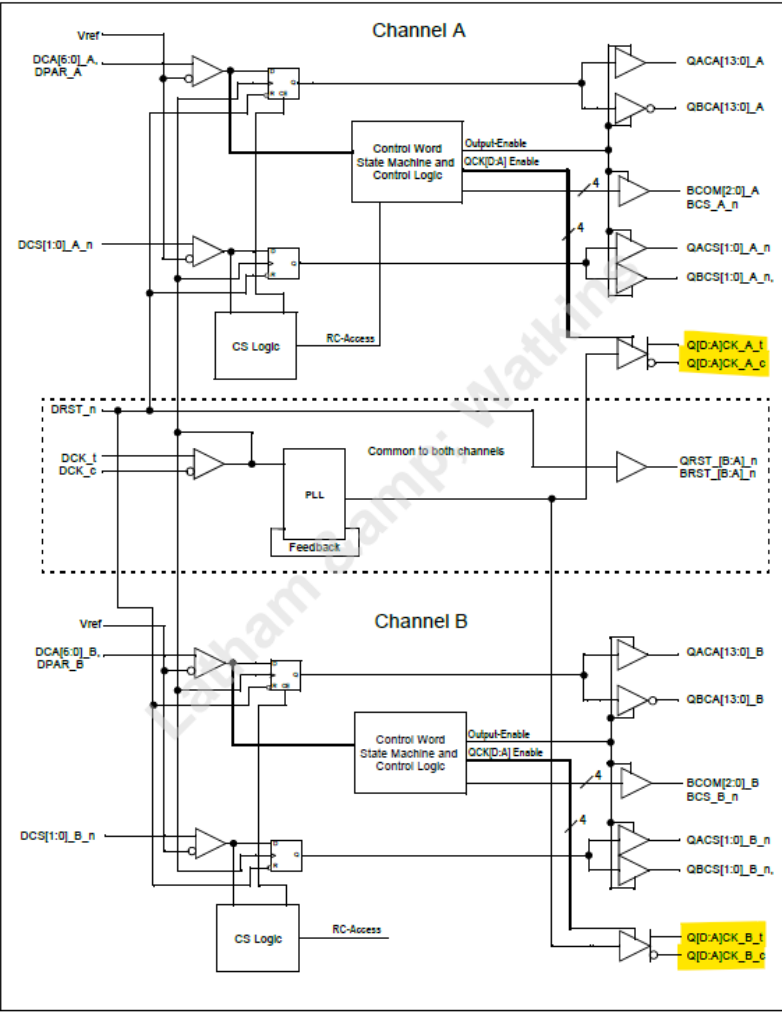
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																																																																																																						
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td>CB[7:0]</td></tr><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td>[31:24]</td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td>CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive corresponding local clocks, e.g., QACK, QBCK, QCK, QDCK, generated by the RCD). Furthermore, the RCD in the Samsung Practicing '407 Products comply with JEDEC DDR5 RCD specification. The JEDEC DDR5 RCD specification explains the RCD generates a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices. <i>See, e.g.,</i> <table><tr><td>Clock outputs</td><td>Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c</td><td>POD differential</td><td>cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>Feedback</td><td>OT RD</td><td>POD</td><td>Feedback outputs to Host</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4 (showing the RCD includes clock outputs for the memory devices).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCK	QACS0	QACA	0	Bottom	Lower	QCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top				[15:8]				[31:24]			Bottom		[7:0]				[23:16]				CB[7:0]	1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top		[7:0]				[23:16]			[31:24]		Bottom				[15:8]				[31:24]		CB[7:0]	Clock outputs	Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.	Feedback	OT RD	POD	Feedback outputs to Host
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	3.2 Logic Diagram Positive Logic  The diagram illustrates a dual-channel system (Channel A and Channel B) for data conversion. Channel A inputs include Vref, DCA[6:0]_A, DPAR_A, DCS[1:0]_A_n, and DRST_n. Channel B inputs include Vref, DCA[6:0]_B, DPAR_B, DCS[1:0]_B_n, and DRST_n. A common PLL block, labeled 'Common to both channels', receives DCK_t and DCK_c inputs and provides a clock signal to both channels. Each channel contains a 'Control Word State Machine and Control Logic' block, which is enabled by 'Output-Enable' and 'QCK[D:A] Enable' signals. The control logic also receives 'RC-Access' signals from 'CS Logic' blocks. The outputs of the control logic are connected to a series of multiplexers and buffers, producing signals such as QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBCS[1:0]_A_n, Q[D:A]CK_A_t, and Q[D:A]CK_A_c. Similar outputs are provided for Channel B. A 'Feedback' block is also shown, connected to the PLL and the control logic. Ex. CC-32 (JEDEC 82-512) at 7.

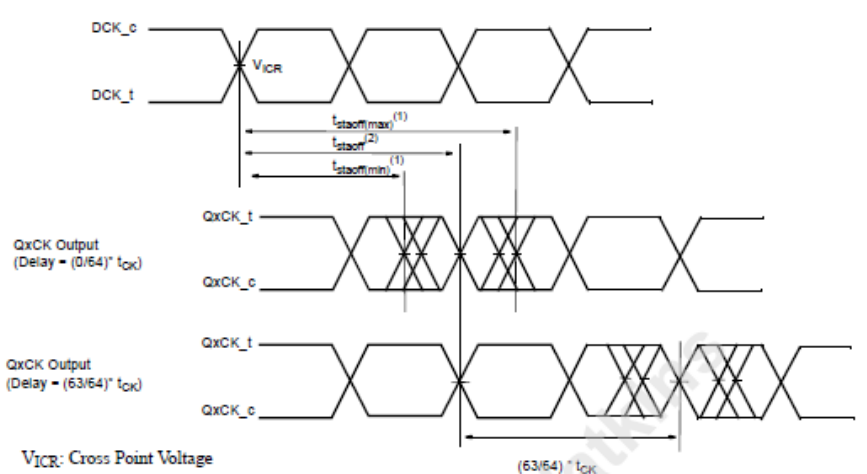
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
[1.c.3.i] the plurality of local clocks having respective phase relationships with the system clock,	In the Samsung Practicing '407 Products, the plurality of local clocks have respective phase relationships with the system clock. As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition demonstrates the RCD of the Samsung Practicing '407 Products supports “output delay control,” where the local clocks have respective phase relationships with the system clock. <i>See, e.g.</i>, 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																								
	Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCCCK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table> Ex. CC-32 (JEDEC 82-512) at 68 (showing local clocks QACK, QBCK, QCCCK, and QDCK that have respective phase relationships with the system clock). For example, local clocks (QxCK) shown below have respective phase relationships with respect to the system clock (DCK).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCCCK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
Register Address	Description																																								
RW0A	QCK Signals Driver Characteristics Control Word																																								
RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word																																								
RW0D	Data Buffer Interface Driver Characteristics Control Word																																								
RW10	IBT Control Word																																								
RW11[3:0]	Command Latency Adder Configuration Control Word																																								
RW12	QACK Output Delay Control Word																																								
RW13	QBCK Output Delay Control Word																																								
RW14	QCCCK Output Delay Control Word																																								
RW15	QDCK Output Delay Control Word																																								
RW17 .. RW1A	QxCS_n Output Delay Control Word																																								
RW1B, RW1C	QxCA Output Delay Control Word																																								
RW1D	BCS_n and BCOM[2:0] Output Delay Control Word																																								
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PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words																																								

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	14.7 Output Clock Driver Characteristics (cont'd)  The diagram illustrates the timing characteristics of the output clock driver in SDR Mode. It shows two input signals, DCK_e and DCK_t, which are differential signals. The output signals are $QxCK_t$ and $QxCK_e$, also differential. The diagram includes annotations for V_{ICR} (Cross Point Voltage), $t_{staoff(max)}$, $t_{staoff(2)}$, $t_{staoff(min)}$, and $(63/64) \cdot t_{CK}$. The output signals are shown with a delay of $(0/64) \cdot t_{CK}$ and $(63/64) \cdot t_{CK}$. Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode Ex. CC-32 (JEDEC 82-512) at 242.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

8.13.2 RW12 - QACK Output Delay Control Word

Table 118 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}
x	x								...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW12[7] = 1.

NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

Ex. CC-32 (JEDEC 82-512) at 146; see also id. at 147-148 (showing tables 119, 120, and 121).

3.3.8 Per-group Output Delay Control

The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 t_{CK}. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable.

3.3.8 Per-group Output Delay Control (cont'd)

The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111).

Ex. CC-30 (JEDEC 82-513) at 11-12.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																										
	Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW33[7:6]</td><td>DFE Tap 5 and Tap 6 Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table> Ex. CC-30 (JEDEC 82-513) at 70 (showing local clocks QACK, QBCK, QCKK, and QDCK that have respective phase relationships with the system clock).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW33[7:6]	DFE Tap 5 and Tap 6 Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
Register Address	Description																																										
RW0A	QCK Signals Driver Characteristics Control Word																																										
RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word																																										
RW0D	Data Buffer Interface Driver Characteristics Control Word																																										
RW10	IBT Control Word																																										
RW11[3:0]	Command Latency Adder Configuration Control Word																																										
RW12	QACK Output Delay Control Word																																										
RW13	QBCK Output Delay Control Word																																										
RW14	QCKK Output Delay Control Word																																										
RW15	QDCK Output Delay Control Word																																										
RW17 .. RW1A	QxCS_n Output Delay Control Word																																										
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RW1D	BCS_n and BCOM[2:0] Output Delay Control Word																																										
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RW48, RW49	Internal VrefCS Control Word																																										
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words																																										
PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver																																										
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words																																										

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

8.13.2 RW12 - QACK Output Delay Control Word

Table 119 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_ _t / QACK_ _c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_ _t /QACK_ _c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW12[7] = 1.

NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODT} for the delay to become stable on the outputs.

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8.13.3 RW13 - QBCK Output Delay Control Word

Table 120 — RW13: QBCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCK_/_ QBCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QBCK_t/QBCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³
NOTE 1 These control bits do not have any effect unless RW13[7] = 1.									
NOTE 2 RW13[7:0] will be sticky, cleared by power cycle not Reset.									
NOTE 3 When feature is enabled the delay settings in RW13[5:0] require a time of t _{ODU} for the delay to become stable on the outputs.									

8.13.4 RW14 - QCK Output Delay Control Word

Table 121 — RW14: QCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QCKCK_t/ QCKCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QCKCK_t/QCKCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³
NOTE 1	These control bits do not have any effect unless RW14[7] = 1.								
NOTE 2	RW14[7:0] will be sticky, cleared by power cycle not Reset.								
NOTE 3	When feature is enabled the delay settings in RW14[5:0] require a time of t _{ODU} for the delay to become stable on the outputs.								

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

8.13.5 RW15 - QDCK Output Delay Control Word

Table 122 — RW15: QDCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QDCK_t/ QDCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QDCK_t/QDCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

NOTE 2

NOTE 3

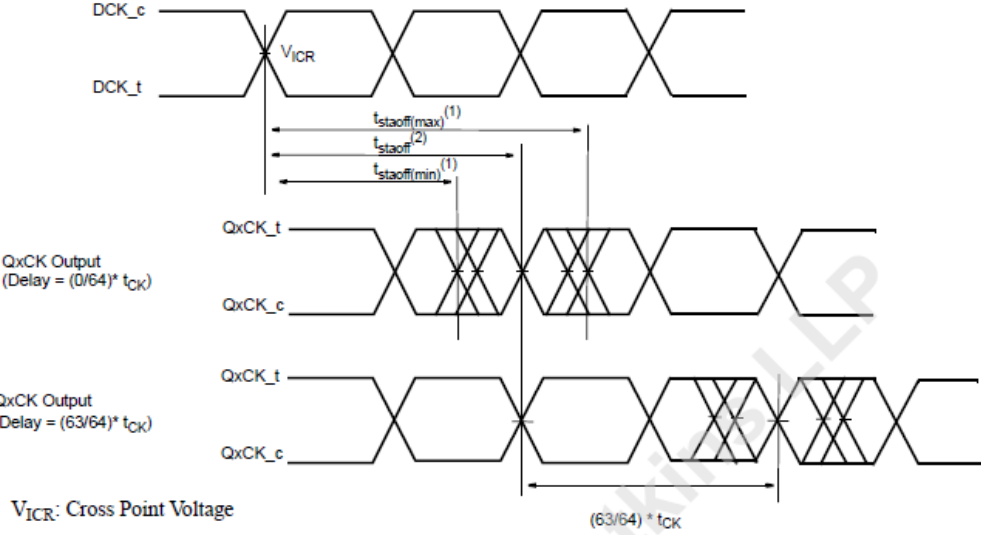
These control bits do not have any effect unless RW15[7] = 1.

RW15[7:0] will be sticky, cleared by power cycle not Reset.

When feature is enabled the delay settings in RW15[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

Ex. CC-30 (JEDEC 82-513) at 150-152 (showing that local clocks QACK, QBCK, QCK, and QDCK have respective phase relationships with the system clock).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	14.7 Output Clock Driver Characteristics (cont'd)  V_{ICR}: Cross Point Voltage 1. $t_{staoff(max/min)}$ = propagation delay specification limits for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles) over process, voltage and temperature. 2. t_{staoff} = measured propagation delay for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles). 3. t_{dynoff} = maximum propagation delay variation over voltage and temperature within t_{staoff} window. This includes all sources of jitter and drift (e.g. thermal noise, supply noise, voltage/temperature drift, SSC tracking, SSO, etc.) except reference clock noise. Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode Ex. CC-30 (JEDEC 82-513) at 253.
[1.c.3.ii] the respective phase relationships being	In the Samsung Practicing '407 Products, the respective phase relationships are programmable independently of each other.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
programmable independently of each other; and	As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition demonstrates the RCD of the Samsung Practicing '407 Products supports “output delay control,” where the local clocks have respective phase relationships with the system clock. The delay control of the local clocks (QxCK) with respect to the system clock (DCK) can be independently programmed. <i>See, e.g.,</i> 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12 (showing “output delay” for output signal groups).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																								
	Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table> Ex. CC-32 (JEDEC 82-512) at 68 (showing delays in QACK, QBCK, QCKK, and QDCK can be set independently).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
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Claim 1

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8.13.2 RW12 - QACK Output Delay Control Word

Table 118 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

These control bits do not have any effect unless RW12[7] = 1.

NOTE 2

RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in RW12[5:0] require a time of t_{QDU} for the delay to become stable on the outputs.

Ex. CC-32 (JEDEC 82-512) at 146; *see also id.* at 147-148 (showing how to program the delays independently, in tables 119, 120, and 121).

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Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1) (cont'd)

Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600		Unit	Note
			Min	Max	Min	Max	Min	Max	Min	Max		
tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5
tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6
tstaoff	Clock delay through the register between the input clock and output clock		Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)								ps	7,8
tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10

NOTE 1 At Down-bin Data Rate ($1980 \text{ MT/s} \leq f \leq 2100 \text{ MT/s}$), the max value of tSTAB01 is 10μs.

NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a tSTAB01 waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., tSTAB02 is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49.

NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.).

NOTE 4 Applies only to Self Refresh Exit without frequency change.

NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD).

NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCK_A_t/QCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCK_B_t/QCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.

NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 199 on page 191. See Figure 111 and Figure 112.

NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock tPD.

NOTE 9 VDD measurement DC bandwidth is limited to 20 MHz.

NOTE 10 See Figure 111 and Figure 112.

Ex. CC-32 (JEDEC 82-512) at 240 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”).

Claim 1

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14.7 Output Clock Driver Characteristics (cont'd)

Table 233 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2)

Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Note
			Min	Max		
t _{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	
t _{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	50	μs	1
t _{CKsk}	Fractional Clock Output Skew		-	100	ps	2
t _{stao}	Clock delay through the register between the input clock and output clock		Min: (t _{PD} (min) + 1/2 tCK); Max: (t _{PD} (max) + 1/2 tCK)		ps	3,4
t _{dyno}	Maximum re-driven dynamic clock offset		-	500	ps	5,6
NOTE 1 This parameter applies to control word writes to RW05 and Bit OP0 in RW00.						
NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.						
NOTE 3 RW05 must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 199 on page 191. See Figure 111 and Figure 112.						
NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t _{PD} .						
NOTE 5 V _{DD} measurement DC bandwidth is limited to 20 MHz.						
NOTE 6 See Figure 111 and Figure 112.						

Ex. CC-32 (JEDEC 82-512) at 241 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”).

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	The figure is a timing diagram illustrating clock output skew. It shows three sets of signals: QACK_c and QACK_t, QCCK_c and QCCK_t, and another QCCK_c and QCCK_t. A horizontal arrow labeled t_{CK} indicates the clock period. Vertical arrows labeled t_{CKsk} indicate the skew between the clock and data signals. Figure 113 — Clock Output (QxCK) Skew Ex. CC-32 (JEDEC 82-512) at 242. 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 t_{CK}. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. 3.3.8 Per-group Output Delay Control (cont'd) The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-30 (JEDEC 82-513) at 11-12.

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Claim 1

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8.13.2 RW12 - QACK Output Delay Control Word

Table 119 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_ _t / QACK_ _c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_ _t /QACK_ _c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW12[7] = 1.

NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODT} for the delay to become stable on the outputs.

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8.13.3 RW13 - QBCK Output Delay Control Word

Table 120 — RW13: QBCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QBCK_/ QBCK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x				...				...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QBCK_v/QBCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless [RW13\[7\]](#) = 1.
 NOTE 2 [RW13\[7:0\]](#) will be sticky, cleared by power cycle not Reset.
 NOTE 3 When feature is enabled the delay settings in [RW13\[5:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

8.13.4 RW14 - QCCK Output Delay Control Word

Table 121 — RW14: QCCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QCCK_/ QCCK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) * t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) * t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) * t_{CK}$
x	x				...				...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) * t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) * t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) * t_{CK}$
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QCCK_v/QCCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless [RW14\[7\]](#) = 1.
 NOTE 2 [RW14\[7:0\]](#) will be sticky, cleared by power cycle not Reset.
 NOTE 3 When feature is enabled the delay settings in [RW14\[5:0\]](#) require a time of t_{ODU} for the delay to become stable on the outputs.

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8.13.5 RW15 - QDCK Output Delay Control Word

Table 122 — RW15: QDCK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QDCK_t/ QDCK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64)* t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x		Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QDCK_t/QDCK_c ²	(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1

NOTE 2

NOTE 3

These control bits do not have any effect unless RW15[7] = 1.

RW15[7:0] will be sticky, cleared by power cycle not Reset.

When feature is enabled the delay settings in RW15[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

Ex. CC-30 (JEDEC 82-513) at 150-152 (showing that local clocks QACK, QBCK, QCK, and QDCK have respective phase relationships with the system clock).

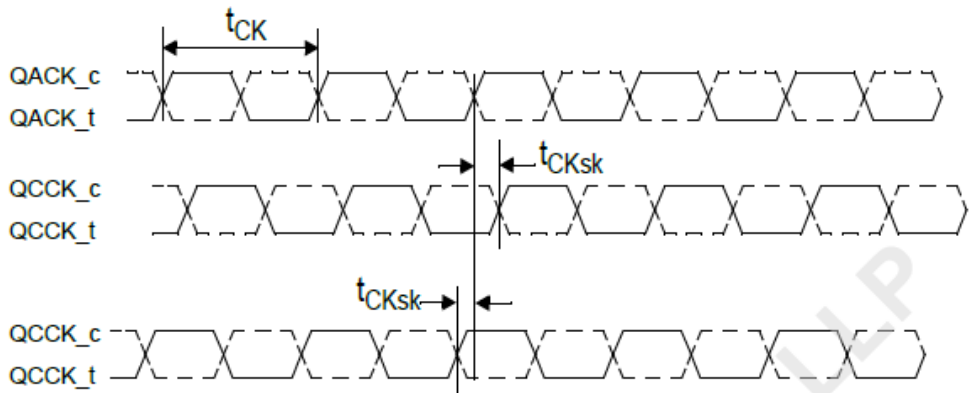
14.7 Output Clock Driver Characteristics

Table 237 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1)

Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000~6400		Unit
			Min	Max	Min	Max	Min	Max	
t_{STAB01}^1	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/ DCK_c stable	-	3.5	-	3.5	-	3.5	μs
t_{STAB02}	RCD Stabilization time from control word write to PLL lock ²	DCK_t/ DCK_c stable	-	25.0	-	25.0	-	25.0	μs
t_{STAB_DB}	Data Buffer Stabilization time	BCK_t/ BCK_c stable	See DDR5DB03 Specification						μs
t_{STAB_DBCMD}	LRDIMM stabilization time for commands not requiring use of data path ^{3,4}	DCK_t/ DCK_c stable	-	3.5	-	3.5	-	3.5	μs
t_{STAB_DBDATA}	LRDIMM stabilization time for commands requiring use of data path ^{5,4}	DCK_t/ DCK_c stable	-	4.14	-	4.14	-	4.14	μs
t_{CKsk}	Fractional Clock Output Skew ⁶		-	20	-	20	-	20	ps
t_{staoff}^7	Clock delay through the register between the input clock and output clock ⁸		Min: ($t_{PDM}(\min) + 1/2 t_{CK}$); Max: ($t_{PDM}(\max) + 1/2 t_{CK}$)						ps
t_{dynoff}^9	Maximum re-driven dynamic clock offset ¹⁰		-	40	-	35	-	35	ps
NOTE 1 At Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s), the max value of t_{STAB01} is 10 μs. NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a t_{STAB01} waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., t_{STAB02} is not required). See timing diagrams in Figure 35 and Figure 36. NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.). NOTE 4 Applies only to Self Refresh Exit without frequency change. NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD). NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QACK_A_t/QACK_B_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QACK_B_t/QACK_A_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only. NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 204. See Figure 111 and Figure 112. NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{PD}. NOTE 9 V_{DD} measurement DC bandwidth is limited to 20 MHz. NOTE 10 See Figure 111 and Figure 112.									

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																						
	Ex. CC-32 (JEDEC 82-513) at 250 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). 14.7 Output Clock Driver Characteristics (cont’d) Table 238 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2) <table><tr><th rowspan="2">Symbol</th><th rowspan="2">Parameter</th><th rowspan="2">Conditions</th><th colspan="2">Test Frequency (140 to 990 MHz)</th><th rowspan="2">Unit</th></tr><tr><th>Min</th><th>Max</th></tr><tr><td>t_{STAB01}</td><td>RCD Stabilization time from start of DCK toggle to PLL lock</td><td>DCK_t/DCK_c stable</td><td>-</td><td>50</td><td>μs</td></tr><tr><td>t_{STAB02}</td><td>RCD Stabilization time from control word write to PLL lock¹</td><td>DCK_t/DCK_c stable</td><td>-</td><td>50</td><td>μs</td></tr><tr><td>t_{CKsk}</td><td>Fractional Clock Output Skew²</td><td></td><td>-</td><td>100</td><td>ps</td></tr><tr><td>t_{stao}^3</td><td>Clock delay through the register between the input clock and output clock⁴</td><td></td><td colspan="2">Min: (t_{PDM}(min) + 1/2 tCK); Max: (t_{PDM}(max) + 1/2 tCK)</td><td>ps</td></tr><tr><td>t_{dynoff}^5</td><td>Maximum re-driven dynamic clock offset⁶</td><td></td><td>-</td><td>500</td><td>ps</td></tr></table> NOTE 1 This parameter applies to control word writes to RW05 and Bit OP0 in RW00. NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c, and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c, and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only. NOTE 3 RW05 must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 204. See Figure 111 and Figure 112. NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{PD}. NOTE 5 V_{DD} measurement DC bandwidth is limited to 20 MHz. NOTE 6 See Figure 111 and Figure 112. Ex. CC-32 (JEDEC 82-513) at 251 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”).	Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Min	Max	t_{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	t_{STAB02}	RCD Stabilization time from control word write to PLL lock ¹	DCK_t/DCK_c stable	-	50	μs	t_{CKsk}	Fractional Clock Output Skew ²		-	100	ps	t_{stao}^3	Clock delay through the register between the input clock and output clock ⁴		Min: (t _{PDM} (min) + 1/2 tCK); Max: (t _{PDM} (max) + 1/2 tCK)		ps	t_{dynoff}^5	Maximum re-driven dynamic clock offset ⁶		-	500	ps
Symbol	Parameter				Conditions	Test Frequency (140 to 990 MHz)		Unit																															
		Min	Max																																				
t_{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs																																		
t_{STAB02}	RCD Stabilization time from control word write to PLL lock ¹	DCK_t/DCK_c stable	-	50	μs																																		
t_{CKsk}	Fractional Clock Output Skew ²		-	100	ps																																		
t_{stao}^3	Clock delay through the register between the input clock and output clock ⁴		Min: (t _{PDM} (min) + 1/2 tCK); Max: (t _{PDM} (max) + 1/2 tCK)		ps																																		
t_{dynoff}^5	Maximum re-driven dynamic clock offset ⁶		-	500	ps																																		

Claim 1	Evidence of Use
	14.7 Output Clock Driver Characteristics (cont'd)  The diagram illustrates clock output characteristics. It shows three pairs of signals: QACK_c and QACK_t, QCCK_c and QCCK_t, and QCCK_c and QCCK_t. The first pair (QACK_c and QACK_t) shows a period t_{CK}. The second pair (QCCK_c and QCCK_t) shows a period t_{CKsk}. The third pair (QCCK_c and QCCK_t) also shows a period t_{CKsk}. A vertical line indicates a reference point for the skew measurements. Figure 113 — Clock Output (QxCK) Skew Ex. CC-32 (JEDEC 82-513) at 254.
[1.c.4] output the module C/A signals to the memory devices; and	See evidence cited for element [1.c.2]. The Samsung Practicing '407 Products comprise circuitry mounted on the PCB and configurable to output the module C/A signals to the memory devices. For example, the module C/A signals (such as QACA, QBCA, QACS and QBCS signals) are output to the SDRAM memory devices. See, e.g.,

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[6:0]_A DCS0_A_n DCS1_A_n DCA[6:0]_B DCS0_B_n DCS1_B_n DCK_t DCK_c REGISTER QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10] QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10] QACS1_A -> CS_n: SDRAMs D [1,2,8,9,10] QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3] QACS0_B -> CS_n: SDRAMs D [7:3] QACS1_B -> CS_n: SDRAMs D [7:3] QACK_A_t/c -> CK_t/c: SDRAMs D [1,2,8,9,10] QACK_A_c -> CK_c: SDRAMs D [7,3] DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7:3] DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7:3] QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3] DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

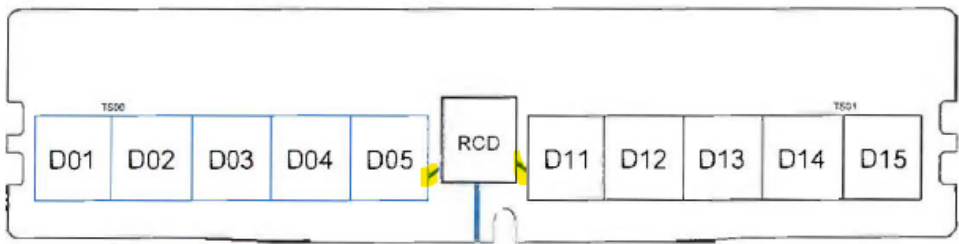
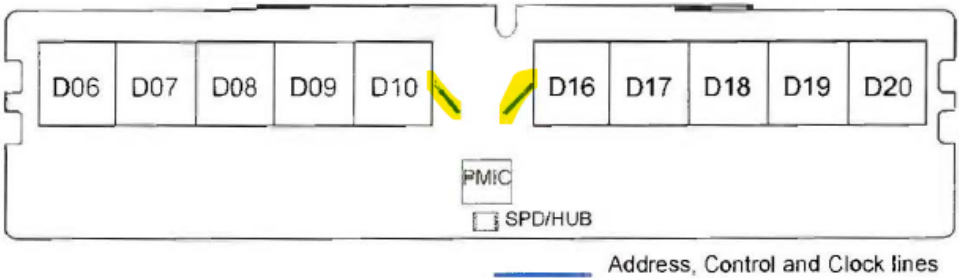
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front Back TS00 PMIC SPD/HUB TS01 Address, Control and Clock lines Ex. CC-89 at 21 (Samsung DDR5 RDIMM Datasheet)

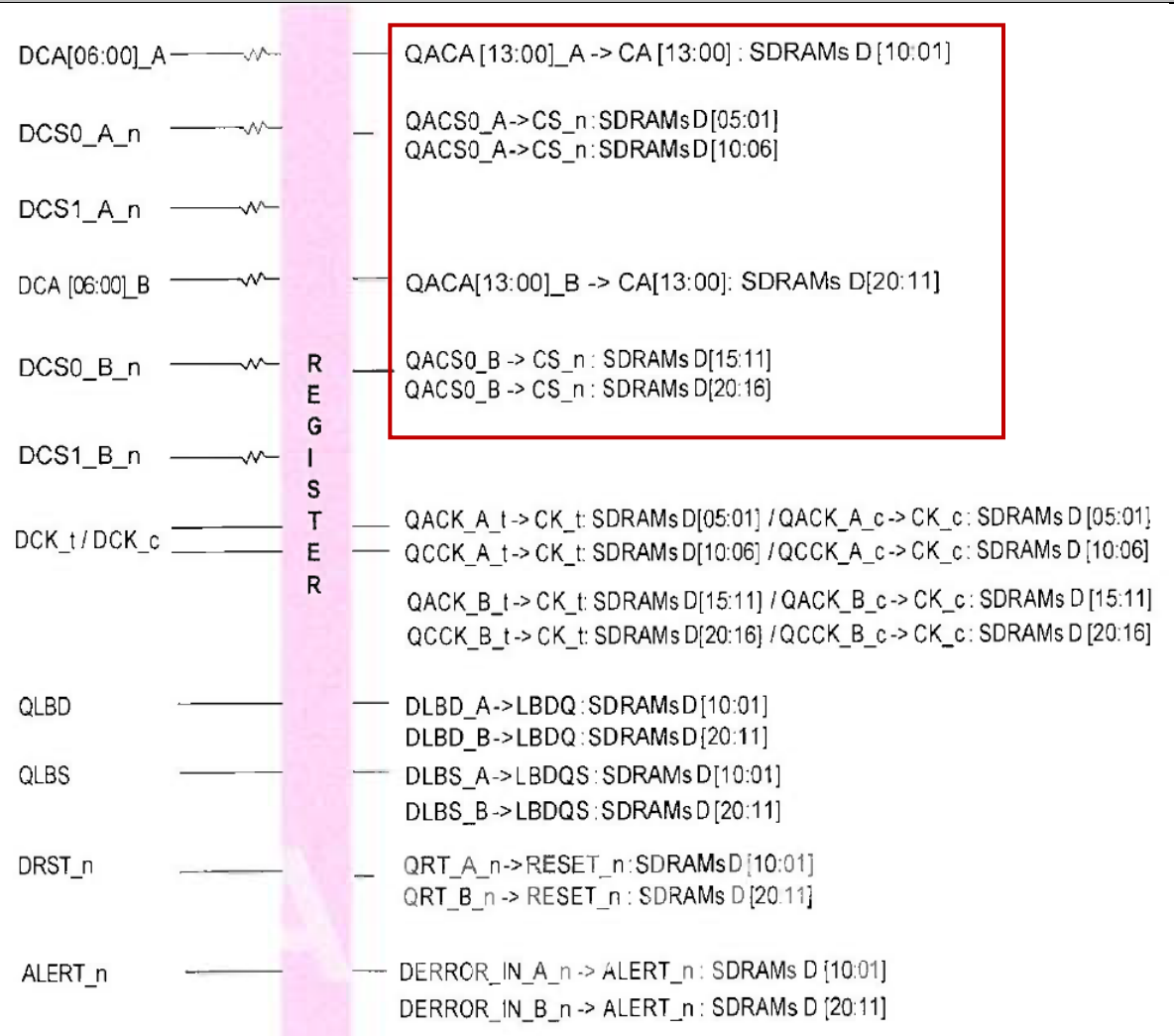
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[06:0]_A — QACA[13:00]_A -> CA[13:00] : SDRAMs D [10:01] DCS0_A_n — QACS0_A -> CS_n : SDRAMs D [05:01] DCS1_A_n — QACS1_A -> CS_n : SDRAMs D [10:06] DCA[06:00]_B — QACA[13:00]_B -> CA[13:00] : SDRAMs D [20:11] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D [15:11] DCS1_B_n — QACS1_B -> CS_n : SDRAMs D [20:16] DCK_t/DCK_c — QACK_A_t -> CK_t : SDRAMs D [05:01] / QACK_A_c -> CK_c : SDRAMs D [05:01] — QCCCK_A_t -> CK_t : SDRAMs D [10:06] / QCCCK_A_c -> CK_c : SDRAMs D [10:06] — QACK_B_t -> CK_t : SDRAMs D [15:11] / QACK_B_c -> CK_c : SDRAMs D [15:11] — QCCCK_B_t -> CK_t : SDRAMs D [20:16] / QCCCK_B_c -> CK_c : SDRAMs D [20:16] QLBD — DLBD_A -> LBDQ : SDRAMs D [10:01] — DLBD_B -> LBDQ : SDRAMs D [20:11] QLBS — DLBS_A -> LBDQS : SDRAMs D [10:01] — DLBS_B -> LBDQS : SDRAMs D [20:11] DRST_n — QRT_A_n -> RESET_n : SDRAMs D [10:01] — QRT_B_n -> RESET_n : SDRAMs D [20:11] ALERT_n — DERROR_IN_A_n -> ALERT_n : SDRAMs D [10:01] — DERROR_IN_B_n -> ALERT_n : SDRAMs D [20:11]

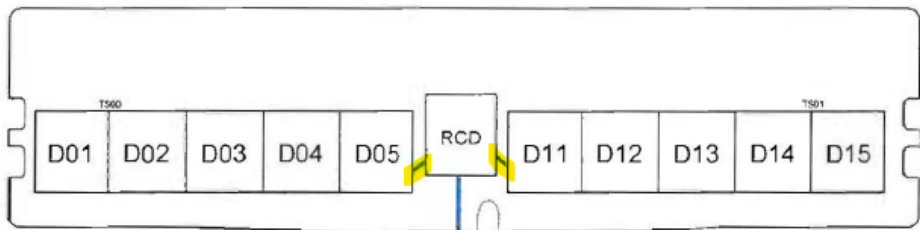
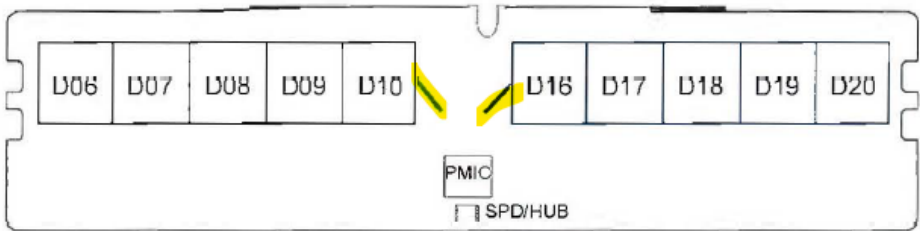
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:00]_A ——— QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n ——— QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n ——— DCA [06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n : SDRAMs D[15:11] QACS0_B -> CS_n : SDRAMs D[20:16] DCS1_B_n ——— DCK_t / DCK_c ——— QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] QCCCK_A_t->CK_t: SDRAMsD[10:06] / QCCCK_A_c->CK_c: SDRAMs D [10:06] QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QCCCK_B_t->CK_t: SDRAMsD[20:16] / QCCCK_B_c->CK_c: SDRAMs D [20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD [10:01] DLBD_B->LBDQ: SDRAMsD [20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD [10:01] DLBS_B->LBDQS: SDRAMsD [20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD [10:01] QRT_B_n-> RESET_n : SDRAMs D [20:11] ALERT_n ——— DERROR_IN_A_n-> ALERT_n : SDRAMs D [10:01] DERROR_IN_B_n-> ALERT_n : SDRAMs D [20:11]

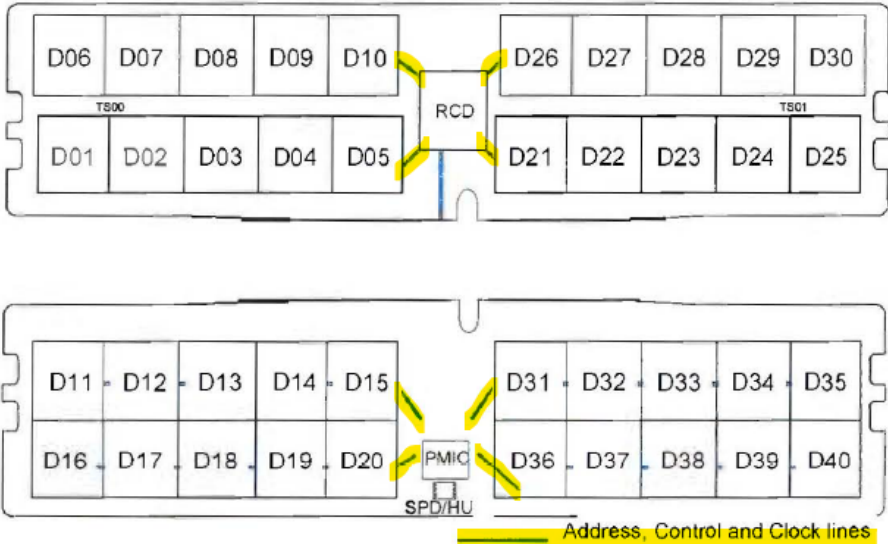
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Front  Back  Address, Control and Clock lines Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet)

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] QACS0_A->CS_n:SDRAMsD[5:1] QBCS0_A->CS_n:SDRAMsD[10:6] QACS1_A->CS_n:SDRAMsD[15:11] QBCS1_A->CS_n:SDRAMsD[20:16] QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] QACS0_B->CS_n:SDRAMsD[25:21] QBCS0_B->CS_n:SDRAMsD[30:26] QACS1_B->CS_n:SDRAMsD[35:31] QBCS1_B->CS_n:SDRAMsD[40:36] QACK_A_t->CK_t:SDRAMsD[5:1]/QACK_A_c->CK_c:SDRAMsD[5:1] QBCK_A_t->CK_t:SDRAMsD[10:6]/QBCK_A_c->CK_c:SDRAMsD[10:6] QCCK_A_t->CK_t:SDRAMsD[15:11]/QCCK_A_c->CK_c:SDRAMsD[15:11] QDCK_A_t->CK_t:SDRAMsD[20:16]/QDCK_A_c->CK_c:SDRAMsD[20:16] QACK_B_t->CK_t:SDRAMsD[25:21]/QACK_B_c->CK_c:SDRAMsD[25:21] QBCK_B_t->CK_t:SDRAMsD[30:26]/QBCK_B_c->CK_c:SDRAMsD[30:26] QCCK_B_t->CK_t:SDRAMsD[35:31]/QCCK_B_c->CK_c:SDRAMsD[35:31] QDCK_B_t->CK_t:SDRAMsD[40:36]/QDCK_B_c->CK_c:SDRAMsD[40:36] DLBD_A->LBDQ:SDRAMsD[20:1] DLBD_B->LBDQ:SDRAMsD[40:21] DLBS_A->LBDQS:SDRAMsD[20:1] DLBS_B->LBDQS:SDRAMsD[40:21] QRT_A_n->RESET_n:SDRAMsD[20:1] QRT_B_n->RESET_n:SDRAMsD[40:21] DERROR_IN_A_n->ALERT_n:SDRAMsD[20:1] DERROR_IN_B_n->ALERT_n:SDRAMsD[40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet) As explained supra, the Samsung Practicing '407 Products are JEDEC compliant. According to the JEDEC RCD specifications, RCD outputs module C/A signals (such as QACA, QBCA, QACS and QBCS signals) in response to the input C/A signals (DCA and DCS) received from the host. 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

Ex. CC-32 (JEDEC 82-512) at 6.

2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input Control bus	DCS[1:0]_[B:A]_n	POD V_{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Input Address and Command bus	DCA[6:0]_[B:A]	POD V_{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Parity input	DPAR_[B:A]	POD V_{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V_{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs	Q[B:A]CK [B:A]_t	POD differential	Clock outputs to the DRAMs. Four copies per channel.

Ex. CC-32 (JEDEC 82-512) at 4.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-32 (JEDEC 82-512) at 7 (showing the RCD circuitry outputs module C/A signals shown in yellow to memory devices).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. <table><caption>Table 8 — DDR CA Mapping DCA to QCA</caption><thead><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr></thead><tbody><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></tbody></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15. 3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled RW01[1] = 0. • When CA Pass-through mode is enabled RW00[2] = 1.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-32 (JEDEC 82-512) at 9. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1

Evidence of Use

2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Control bus			
Input	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Address and Command bus			
Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V _{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Control bus			
Output	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Address and Command bus			

Ex. CC-30 (JEDEC 82-513) at 4.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-30 (JEDEC 82-513) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		
	Ex. CC-30 (JEDEC 82-513) at 15.																								

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled <code>RW01[1] = 0</code>. • When CA Pass-through mode is enabled <code>RW00[2] = 1</code>. Ex. CC-30 (JEDEC 82-513) at 9. The JEDEC DDR5 RDIMM standard further demonstrates that the Samsung Practicing '407 Products output module C/A signals (e.g., “QACA[]_A,” “QBCA[]_A,” “QACA[]_B,” and “QBCA[]_B”) to memory devices. See, e.g.,

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Claim 1	Evidence of Use
	6.1 Signal Groups This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are: Pre-RCD - Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A] - Pre-RCD Clock group includes: DCK_t and DCK_c - Pre-RCD Other group includes: ALERT_n, DRST_n - Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A] Post-RCD to SDRAM - Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A] - Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n - Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c - Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n - Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A] DQ and DQS - edge connector pin to SDRAM with series resistor SideBand - Host to Hub group: HSDA, HSCL, HSA - Hub local bus group: LSDA, LSCL (star topology from Hub) Ex. CC-88 (JEDEC 305B) at 12 (that RCD outputs Post-RCD command and address signals to memory devices).
[1.c.5] output the plurality of local clocks to the memory devices,	See evidence cited for element [1.c.3]. The Samsung Practicing '407 Products comprise circuitry mounted on the PCB and configurable to output the plurality of local clocks to the memory devices. For example: The DDR5 RCD circuitry in the Samsung Practicing '407 Products is configurable to output the plurality of local clocks (e.g., “QACK_A_t,” “QBCK_A_t,” “QCK_A_t,” “QDCK_A_t,” “QACK_B_t,” “QBCK_B_t,” “QCK_B_t,” “QDCK_B_t,” “QACK_A_c,” “QBCK_A_c,” “QCK_A_c,” “QDCK_A_c,” “QACK_B_c,” “QBCK_B_c,” “QCK_B_c,” and “QDCK_B_c”) to SDRAM and DRAM memory devices as shown below. See, e.g.,

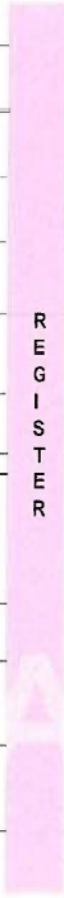
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

DCA[6:0]_A	QACA [13:0] _A -> CA [13:0]: SDRAMs D [1,2,8,9,10]
DCS0_A_n	QACS0_A -> CS_n: SDRAMs D [1,2,8,9,10]
DCS1_A_n	
DCA[6:0]_B	QACA [13:0] _B -> CA [13:0]: SDRAMs D [7:3]
DCS0_B_n	QACS0_B -> CS_n: SDRAMs D [7:3]
DCS1_B_n	
DCK_t	QACK_A t/c -> CK t/c: SDRAMs D [1,2,8,9,10]
DCK_c	QACK_A t/c -> CK t/c: SDRAMs D [7,3]
DLBD	DLBD_A -> LBDQ: SDRAMs D [1,2,8,9,10] DLBD_B -> LBDQ: SDRAMs D [7:3]
QLBS	DLBS_A -> LBDQS: SDRAMs D [1,2,8,9,10] DLBS_B -> LBDQS: SDRAMs D [7,3]
DRST_n	QRT_A_n -> RESET_n: SDRAMs D[1,2,8,9,10] QRT_B_n -> RESET_n: SDRAMs D[7,3]
ALERT_n	DERROR_IN_A_n -> ALERT_n: SDRAMs D [1,2,8,9,10] DERROR_IN_B_n -> ALERT_n: SDRAMs D [40:21]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	The diagram illustrates the internal architecture of a DDR5 RDIMM. The Front view shows a sequence of memory devices D1, D2, RCD (Refresh Control Driver), D3, and D4 connected by yellow lines representing Address, Control and Clock lines. The Back view shows memory devices D10, D9, D8, D7, D6, and D5, along with a PMIC (Power Management IC) and SPD/HUB (Serial Presence Detect/Hub). A legend indicates that yellow lines represent Address, Control and Clock lines. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 21 (RCD generates two sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c and QACK_B_t/c).

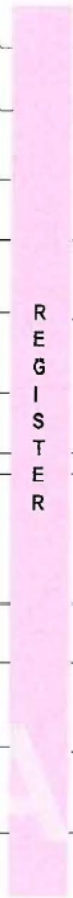
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	 DCA[06:0]_A ——— QACA[13:00]_A -> CA[13:00]: SDRAMs D[10:01] DCS0_A_n ——— QACS0_A -> CS_n: SDRAMs D[05:01] DCS1_A_n ——— QACS1_A -> CS_n: SDRAMs D[10:06] DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMs D[15:11] DCS1_B_n ——— QACS1_B -> CS_n: SDRAMs D[20:16] DCK_t/DCK_c ——— QACK_A_t -> CK_t: SDRAMs D[05:01] / QACK_A_c -> CK_c: SDRAMs D[05:01] — QCCK_A_t -> CK_t: SDRAMs D[10:06] / QCCK_A_c -> CK_c: SDRAMs D[10:06] — QACK_B_t -> CK_t: SDRAMs D[15:11] / QACK_B_c -> CK_c: SDRAMs D[15:11] — QCCK_B_t -> CK_t: SDRAMs D[20:16] / QCCK_B_c -> CK_c: SDRAMs D[20:16] QLBD ——— DLBD_A -> LBDQ: SDRAMs D[10:01] — DLBD_B -> LBDQ: SDRAMs D[20:11] QLBS ——— DLBS_A -> LBDQS: SDRAMs D[10:01] — DLBS_B -> LBDQS: SDRAMs D[20:11] DRST_n ——— QRT_A_n -> RESET_n: SDRAMs D[10:01] — QRT_B_n -> RESET_n: SDRAMs D[20:11] ALERT_n ——— DERROR_IN_A_n -> ALERT_n: SDRAMs D[10:01] — DERROR_IN_B_n -> ALERT_n: SDRAMs D[20:11]

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Claim 1	Evidence of Use
	The diagram illustrates the internal layout of a DDR5 RDIMM. The Front view shows two memory banks: TS00 containing devices D01 through D05, and TS01 containing devices D11 through D15. A Refresh Control Driver (RCD) is positioned between these banks. The Back view shows devices D06 through D10 on the left and D16 through D20 on the right. A Power Management IC (PMIC) and a Serial Presence Detect/Hub (SPD/HUB) are located at the bottom center. A legend at the bottom indicates that blue lines represent Address, Control and Clock lines. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 25 (RCD generates four sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QCCK_A_t/c, QACK_B_t/c, and QCCK_B_t/c).

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Claim 1	Evidence of Use
	 DCA[06:00]_A ——— QACA[13:00]_A -> CA[13:00] : SDRAMs D[10:01] DCS0_A_n ——— QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n ——— DCA[06:00]_B ——— QACA[13:00]_B -> CA[13:00] : SDRAMs D[20:11] DCS0_B_n ——— QACS0_B -> CS_n: SDRAMsD[15:11] QACS0_B -> CS_n: SDRAMsD[20:16] DCS1_B_n ——— DCK_t / DCK_c ——— QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMsD[05:01] QACK_A_t->CK_t: SDRAMsD[10:06] / QACK_A_c->CK_c: SDRAMsD[10:06] QACK_B_t->CK_t: SDRAMsD[15:11] / QACK_B_c->CK_c: SDRAMsD[15:11] QACK_B_t->CK_t: SDRAMsD[20:16] / QACK_B_c->CK_c: SDRAMsD[20:16] QLBD ——— DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS ——— DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMsD[10:01] QRT_B_n->RESET_n: SDRAMsD[20:11] ALERT_n ——— DERROR_IN_A_n->ALERT_n: SDRAMsD[10:01] DERROR_IN_B_n->ALERT_n: SDRAMsD[20:11]

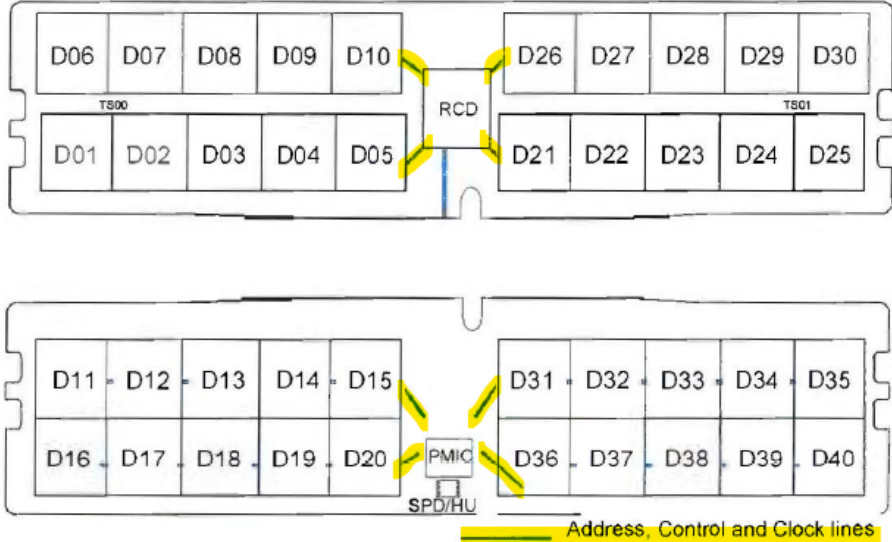
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

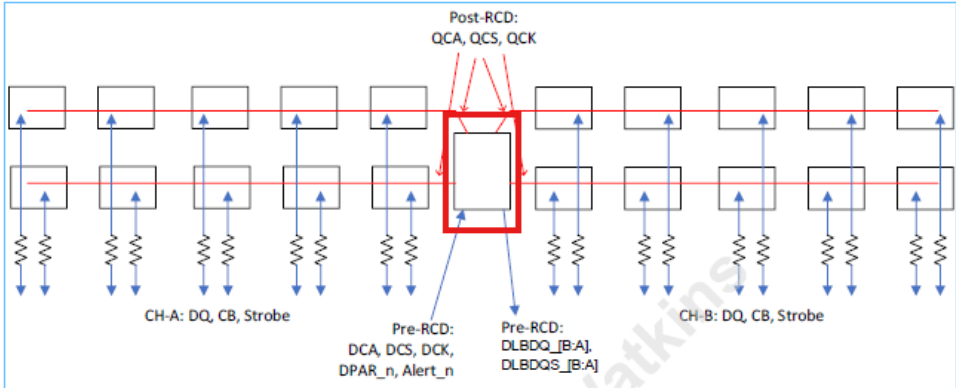
Claim 1	Evidence of Use
	The diagram illustrates the internal architecture of a DDR5 Registered Double In-Line Memory Module (RDIMM). It is divided into two main sections: the Front and the Back. The Front section contains two groups of memory devices, D01-D05 (labeled T500) and D11-D15 (labeled T501), which are connected to a central Register/Control Device (RCD). The Back section contains two groups of memory devices, D06-D10 and D16-D20, connected to a Power Management IC (PMIC) and a Serial Presence Detect/Hub (SPD/HUB). Yellow lines represent the Address, Control, and Clock lines connecting the RCD and PMIC to the memory devices. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29 (RCD generates four sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QCKK_A_t/c, QACK_B_t/c, QCKK_B_t/c).

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Claim 1	Evidence of Use
	DCA[6:0]_A — QACA [13:0]_A -> CA [13:0] : SDRAMs D [5:1, 15:11] QBCA [13:0]_A -> CA [13:0] : SDRAMs D [10:6, 20:16] DCS0_A_n — QACS0_A -> CS_n : SDRAMs D [5:1] QBCS0_A -> CS_n : SDRAMs D [10:6] DCS1_A_n — QACS1_A -> CS_n : SDRAMs D [15:11] QBCS1_A -> CS_n : SDRAMs D [20:16] DCA[6:0]_B — QACA [13:0]_B -> CA [13:0] : SDRAMs D [25:21, 35:31] QBCA [13:0]_B -> CA [13:0] : SDRAMs D [30:26, 40:36] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D [25:21] QBCS0_B -> CS_n : SDRAMs D [30:26] DCS1_B_n — QACS1_B -> CS_n : SDRAMs D [35:31] QBCS1_B -> CS_n : SDRAMs D [40:36] DCK_t/DCK_c — QACK_A_t -> CK_t : SDRAMs D [5:1] / QACK_A_c -> CK_c : SDRAMs D [5:1] QBCK_A_t -> CK_t : SDRAMs D [10:6] / QBCK_A_c -> CK_c : SDRAMs D [10:6] QCCK_A_t -> CK_t : SDRAMs D [15:11] / QCCK_A_c -> CK_c : SDRAMs D [15:11] QDCK_A_t -> CK_t : SDRAMs D [20:16] / QDCK_A_c -> CK_c : SDRAMs D [20:16] QACK_B_t -> CK_t : SDRAMs D [25:21] / QACK_B_c -> CK_c : SDRAMs D [25:21] QBCK_B_t -> CK_t : SDRAMs D [30:26] / QBCK_B_c -> CK_c : SDRAMs D [30:26] QCCK_B_t -> CK_t : SDRAMs D [35:31] / QCCK_B_c -> CK_c : SDRAMs D [35:31] QDCK_B_t -> CK_t : SDRAMs D [40:36] / QDCK_B_c -> CK_c : SDRAMs D [40:36] QLBD — DLBD_A -> LBDQ : SDRAMs D [20:1] DLBD_B -> LBDQ : SDRAMs D [40:21] QLBS — DLBS_A -> LBDQS : SDRAMs D [20:1] DLBS_B -> LBDQS : SDRAMs D [40:21] DRST_n — QRT_A_n -> RESET_n : SDRAMs D [20:1] QRT_B_n -> RESET_n : SDRAMs D [40:21] ALERT_n — DERROR_IN_A_n -> ALERT_n : SDRAMs D [20:1] DERROR_IN_B_n -> ALERT_n : SDRAMs D [40:21]

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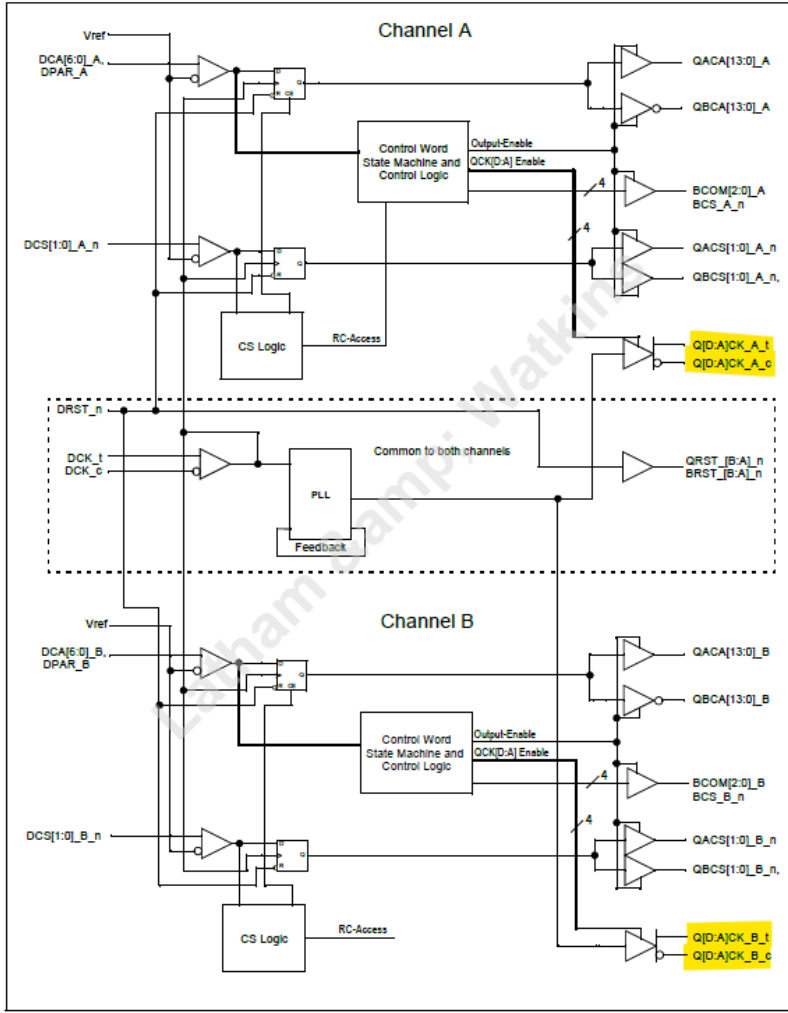
Claim 1	Evidence of Use
	 The diagram illustrates the internal structure of a Samsung DDR5 RDIMM. It shows two main sections of memory devices, each connected to a central RCD (Row Address Decoder) and a PMIC (Power Management IC). The top section contains two groups of memory devices: D06-D10 and D26-D30, connected to the RCD. The bottom section contains two groups: D11-D15 and D31-D35, connected to the PMIC. The RCD is connected to the PMIC via a SPD/HU (Serial Presence Detect/Hot Swap) interface. A legend indicates that yellow lines represent Address, Control and Clock lines. Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 29 (RCD generates eight sets of local clocks from DCK_t/c corresponding to groups of memory devices: QACK_A_t/c, QBCK_A_t/c, QCCCK_A_t/c, QDCK_A_t/c, QACK_B_t/c, QBCK_B_t/c, QCCCK_B_t/c, and QDCK_B_t/c). As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates each Samsung Practicing '407 Products' RCD outputs a plurality of local clocks to the memory devices.

Claim 1	Evidence of Use								
	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” clock (“DCK”) and generating “Post-RCD” control and address signals (“QCK”). Furthermore, the RCD in the Samsung Practicing ’407 Products comply with JEDEC DDR5 RCD specification. The JEDEC DDR5 RCD specification explains the RCD outputs the plurality of local clocks to the memory devices. <i>See, e.g.,</i> <table><tr><td>Clock outputs</td><td>Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>Feedback</td><td>OT RD</td><td>PDN</td><td>Feedback outputs to Host</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4 (showing the RCD includes clock outputs for the memory devices).	Clock outputs	Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.	Feedback	OT RD	PDN	Feedback outputs to Host
Clock outputs	Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.						
Feedback	OT RD	PDN	Feedback outputs to Host						

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Claim 1	Evidence of Use
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

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Claim 1	Evidence of Use
	3.2 Logic Diagram Positive Logic  The diagram illustrates a dual-channel system (Channel A and Channel B) for data conversion. Channel A inputs include Vref, DCA[6:0]_A, DPAR_A, DCS[1:0]_A_n, and DRST_n. Channel B inputs include Vref, DCA[6:0]_B, DPAR_B, DCS[1:0]_B_n, and DRST_n. A common PLL block receives DCK_t and DCK_c inputs and provides a clock signal to both channels. Each channel contains a CS Logic block, a Control Word State Machine and Control Logic block, and a Feedback loop. The Control Word State Machine and Control Logic block outputs QCK[D:A] Enable signals to the CS Logic and the output multiplexers. The CS Logic outputs RC-Access signals to the output multiplexers. The output multiplexers produce QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBCS[1:0]_A_n, Q[D:A]CK_A_t, and Q[D:A]CK_A_c for Channel A, and corresponding signals for Channel B. The Q[D:A]CK_A_t and Q[D:A]CK_A_c signals are highlighted in yellow. Ex. CC-32 (JEDEC 82-512) at 7.

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Claim 1	Evidence of Use
[1.c.5.i] wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices,	In the Samsung Practicing '407 Products, a respective local clock has a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices. For example: The local clocks generated by the RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices. <i>See, e.g.,</i>

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	DCA[06:0]_A ——— QACA[13:00]_A -> CA[13:00]: SDRAMs D [10:01] DCS0_A_n ——— QACS0_A -> CS_n : SDRAMs D [05:01] DCS1_A_n ——— QACS1_A -> CS_n : SDRAMs D [10:06] DCA[06:0]_B ——— QACA[13:00]_B -> CA[13:00]: SDRAMs D [20:11] DCS0_B_n ——— QACS0_B -> CS_n : SDRAMs D [15:11] DCS1_B_n ——— QACS1_B -> CS_n : SDRAMs D [20:16] DCK_t/DCK_c ——— QACK_A_t->CK_t: SDRAMs D [05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] QACK_B_t->CK_t: SDRAMs D [15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QACK_C_t->CK_t: SDRAMs D [20:16] / QACK_C_c->CK_c: SDRAMs D [20:16] QLBD ——— DLBD_A->LBDQ: SDRAMs D [10:01] DLBD_B->LBDQ: SDRAMs D [20:11] QLBS ——— DLBS_A->LBDQS: SDRAMs D [10:01] DLBS_B->LBDQS: SDRAMs D [20:11] DRST_n ——— QRT_A_n->RESET_n: SDRAMs D [10:01] QRT_B_n->RESET_n: SDRAMs D [20:11] ALERT_n ——— DERROR_IN_A_n -> ALERT_n : SDRAMs D [10:01] DERROR_IN_B_n -> ALERT_n : SDRAMs D [20:11]
	Ex. CC-89 at 25 (Samsung DDR5 RDIMM Datasheet) (illustrating, e.g., that (1) local clock signals “QACK_A_t” and “QACK_A_c” are output from the RCD to the “CK_t” and “CK_c” inputs of memory device “SDRAMs D [05:01]” but no other memory devices and (2) local clock signals “QACK_B_t” and “QACK_B_c” are output from the RCD to the “CK_t” and “CK_c” inputs of memory devices “SDRAMs D [15:11]” but no other memory devices).

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Claim 1	Evidence of Use
	DCA[06:00]_A — QACA [13:00]_A -> CA [13:00] : SDRAMs D [10:01] DCS0_A_n — QACS0_A->CS_n: SDRAMsD[05:01] QACS0_A->CS_n: SDRAMsD[10:06] DCS1_A_n — DCA [06:00]_B — QACA[13:00]_B -> CA[13:00]: SDRAMs D[20:11] DCS0_B_n — QACS0_B -> CS_n : SDRAMs D[15:11] QACS0_B -> CS_n : SDRAMs D[20:16] DCS1_B_n — DCK_t / DCK_c — QACK_A_t->CK_t: SDRAMsD[05:01] / QACK_A_c->CK_c: SDRAMs D [05:01] QCCK_A_t->CK_t: SDRAMsD[10:06] / QCCK_A_c->CK_c: SDRAMs D [10:06] QACK_B_t->CK_t: SDRAMs D[15:11] / QACK_B_c->CK_c: SDRAMs D [15:11] QCCK_B_t->CK_t: SDRAMsD[20:16] / QCCK_B_c->CK_c: SDRAMs D [20:16] QLBD — DLBD_A->LBDQ: SDRAMsD[10:01] DLBD_B->LBDQ: SDRAMsD[20:11] QLBS — DLBS_A->LBDQS: SDRAMsD[10:01] DLBS_B->LBDQS: SDRAMsD[20:11] DRST_n — QRT_A_n->RESET_n: SDRAMsD [10:01] QRT_B_n-> RESET_n : SDRAMs D [20:11] ALERT_n — DERROR_IN_A_n-> ALERT_n : SDRAMs D [10:01] DERROR_IN_B_n-> ALERT_n : SDRAMs D [20:11]
	Ex. CC-89 at 29 (Samsung DDR5 RDIMM Datasheet) (illustrating, e.g., that (1) local clock signals “QACK_A t” and “QACK_A c” are output from the RCD to the “CK_t” and “CK_c” inputs of memory devices

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Claim 1	Evidence of Use
	“SDRAMs D [05:01]” but no other memory devices and (2) local clock signals “QACK_B_t” and “QACK_B_c” are output from the RCD to the “CK_t” and “CK_c” inputs of memory devices “SDRAMs D [15:11]” but no other memory devices). Ex. CC-89 at 33 (Samsung DDR5 RDIMM Datasheet) (illustrating, e.g., that (1) local clock signals “QACK_A_t” and “QACK_A_c” are output from the RCD to the “CK_t” and “CK_c” inputs of memory devices “SDRAMs D [5:1]” but no other memory devices and (2) local clock signals “QACK_B_t” and “QACK_B_c”

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Claim 1	Evidence of Use
	are output from the RCD to the “CK_t” and “CK_c” inputs of memory devices “SDRAMs D [25:21]” but no other memory devices). As explained <i>supra</i>, the Samsung Practicing ’407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates the local clocks generated by the Samsung Practicing ’407 Products’ RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices.

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Claim 1

Evidence of Use

6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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Claim 1	Evidence of Use																																																																																																																																																														
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td>CB[7:0]</td></tr><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td>CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing local clocks, e.g., QACK, QBCK, QCCK, QDCK, being output to certain groups of memory devices but not others).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top				[15:8]				[31:24]			Bottom		[7:0]				[23:16]				CB[7:0]	1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top		[7:0]				[23:16]					Bottom				[15:8]				[31:24]		CB[7:0]
1Rx8 (less than 8000 Mbps)																																																																																																																																																															
DIMM Side	Row	Clock	CS	CA	Rank																																																																																																																																																										
Top	Lower	QACK	QACS0	QACA	0																																																																																																																																																										
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	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
Top				[15:8]				[31:24]																																																																																																																																																							
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1Rx8 Sub Channel 1 (Channel B)																																																																																																																																																															
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																					
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
Top		[7:0]				[23:16]																																																																																																																																																									
Bottom				[15:8]				[31:24]		CB[7:0]																																																																																																																																																					
[1.c.5.ii] wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock.	In the Samsung Practicing '407 Products, the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock. For example, a group of memory devices in the Samsung Practicing '407 Products associated with Channel A perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines (e.g., “DQ[]_A”), and a group of memory devices in the Samsung Practicing '407 Products associated with Channel B perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines (e.g., “DQ[]_B”). <i>See, e.g.:</i>																																																																																																																																																														

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Claim 1	Evidence of Use																																																		
	[Table 5] PIN Description <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>CA[6:0]_A CA[6:0]_B</td><td>SDRAM Address and Command Bus</td></tr> <tr> <td>CS[1:0]_A CS[1:0]_B</td><td>SDRAM Chip Select</td></tr> <tr> <td>PAR_A PAR_B</td><td>SDRAM Parity input</td></tr> <tr> <td>CK_t</td><td>SDRAM Clocks (true/positive)</td></tr> <tr> <td>CK_c</td><td>SDRAM Clocks (complement/negative)</td></tr> <tr> <td>ALERT_n</td><td>SDRAM alert for CRC error</td></tr> <tr> <td>RESET_n</td><td>Set DRAM to known state</td></tr> <tr> <td>PCAMP</td><td>Control and Monitor Port</td></tr> <tr> <td>HSCL</td><td>I2C/I3C-Basic Host Sideband Bus Clock</td></tr> <tr> <td>HSDA</td><td>I2C/I3C-Basic Host Sideband Bus Data</td></tr> <tr> <td>HSA</td><td>I2C/I3C-Basic Host Sideband Bus Address</td></tr> <tr> <td>LBDQ</td><td>Loopback data output:</td></tr> </table> <table> <tr> <th>Pin Name</th><th>Description</th></tr> <tr> <td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory data bus channel A and B</td></tr> <tr> <td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC check bits(CB) channel A and B</td></tr> <tr> <td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>SDRAM data strobes (positive line of differential pair)</td></tr> <tr> <td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>SDRAM data strobes (negative line of differential pair)</td></tr> <tr> <td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr> <tr> <td>VIN_BULK</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VIN_MGMT</td><td>DIMM Power Supply from system to PMIC</td></tr> <tr> <td>VSS</td><td>Power supply return (ground)</td></tr> <tr> <td>RFU</td><td>Reserved for future use</td></tr> <tr> <td>LBDQS</td><td>Loopback data strobe output</td></tr> </table> Ex. CC-89 at 9.	Pin Name	Description	CA[6:0]_A CA[6:0]_B	SDRAM Address and Command Bus	CS[1:0]_A CS[1:0]_B	SDRAM Chip Select	PAR_A PAR_B	SDRAM Parity input	CK_t	SDRAM Clocks (true/positive)	CK_c	SDRAM Clocks (complement/negative)	ALERT_n	SDRAM alert for CRC error	RESET_n	Set DRAM to known state	PCAMP	Control and Monitor Port	HSCL	I2C/I3C-Basic Host Sideband Bus Clock	HSDA	I2C/I3C-Basic Host Sideband Bus Data	HSA	I2C/I3C-Basic Host Sideband Bus Address	LBDQ	Loopback data output:	Pin Name	Description	DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B	CB[7:0]_A CB[7:0]_B	DIMM ECC check bits(CB) channel A and B	DQS[9:0]_A_t DQS[9:0]_B_t	SDRAM data strobes (positive line of differential pair)	DQS[9:0]_A_c DQS[9:0]_B_c	SDRAM data strobes (negative line of differential pair)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Not valid for x4 operation. Enabled via Mode Register.	VIN_BULK	DIMM Power Supply from system to PMIC	VIN_MGMT	DIMM Power Supply from system to PMIC	VSS	Power supply return (ground)	RFU	Reserved for future use	LBDQS	Loopback data strobe output
Pin Name	Description																																																		
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DQ[31:0]_A DQ[31:0]_B	DIMM memory data bus channel A and B																																																		
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VIN_BULK	DIMM Power Supply from system to PMIC																																																		
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LBDQS	Loopback data strobe output																																																		

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[Table 4] PIN Configuration

Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side	Pin	Front Side	Pin	Back Side
1	VIN_BULK	145	VIN_BULK	50	VSS	194	DQ31_A	96	CB0_B	240	VSS
2	RFU	146	VIN_BULK	51	CB0_A	195	VSS	97	VSS	241	CB2_B
3	VIN_MGMT	147	PWR_GOOD(FAIL_n)	52	VSS	196	CB2_A	98	CB1_B	242	VSS
4	SCL	148	SAA	53	CB1_A	197	VSS	99	VSS	243	CB3_B
5	SDA	149	RFU	54	VSS	198	CB3_A	100	DQ0_B	244	VSS
6	VSS	150	RFU	55	DQ64_A_t	199	VSS	101	VSS	245	DQ2_B
7	DQ9_A	151	VSS	56	DQ64_A_c	200	DQ69_A_cTDQ69_A_c	102	DQ1_B	246	VSS
8	VSS	152	DQ2_A	57	VSS	201	DQ69_A_tTDQ69_A_t	103	VSS	247	DQ3_B
9	DQ1_A	153	VSS	58	CB4_A	202	VSS	104	DQ60_B_t	248	VSS
10	VSS	154	DQ3_A	59	VSS	203	CB6_A	105	DQ60_B_c	249	DQ65_B_cTDQ65_B_c
11	DQ60_A_t	155	VSS	60	CB5_A	204	VSS	106	VSS	250	DQ65_B_tTDQ65_B_t
12	DQ60_A_c	156	DQ65_A_cTDQ65_A_c	61	VSS	205	CB7_A	107	DQ4_B	251	VSS
13	VSS	157	DQ65_A_tTDQ65_A_t	62	ALERT_n	206	VSS	108	VSS	252	DQ6_B
14	DQ4_A	158	VSS	63	VSS	207	RESET_n	109	DQ6_B	253	VSS
15	VSS	159	DQ6_A	64	CS0_A_n	208	VSS	110	VSS	254	DQ7_B
16	DQ5_A	160	VSS	65	VSS	209	CS1_A_n	111	DQ8_B	255	VSS
17	VSS	161	DQ7_A	66	CA0_A	210	VSS	112	VSS	256	DQ10_B
18	DQ6_A	162	VSS	67	VSS	211	CA1_A	113	DQ9_B	257	VSS
19	VSS	163	DQ10_A	68	CA2_A	212	VSS	114	VSS	258	DQ11_B
20	DQ9_A	164	VSS	69	VSS	213	CA3_A	115	DQ61_B_t	259	VSS
21	VSS	165	DQ11_A	70	CA4_A	214	VSS	116	DQ61_B_c	260	DQ66_B_cTDQ66_B_c
22	DQ61_A_t	166	VSS	71	VSS	215	CA5_A	117	VSS	261	DQ66_B_tTDQ66_B_t
23	DQ61_A_c	167	DQ66_A_cTDQ66_A_c	72	CA6_A	216	VSS	118	DQ12_B	262	VSS
24	VSS	168	DQ66_A_tTDQ66_A_t	73	VSS	217	CK_t	119	VSS	263	DQ14_B
25	DQ12_A	169	VSS	74	PAR_A	218	CK_c	120	DQ13_B	264	VSS
26	VSS	170	DQ14_A	75	VSS	219	VSS	121	VSS	265	DQ15_B
27	DQ13_A	171	VSS	KEY				122	DQ16_B	266	VSS
28	VSS	172	DQ15_A					123	VSS	267	DQ18_B
29	DQ16_A	173	VSS					124	DQ17_B	268	VSS
30	VSS	174	DQ18_A					125	VSS	269	DQ19_B
31	DQ17_A	175	VSS	76	CA0_B	220	RFU	126	DQ62_B_t	270	VSS
32	VSS	176	DQ19_A	77	VSS	221	CA1_B	127	DQ62_B_c	271	DQ67_B_cTDQ67_B_c
33	DQ62_A_t	177	VSS	78	CA2_B	222	VSS	128	VSS	272	DQ67_B_tTDQ67_B_t
34	DQ62_A_c	178	DQ67_A_cTDQ67_A_c	79	VSS	223	CA3_B	129	DQ20_B	273	VSS
35	VSS	179	DQ67_A_tTDQ67_A_t	80	CA4_B	224	VSS	130	VSS	274	DQ22_B
36	DQ20_A	180	VSS	81	VSS	225	CA5_B	131	DQ21_B	275	VSS
37	VSS	181	DQ22_A	82	CA6_B	226	VSS	132	VSS	276	DQ23_B
38	DQ21_A	182	VSS	83	VSS	227	PAR_B	133	DQ24_B	277	VSS
39	VSS	183	DQ23_A	84	CS0_B_n	228	VSS	134	VSS	278	DQ26_B
40	DQ24_A	184	VSS	85	VSS	229	CS1_B_n	135	DQ25_B	279	VSS
41	VSS	185	DQ26_A	86	LBD, RSP_A_n	230	VSS	136	VSS	280	DQ27_B
42	DQ25_A	186	VSS	87	LBS, RSP_B_n	231	RFU	137	DQ63_B_t	281	VSS
43	VSS	187	DQ27_A	88	VSS	232	RFU	138	DQ63_B_c	282	DQ68_B_cTDQ68_B_c
44	DQ63_A_t	188	VSS	89	CB4_B	233	VSS	139	VSS	283	DQ68_B_tTDQ68_B_t
45	DQ63_A_c	189	DQ68_A_cTDQ68_A_c	90	VSS	234	CB6_B	140	DQ28_B	284	VSS
46	VSS	190	DQ68_A_tTDQ68_A_t	91	CB5_B	235	VSS	141	VSS	285	DQ30_B
47	DQ28_A	191	VSS	92	VSS	236	CB7_B	142	DQ29_B	286	VSS
48	VSS	192	DQ30_A	93	DQ69_B_tTDQ69_B_t	237	VSS	143	VSS	287	DQ31_B
49	DQ29_A	193	VSS	94	DQ69_B_cTDQ69_B_c	238	DQ64_B_c	144	RFU	288	VSS
				95	VSS	239	DQ64_B_t				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

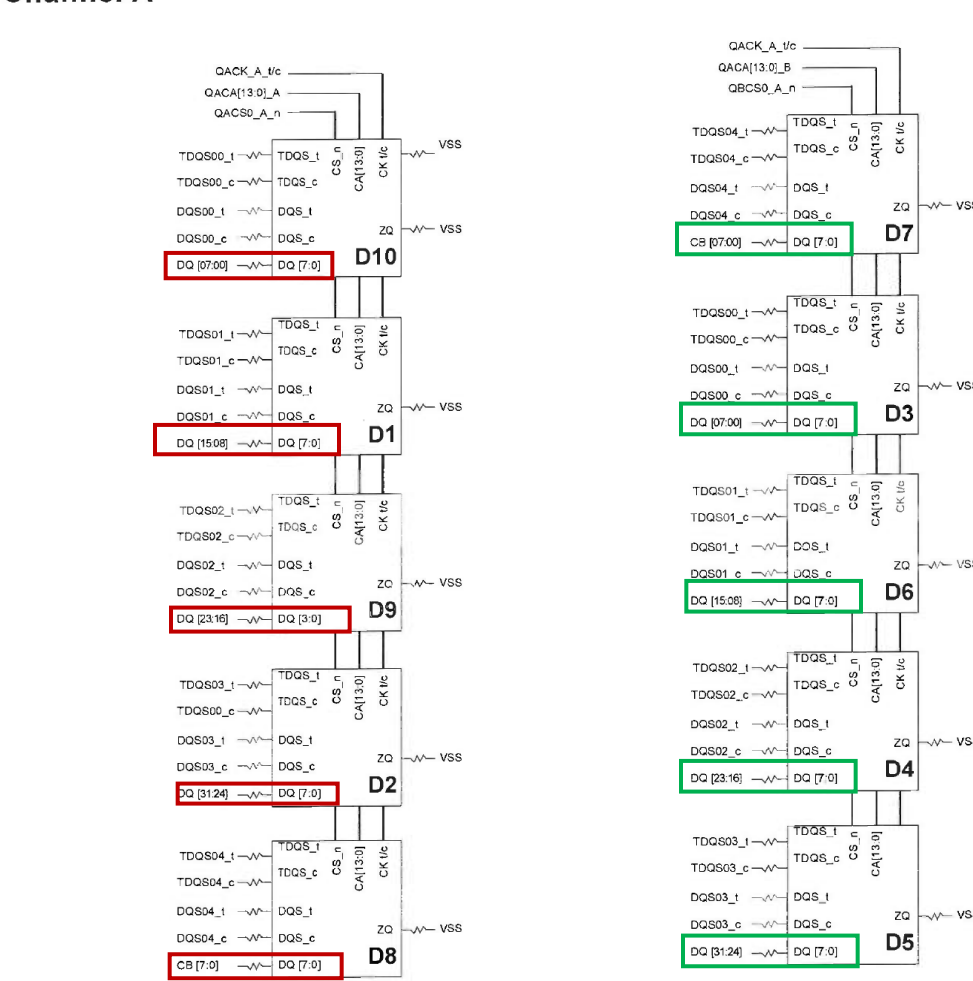
Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	V33	75	219	V33
VIN_MGMT	3	147	PWR_GOOD/FAIL_n	Key			
HSCL	4	148	HSA	CA0_B	76	220	RFU
HSDA	5	149	RFU	VSS	77	221	CA1_B
VSS	6	150	RFU	CA2_B	78	222	VSS
DQ0_A	7	151	VSS	VSS	79	223	CA3_B
VSS	8	152	DQ2_A	CA4_B	80	224	VSS
DQ1_A	9	153	VSS	VSS	81	225	CA5_B
VSS	10	154	DQ3_A	CA6_B	82	226	VSS
DQS0_A_t	11	155	VSS	VSS	83	227	PAR_B
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	CS0_B_n	84	228	VSS
VSS	13	157	DU55_A_t, DU55_A_t DBI0_A_n	VSS	85	229	CS1_B_n
DQ4_A	14	158	VSS	LBD, RSP_A_n	86	230	VSS
VSS	15	159	DQ6_A	LBS, RSP_B_n	87	231	RFU
DQ5_A	16	160	VSS	VSS	88	232	RFU
VSS	17	161	DQ7_A	CB4_B	89	233	VSS
DQ8_A	18	162	VSS	VSS	90	234	CB6_B
VSS	19	163	DQ10_A	CB5_B	91	235	VSS
DQ9_A	20	164	VSS	VSS	92	236	CB7_B
VSS	21	165	DQ11_A	DQS9_B_t, TDQS9_B_t DBI4_B_n	93	237	VSS
DQS1_A_t	22	166	VSS	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	VSS	95	239	DQS4_B_t
VSS	24	168	DQS6_A_t, TDQS6_A_t DBI1_A_n	CB0_B	96	240	VSS
DQ12_A	25	169	VSS	V33	97	241	OD2_D
VSS	26	170	DQ14_A	CB1_B	98	242	VSS
DQ13_A	27	171	VSS	VSS	99	243	CB3_B
VSS	28	172	DQ15_A	DQ0_B	100	244	VSS
DQ16_A	29	173	VSS	VSS	101	245	DQ2_B
VSS	30	174	DQ18_A	DQ1_B	102	246	VSS
DQ17_A	31	175	VSS	VSS	103	247	DQ3_BDQ3_B
VSS	32	176	DQ19_A	DQS0_B_t	104	248	VSS
DQS2_A_t	33	177	VSS	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	VSS	106	250	DQS5_B_t, TDQS5_B_t DBI0_B_n
VSS	35	179	DQS7_A_t, TDQS7_A_t DBI2_n	DQ4_B	107	251	VSS
DQ20_A	36	180	VSS	VSS	108	252	DQ6_B
VSS	37	181	DQ22_A	DQ5_B	109	253	VSS
DQ21_A	38	182	VSS				

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use											
	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B				
	DQ24_A	40	184	V33	DQ8_B	111	255	VSS				
	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B				
	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS				
	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B				
	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS				
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c				
	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t				
				DBI3_A_n				DBI1_B_n				
	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS				
	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B				
	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS				
	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B				
	CB0_A	51	195	VSS	DQ16_B	122	266	VSS				
	VSS	52	196	CB2_A	VSS	123	267	DQ18_B				
	CB1_A	53	197	VSS	DQ17_B	124	268	VSS				
	VSS	54	198	CB3_A	VSS	125	269	DQ19_B				
	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	V33				
	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c				
	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t				
				DBI4_A_n				DBI2_B_n				
	CB4_A	58	202	VSS	DQ20_B	129	273	VSS				
	VSS	59	203	CB6_A	VSS	130	274	DQ22_B				
	CB5_A	60	204	VSS	DQ21_B	131	275	VSS				
	VSS	61	205	CB7_A	VSS	132	276	DQ23_B				
	AI FFT_n	62	206	VSS	DQ24_B	133	277	VSS				
	VSS	63	207	RESET_n	VSS	134	278	DQ26_B				
	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS				
	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B				
	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS				
	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c				
	CA2	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t				
	VSS	69	213	CA3_A				DBI3_B_n				
	CA4_A	70	214	VSS	DQ28_B	140	284	VSS				
	VSS	71	215	CA5_A	VSS	141	285	DQ30_B				
	CA6_A	72	216	VSS	DQ29_B	142	286	VSS				
	VSS	73	217	CK_t	VSS	143	287	DQ31_B				
					RFU	144	288	VSS				

Figure 1. Wiring Assignments

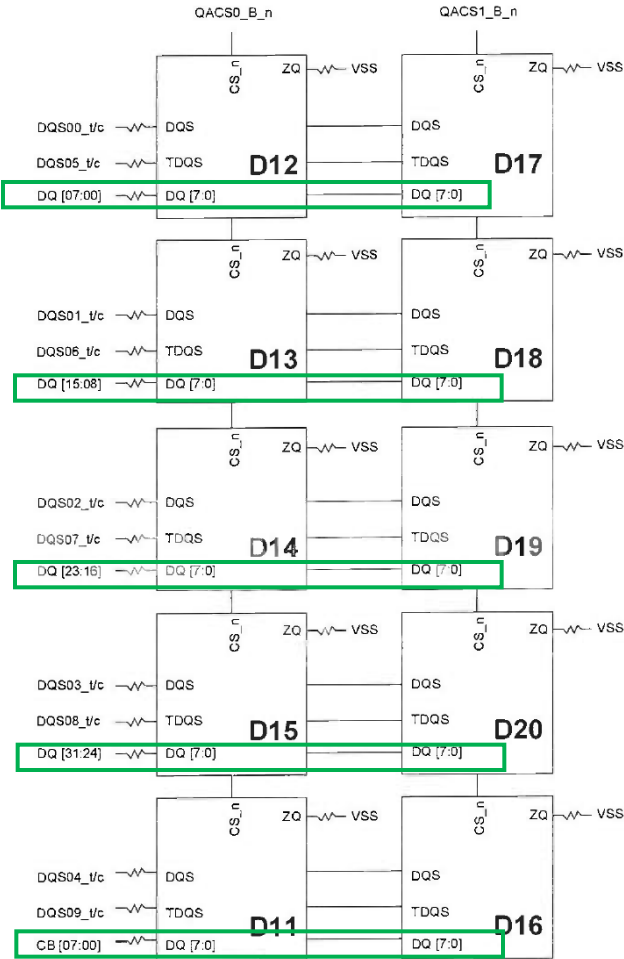
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Ex. CC-89 (Samsung DDR5 RDIMM Datasheet) at 7-8. Channel A Channel B  The diagram illustrates the internal structure of a DDR5 RDIMM, divided into Channel A and Channel B. Channel A contains modules D1 through D10, while Channel B contains modules D3 through D7. Each module is connected to a set of signals including QACK_A, QACA, QACS, TDQS, CS, CA, CK, DQS, and DQ. The DQ signals are highlighted in red boxes in Channel A and green boxes in Channel B, indicating the specific data paths used in the evidence of use.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use Channel A The diagram illustrates the Channel A architecture, showing 10 DRAM components (D01-D10) arranged in a 5x2 grid. Each component has inputs for QACS0_A_n, QACS1_A_n, DQS, TDQS, and DQ. The DQ inputs are highlighted with red boxes and labeled with bit ranges: D01-D05 have DQ [7:0] and D06-D10 have DQ [7:0].
	Ex. CC-89 at 26

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 27

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A Ex. CC-89 at 30

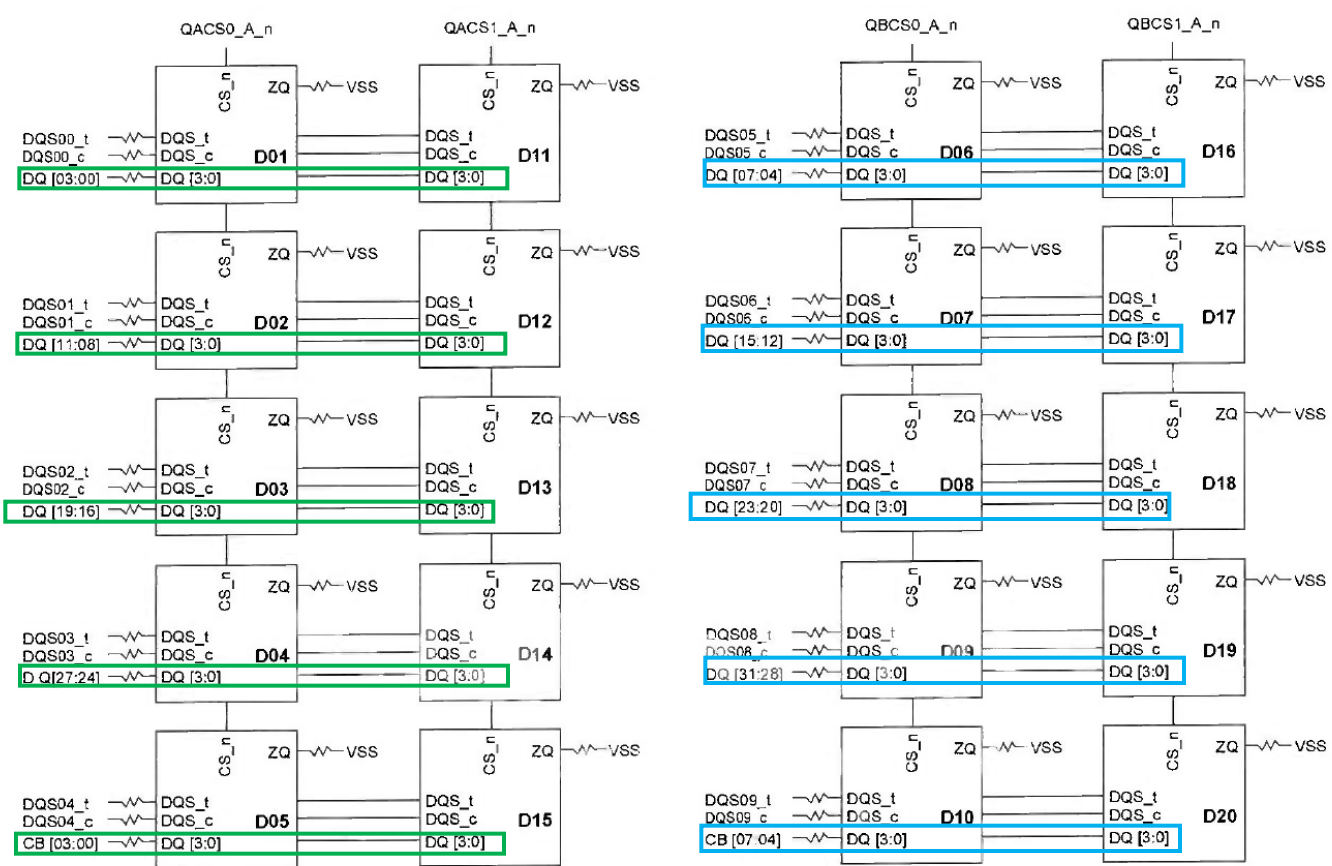
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B Ex. CC-89 at 31

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel A The diagram illustrates the internal structure of Channel A, showing two columns of data storage elements (D01-D05 and D11-D15 on the left; D06-D10 and D16-D20 on the right). Each element is connected to a word line (QACS0_A_n, QACS1_A_n, QBCS0_A_n, QBCS1_A_n) and a bit line (ZQ). The diagram shows data paths for various data words (DQS00_t, DQS00_c, etc.) and their corresponding bit lines (DQ [3:0], DQ [11:08], etc.). Red and yellow boxes highlight specific data paths and bit lines. Ex. CC-89 at 34

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use
	Channel B  Ex. CC-89 at 35 As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates groups of memory devices perform memory read and write operations by

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
	communicating data signals with the memory controller via corresponding subsets of data signal lines (e.g., DQS[]). 6.9.1 DIMM Connectivity Wiring <table><tr><th colspan="6">2Rx4</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Top</td><td>Upper</td><td>QBCK</td><td>QBCS0</td><td>QBCA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS1</td><td>QACA</td><td>1</td></tr><tr><td>Bottom</td><td>Upper</td><td>QDCK</td><td>QBCS1</td><td>QBCA</td><td>1</td></tr></table> <table><tr><th colspan="11">2Rx4 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td>[3:0]</td><td></td><td>[11:8]</td><td></td><td>[19:16]</td><td></td><td>[27:24]</td><td></td><td>CB[3:0]</td><td></td></tr><tr><td>Top</td><td></td><td>[7:4]</td><td></td><td>[15:12]</td><td></td><td>[23:20]</td><td></td><td>[31:28]</td><td></td><td>CB[7:4]</td></tr><tr><td>Bottom</td><td>[3:0]</td><td></td><td>[11:8]</td><td></td><td>[19:16]</td><td></td><td>[27:24]</td><td></td><td>CB[3:0]</td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:4]</td><td></td><td>[15:12]</td><td></td><td>[23:20]</td><td></td><td>[31:28]</td><td></td><td>CB[7:4]</td></tr></table> <table><tr><th colspan="11">2Rx4 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:4]</td><td></td><td>[15:12]</td><td></td><td>[23:20]</td><td></td><td>[31:28]</td><td>CB[3:0]</td><td></td></tr><tr><td>Top</td><td>[3:0]</td><td></td><td>[11:8]</td><td></td><td>[19:16]</td><td></td><td>[27:24]</td><td></td><td></td><td>CB[7:4]</td></tr><tr><td>Bottom</td><td></td><td>[7:4]</td><td></td><td>[15:12]</td><td></td><td>[23:20]</td><td></td><td>[31:28]</td><td>CB[3:0]</td><td></td></tr><tr><td>Bottom</td><td>[3:0]</td><td></td><td>[11:8]</td><td></td><td>[19:16]</td><td></td><td>[27:24]</td><td></td><td></td><td>CB[7:4]</td></tr></table> <table><tr><th colspan="6">1Rx4</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QBCS0</td><td>QACA</td><td>0</td></tr></table> <table><tr><th colspan="11">1Rx4 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td>[3:0]</td><td></td><td>[11:8]</td><td></td><td>[19:16]</td><td></td><td>[27:24]</td><td></td><td>CB[3:0]</td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:4]</td><td></td><td>[15:12]</td><td></td><td>[23:20]</td><td></td><td>[31:28]</td><td></td><td>CB[7:4]</td></tr></table> <table><tr><th colspan="11">1Rx4 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 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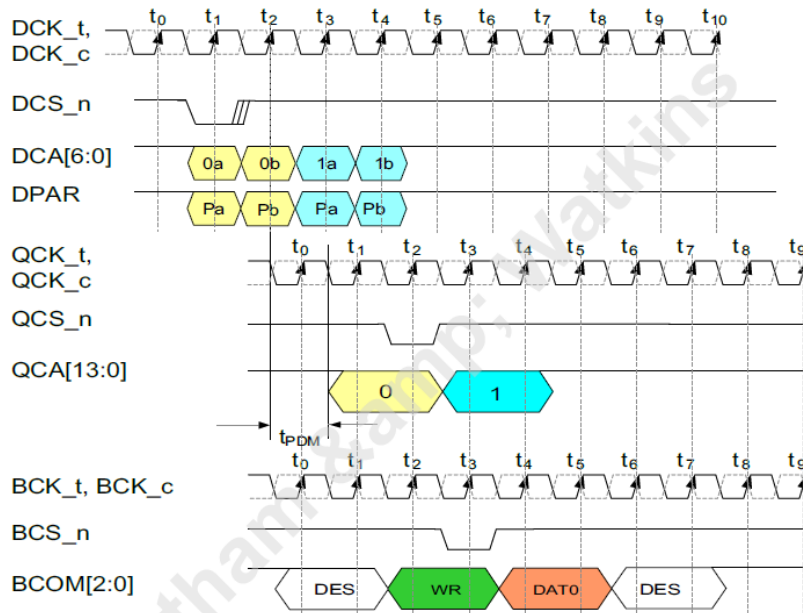
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart (Samsung)

Claim 1	Evidence of Use																																																																																																																																																														
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr></table> <table><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:0]</td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td></td><td>CB[7:0]</td></tr></table> <table><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:0]</td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td>[31:24]</td><td></td><td></td><td>CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing memory devices receiving data from the memory controller via subsets of data signal lines, e.g., [3:0], [11:8], [15:12]). Furthermore, the memory devices in the Samsung Practicing '407 Products are configurable to perform memory read or write operations by communicating data signals in accordance with their respective local clock signals (e.g., “QACK_A_t,” “QBCK_A_t,” “QCCK_A_t,” “QDCK_A_t,” “QACK_B_t,” “QBCK_B_t,” “QCCK_B_t,” “QDCK_B_t,” “QACK_A_c,” “QBCK_A_c,” “QCCK_A_c,” “QDCK_A_c,” “QACK_B_c,” “QBCK_B_c,” “QCCK_B_c,” and “QDCK_B_c”) as those signals are delivered to the “CK_t” and “CK_c” inputs of their respective memory devices. As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition further demonstrates the chip select signal (QCS) and command and address signal (QCA) received by each DDR memory device is synchronized with the local clock (QCK):	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top				[15:8]				[31:24]			Bottom		[7:0]			[23:16]					CB[7:0]	1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top		[7:0]			[23:16]						Bottom				[15:8]			[31:24]			CB[7:0]
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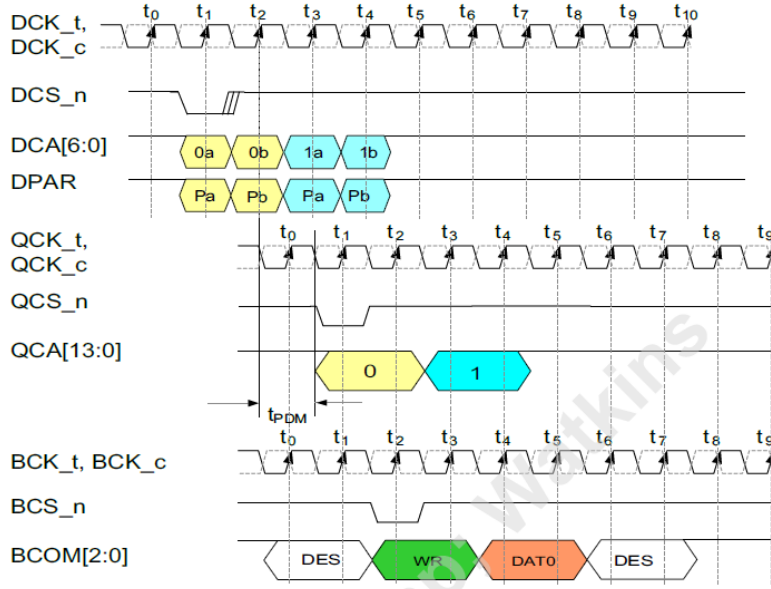
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	3.4.1 Double Data Rate DCA (cont'd) DCK_t DCK_c DCS_n DCA[6:0] DPAR QCK_t QCK_c QCS_n QCA[13:0] t₀ t₁ t₂ t₃ t₄ t₅ t₆ t₇ t₈ t₉ t_{PDM} NOTE Timing Diagram for 0 nCK command latency adder. Figure 7 — One UI DRAM Command Timing Diagram DCK_t DCK_c DCS_n DCA[6:0] DPAR QCK_t QCK_c QCS_n QCA[13:0] t₀ t₁ t₂ t₃ t₄ t₅ t₆ t₇ t₈ t₉ t_{PDM} NOTE Timing Diagram for 0 nCK command latency adder. Figure 8 — Two UI DRAM Command Timing Diagram Ex. CC-32 (JEDEC 82-512) at 16. See also Ex. CC-30 (JEDEC 82-513) at 16.

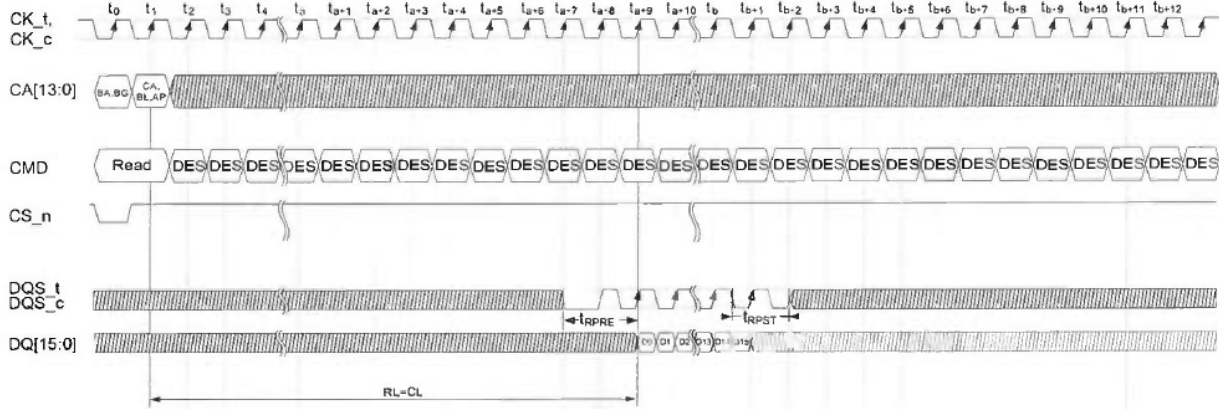
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Claim 1	Evidence of Use
	The JEDEC DDR5 RCD definition explains the Samsung Practicing '407 Products are configurable to perform READ and WRITE operations synchronized with the local clock QCK. <i>See, e.g.,</i>:  Figure 11 — SDR1 Mode Example of DRAM WR Command with RW00[1] = 0

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	 Figure 12 — SDR2 Mode Example of DRAM WR Command with RW00[1] = 1 Ex. CC-32 (JEDEC 82-512) at 18-19. See also Ex. CC-30 (JEDEC 82-513) at 18-19. Further, the timing diagrams for read and write operations show that DQ signals appear after a set latency in terms of clock cycles after the module C/A signals are received. As another example, DQ is output/received in relation to DQS, and DQS is aligned with clock signals in predetermined manner. So the DQ output/input is in accordance with local clock signals.

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	4.7 Read Operation The Read Operation causes the DRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and bank/group address for the data to be retrieved from the array is provided. The data is driven by the DRAM on its DQ pins RL (CL) cycles after the Read Command along with the proper waveform on the DQS inputs. Read Latency (RL or CL) is defined from the Read command to data and is not affected by the Read DQS offset timing (MR40 OP[2:0]). 4.7.1 READ Burst Operation During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode.  NOTES: 1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. In this example, Read DQS Offset Timing is set to 0 Clocks. Figure 35 — READ Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 135.

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	4.8.2 Write Burst Operation The following write timing diagrams are to help understand the meaning of each write parameter; the diagrams are just examples. The details of each parameter are defined separately. In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate. NOTES: 1. BL=16, 2tCK Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization. Figure 49 — WRITE Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 147.